Object Detection and Semantic Segmentation

Danna Gurari
University of Colorado Boulder
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https://home.cs.colorado.edu/~DrG/Courses/NeuralNetworksAndDeepLearning/AboutCourse.html

Review

- Last lecture:
  - Representation learning
  - Pretrained features
  - Fine-tuning
  - Training neural networks: hardware & software
  - Programming tutorial

- Assignments (Canvas)
  - Lab assignment 2 due earlier today
  - Problem set 3 due next week

- Questions?

Today’s Topics

- Problems
- Applications
- PASCAL VOC detection challenge: R-CNNs
- PASCAL VOC semantic segmentation challenge: fully convolutional networks

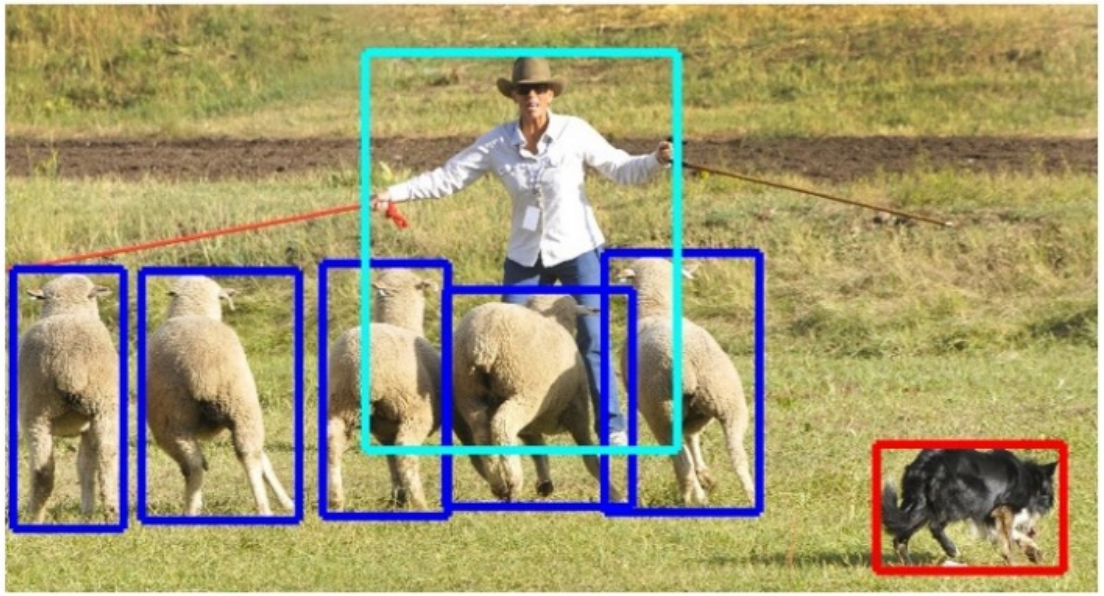
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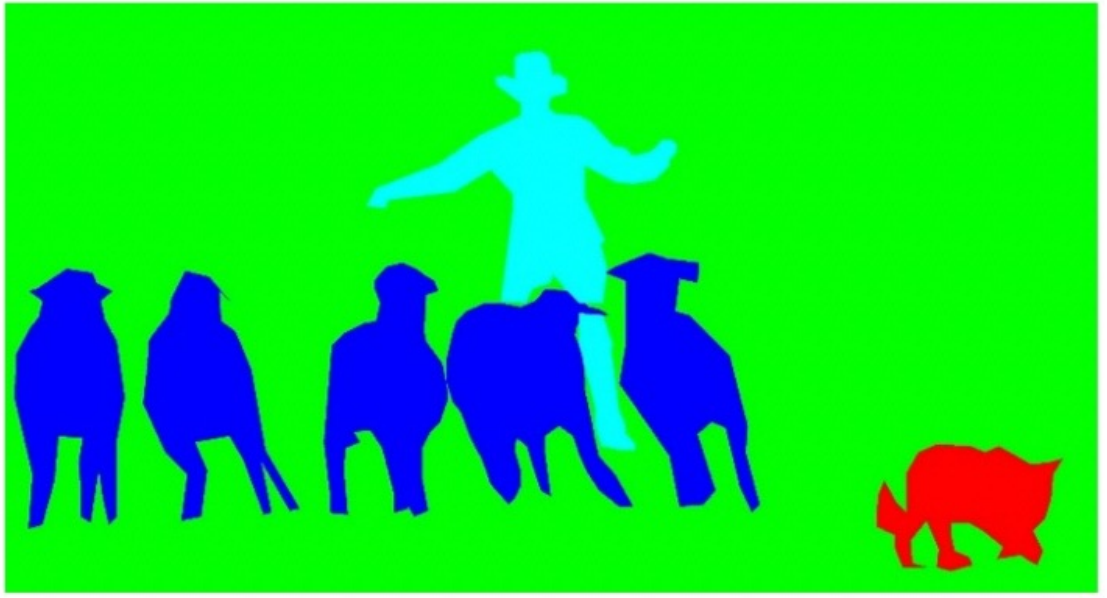
Problem Definition

- Localize content of interest

Object detection



Semantic segmentation

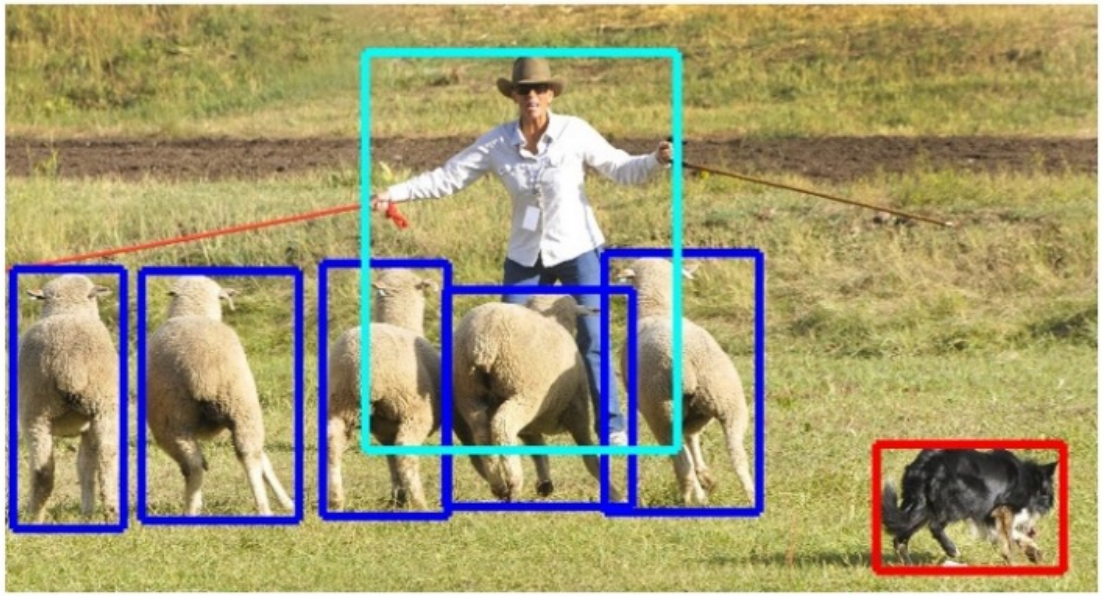


Lin et al. Microsoft COCO: Common Objects in Context. ECCV 2014.

Problem Definition

- Localize content of interest

Object detection



Use bounding box to locate every instance of an object from pre-specified categories

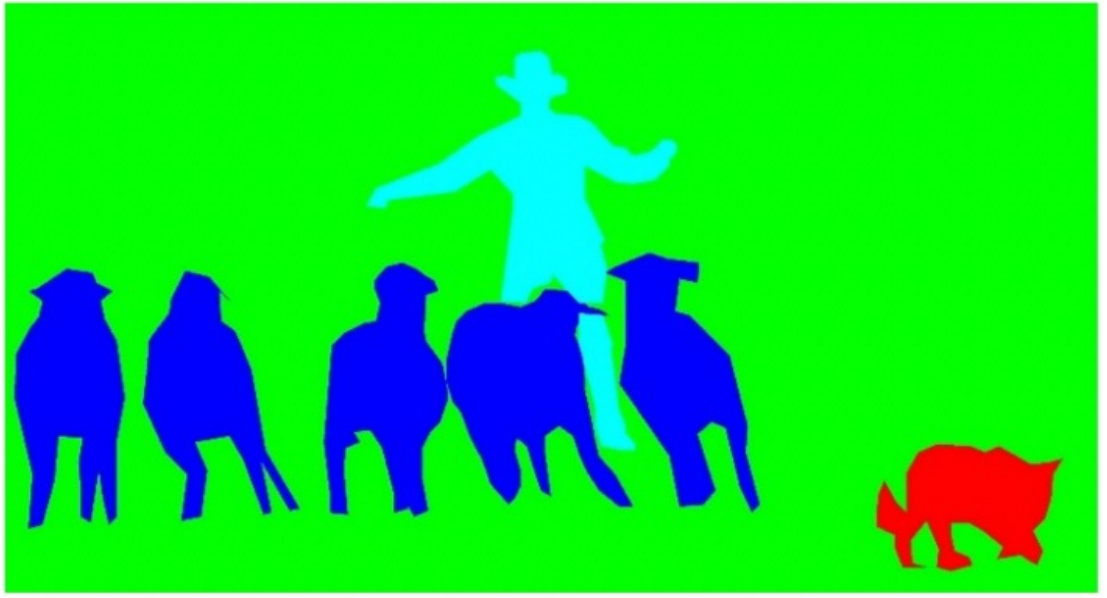
Lin et al. Microsoft COCO: Common Objects in Context. ECCV 2014.

Problem Definition

- Localize content of interest

Locate all pixels that belong to pre-specified categories

Semantic segmentation



Lin et al. Microsoft COCO: Common Objects in Context. ECCV 2014.

Problem Definition

- Localize content of interest

Locate all pixels that belong to pre-specified categories

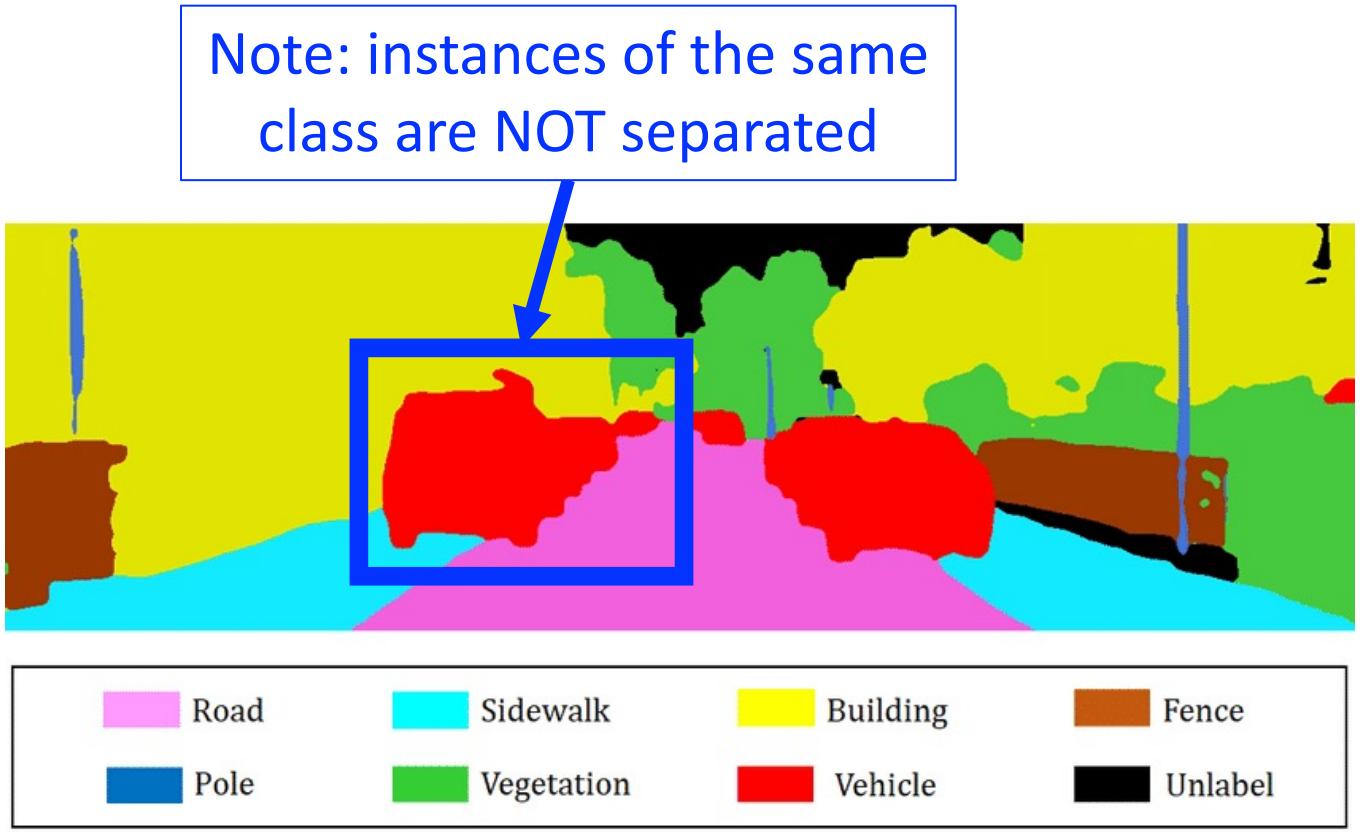


Image Source: Jeong, Yoon, and Park. Sensors 2018.

Object Segmentation vs Detection

- Why choose object “segmentation” vs “detection”?

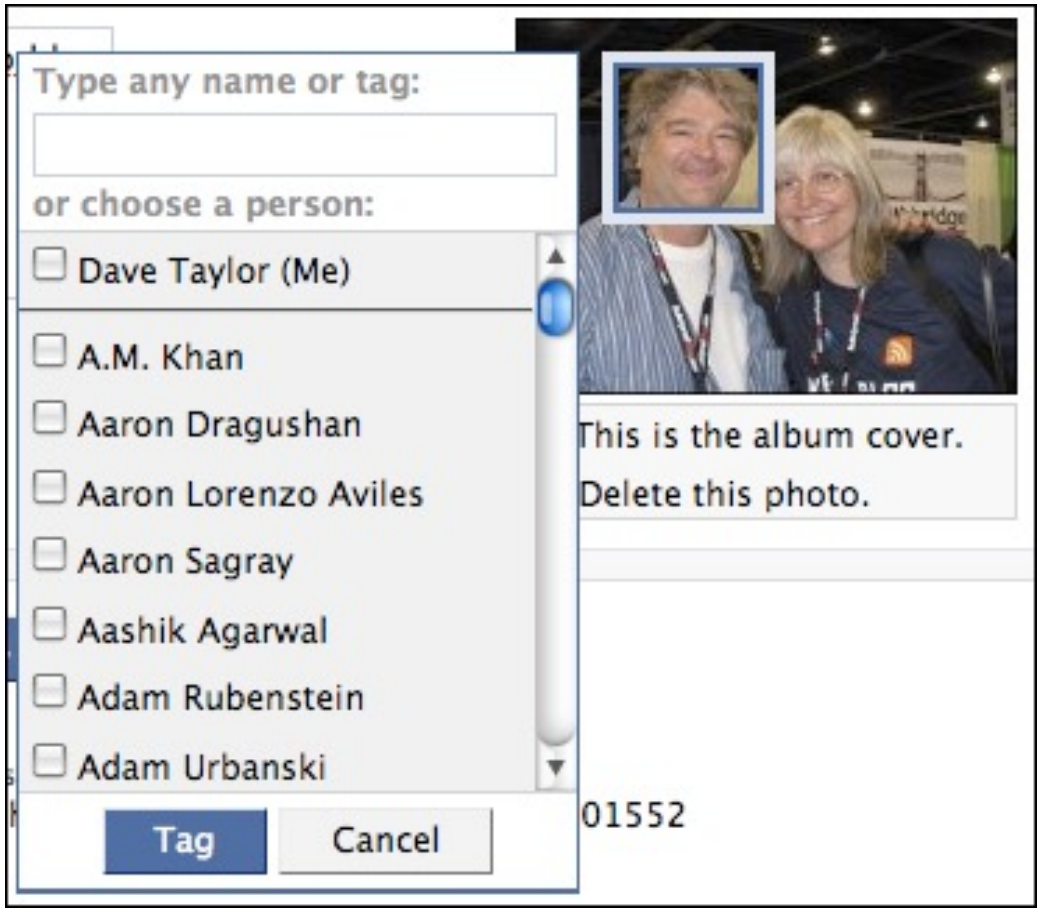


http://mmcheng.net/msra10k/

Today's Topics

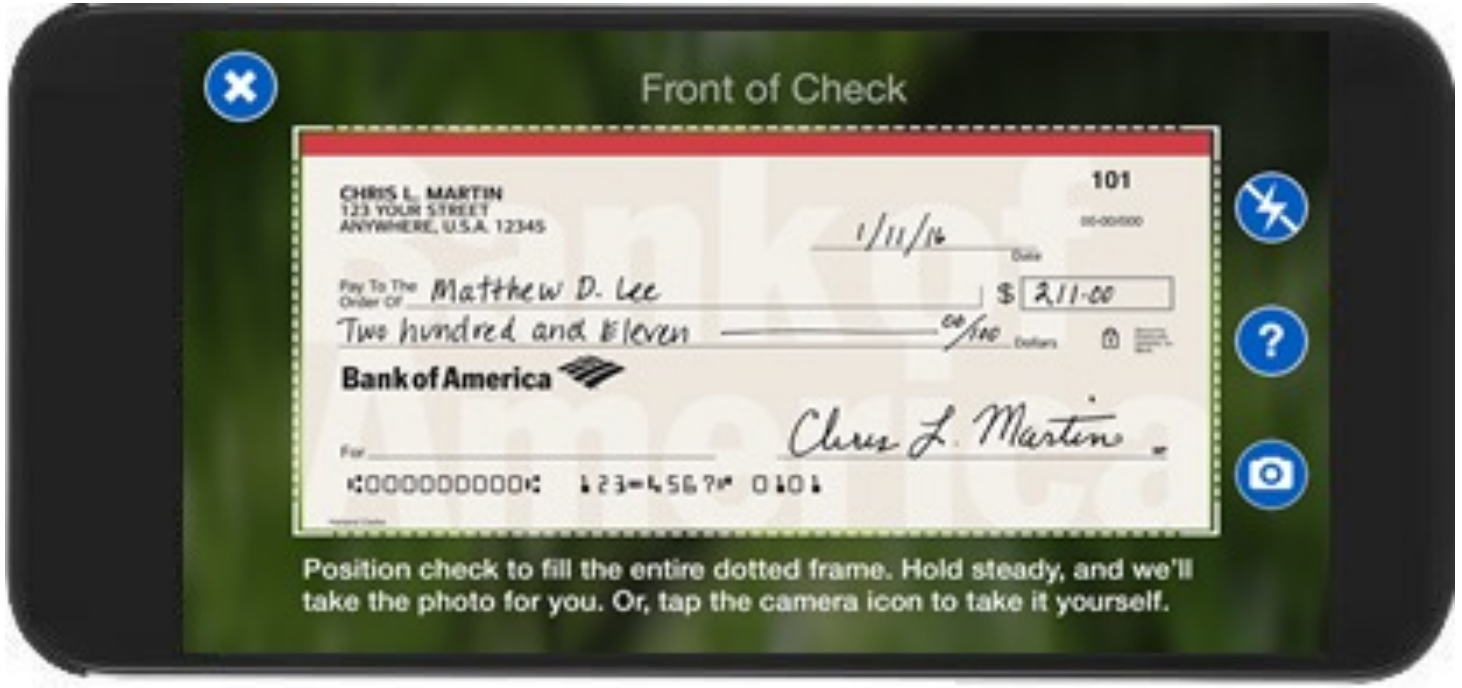
- Problems
- Applications
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Social Media



Face detection
(e.g., Facebook)

Banking



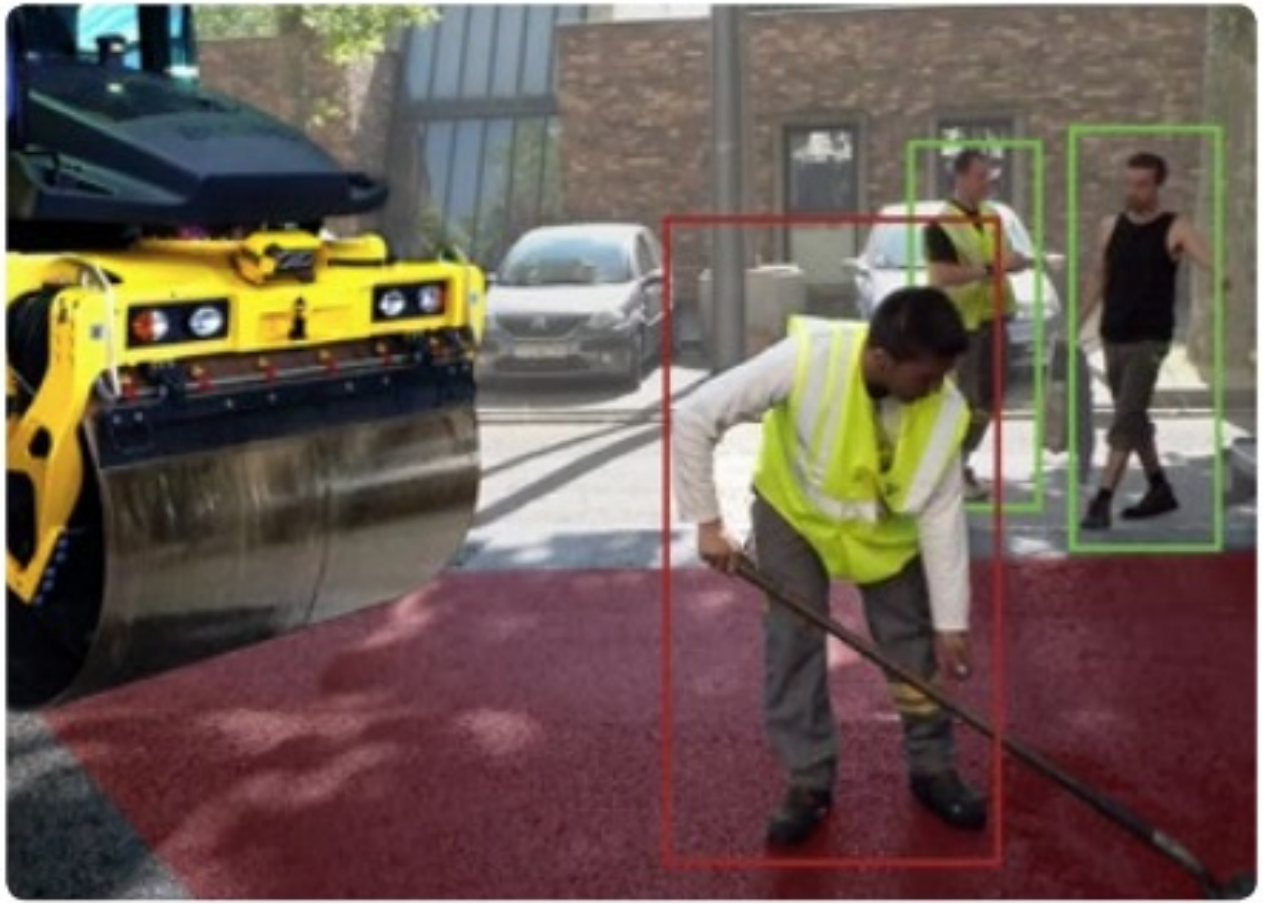
Mobile check deposit
(e.g., Bank of America)

Transportation



License Plate Detection (e.g., AllGoVision)

Construction Safety



Pedestrian Detection
(e.g., Blaxtair)

http://media.brintex.com/Occurrence/121/Brochure/3435/brochure.pdf

Counting



Counting Fish (e.g., SalmonSoft)
http://www.wecountfish.com/?page_id=143



Business Traffic Analytics

Remodeling Inspiration



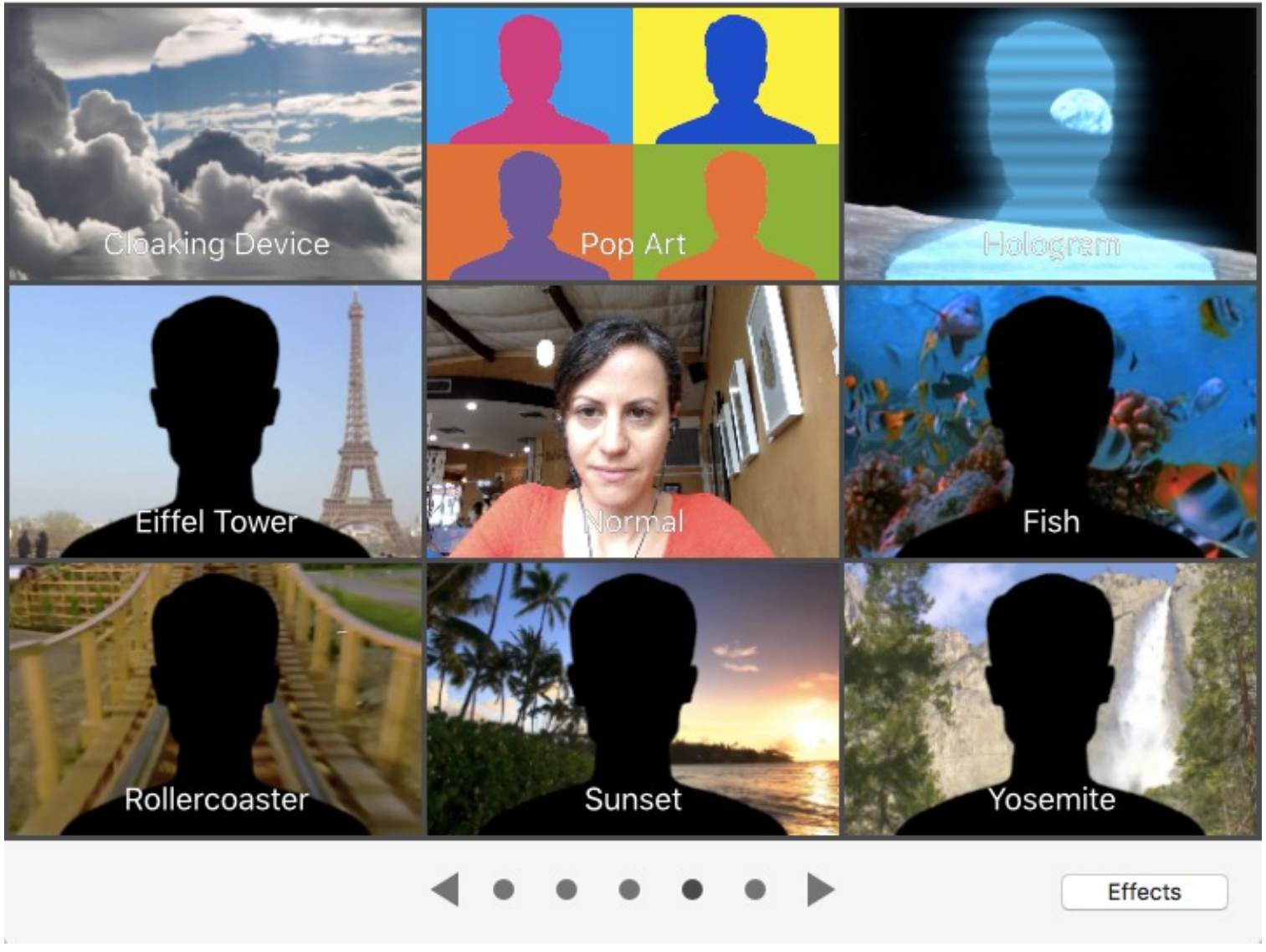
(a) Target photo



(b) Retextured

Bell et al; SIGGRAPH; 2013

Rotoscoping (many examples on Wikipedia)





https://www.starnow.co.uk/ahmedmohammed1/photos/4650871/before-and-after-rotoscopinggreen-screening

Disease Diagnosis; e.g., PathAI

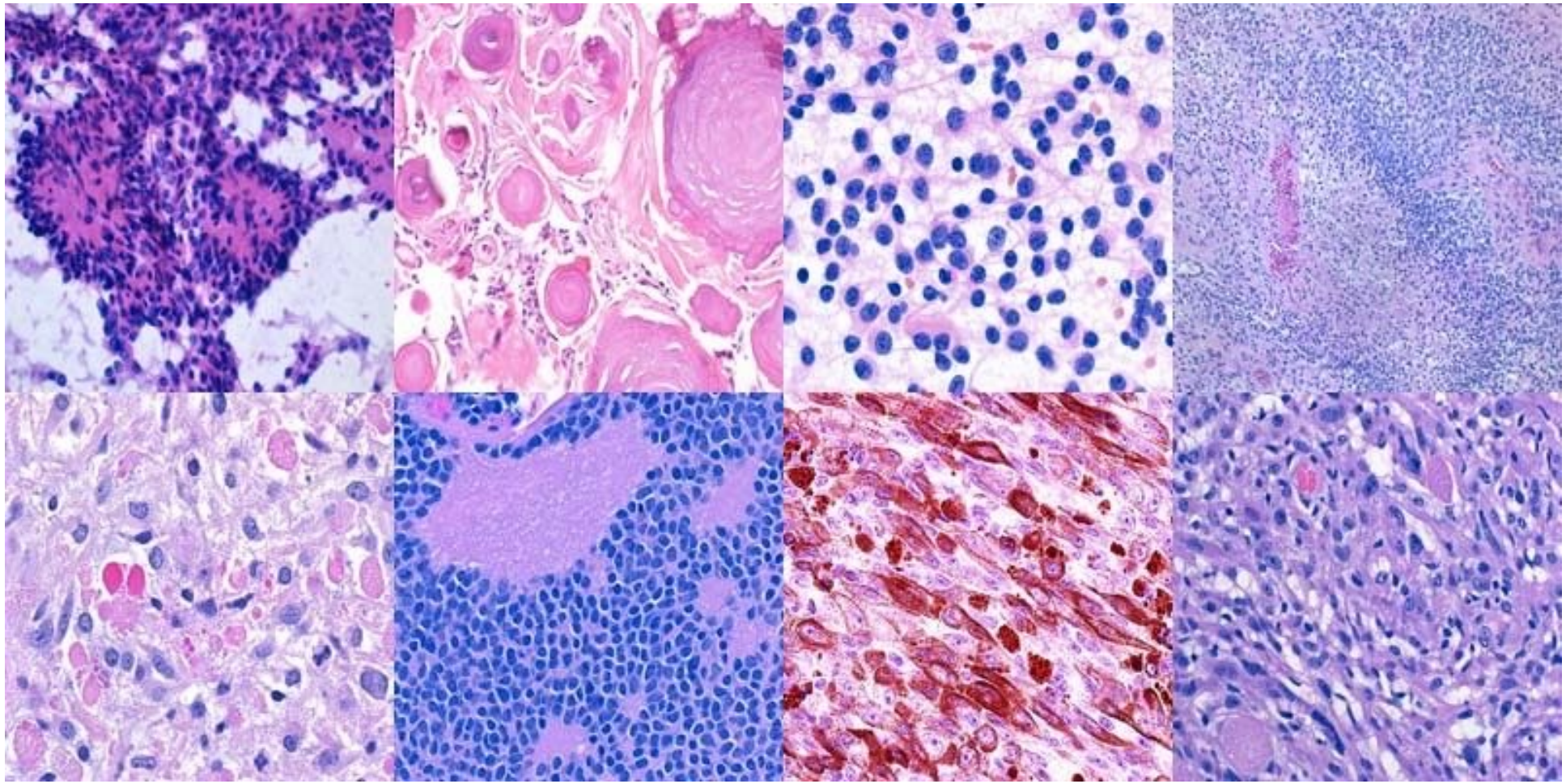
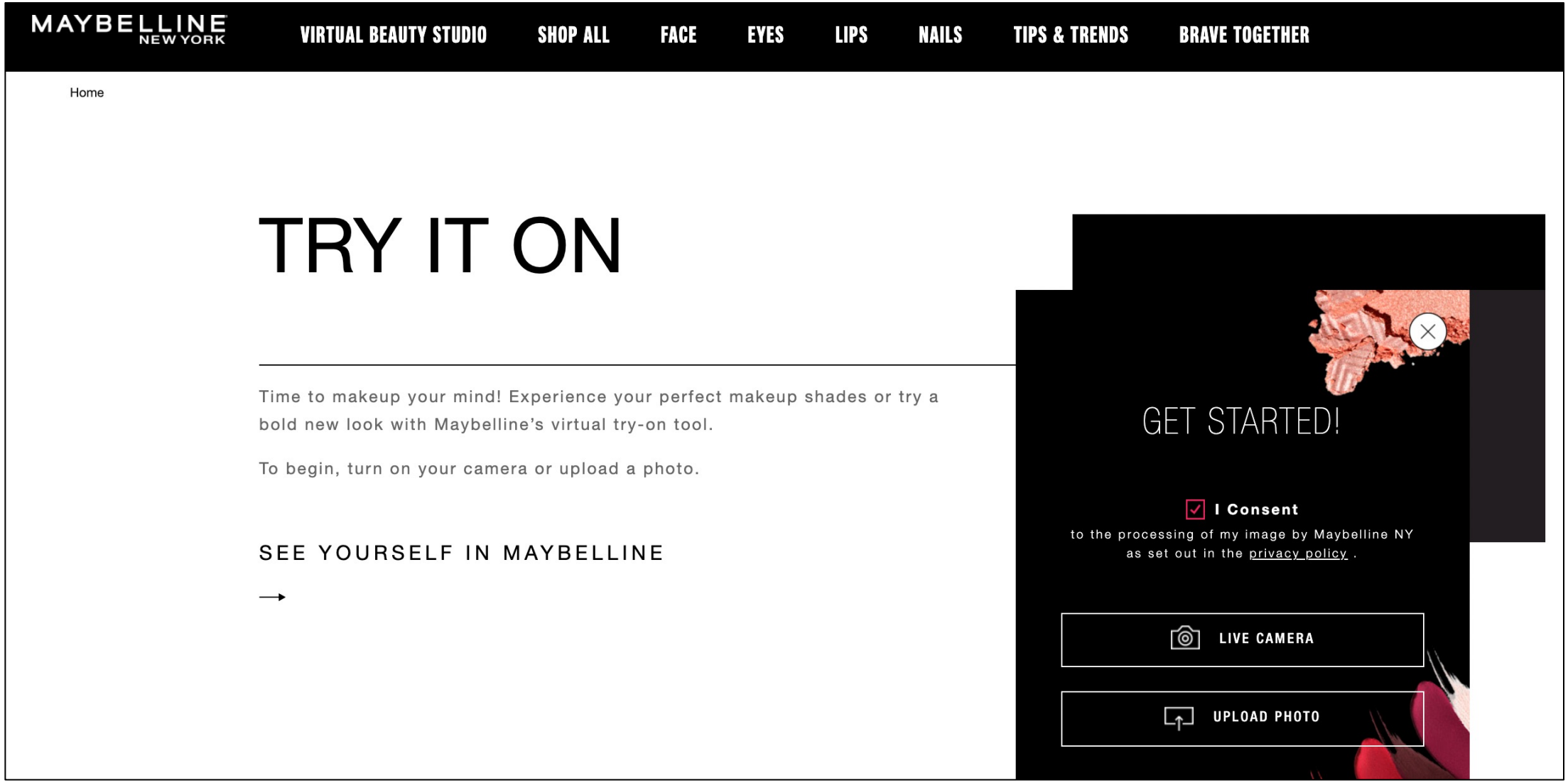


Figure Source: https://pathology.jhu.edu/brain-tumor/grading-classification

Face Makeover



Demo: https://www.maybelline.com/virtual-try-on-makeup-tools

Self-Driving Vehicles



Figure Source: https://www.inc.com/kevin-j-ryan/self-driving-cars-powered-by-people-playing-games-mighty-ai.html

Can you think of any other applications?

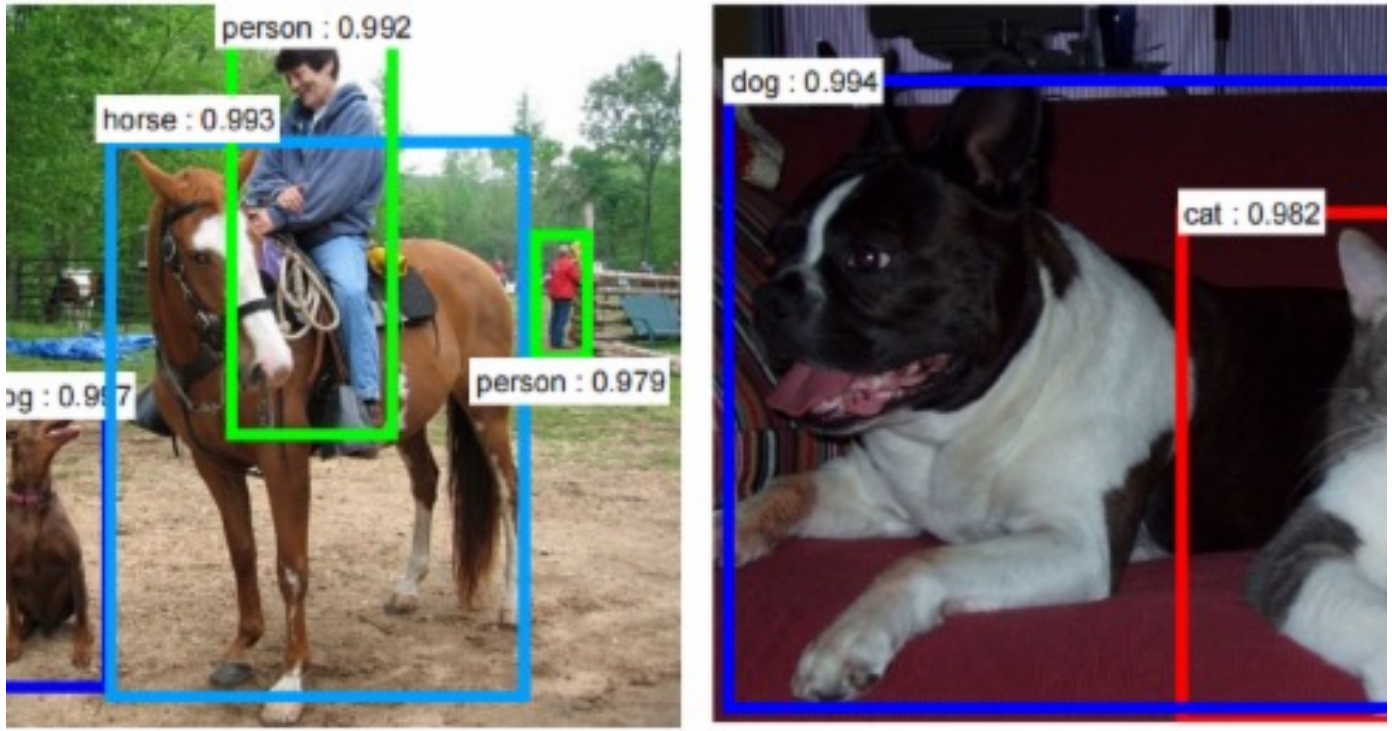
How to solve problems like these with neural networks?

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VOC Challenge

- Goal: locate all instances of 20 object categories with BBs
- Dataset: 11,530 images collected from Flickr and annotated by annotators at University of Leeds

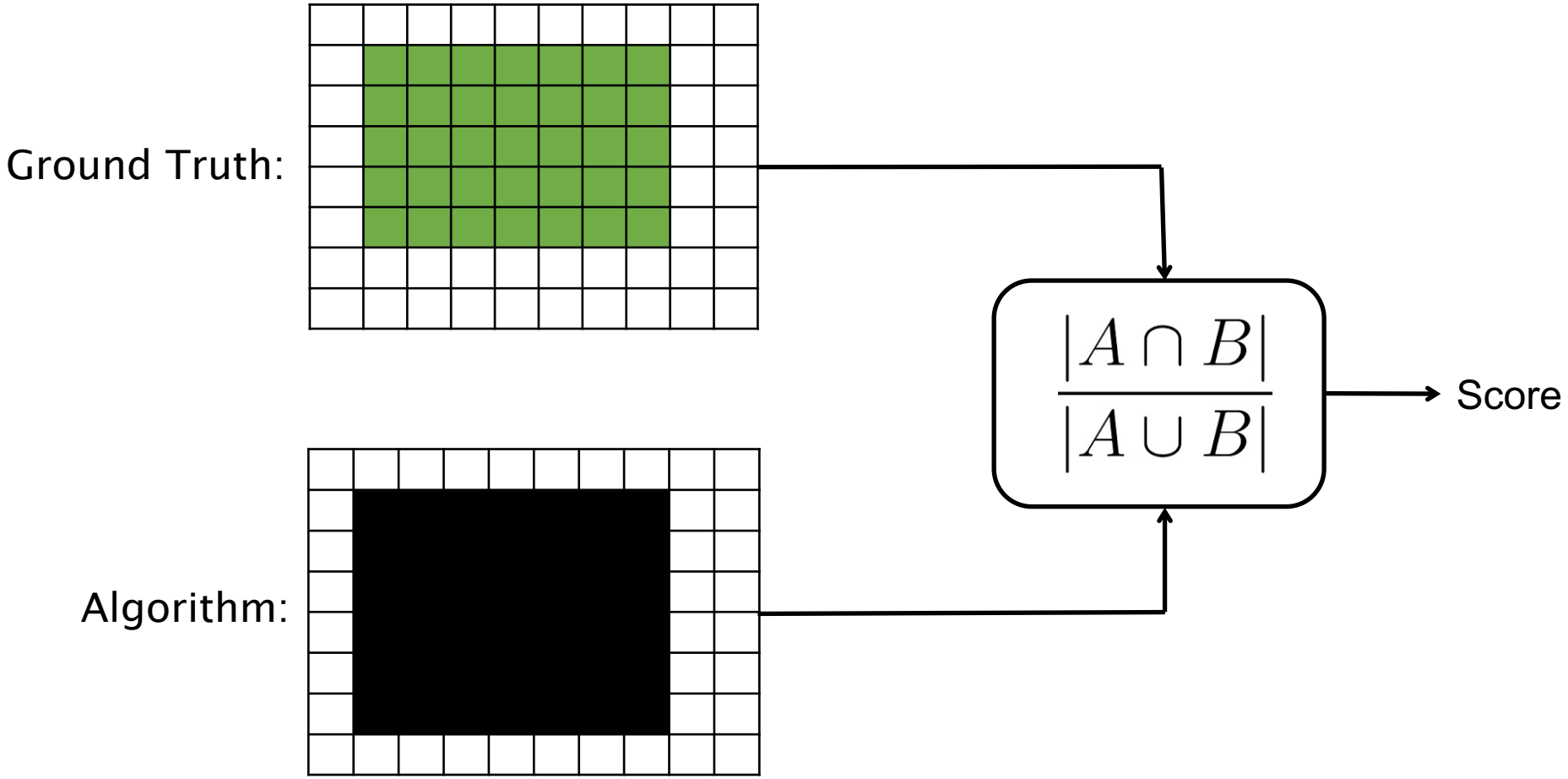


https://cv-tricks.com/artificial-intelligence/object-detection-using-deep-learning-for-advanced-users-part-1/

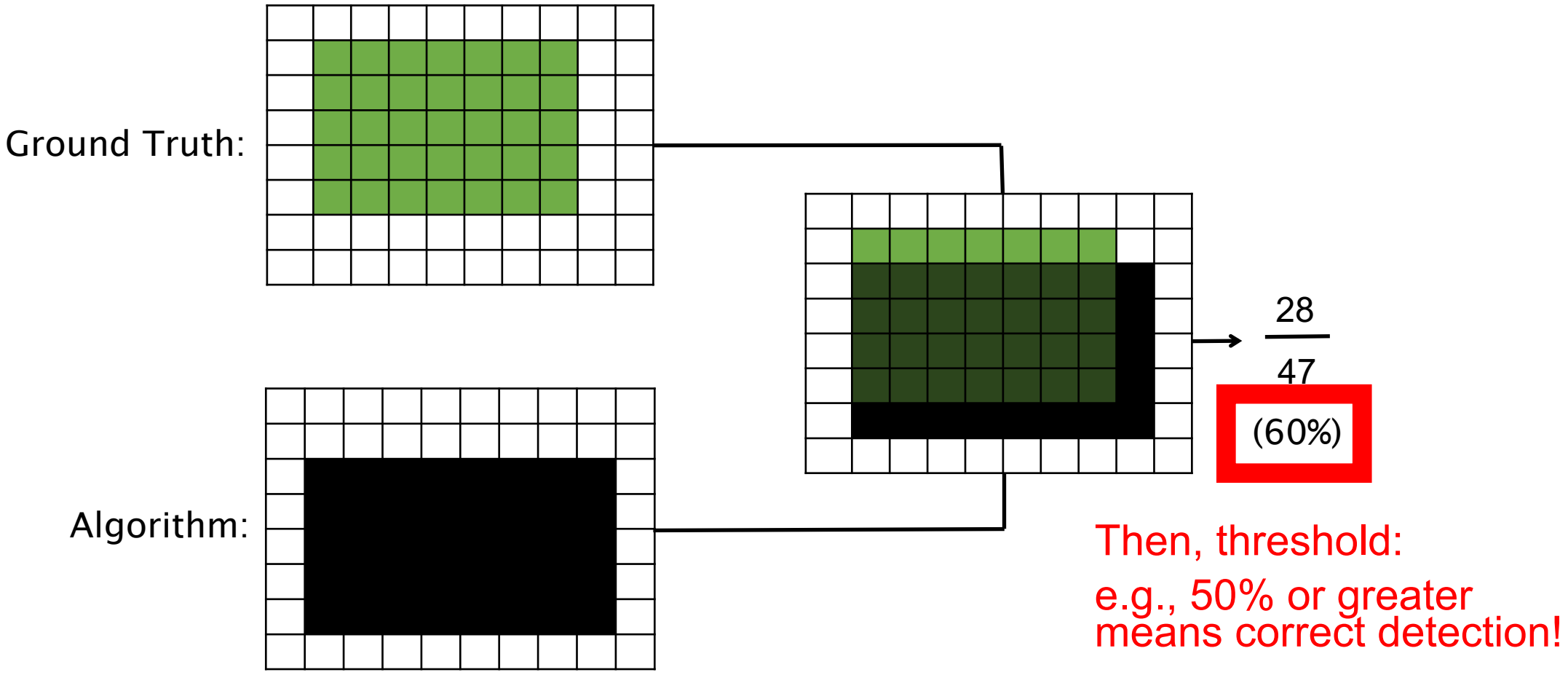
Dataset location: http://host.robots.ox.ac.uk/pascal/VOC/index.html

Mark Everingham, Luc Van Gool, Christopher K. I. Williams, John Winn, and Andrew Zisserman. The PASCAL Visual Object Classes (VOC) Challenge. IJCV 2010.

MSCOCO Challenge: Evaluation Metric (IoU)

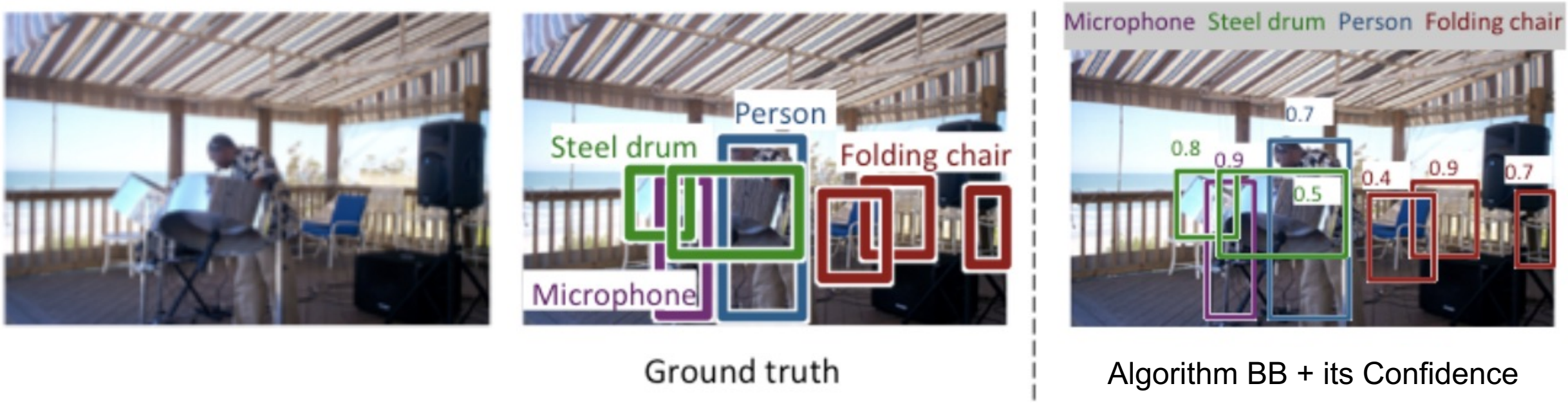


MSCOCO Challenge: Evaluation Metric (IoU)



MSCOCO Challenge: Evaluation Metric (mAP)

- For each object class (e.g., cat, dog, …), compute:
  - Precision: fraction of correct detections from all detections when using a 0.5 IoU



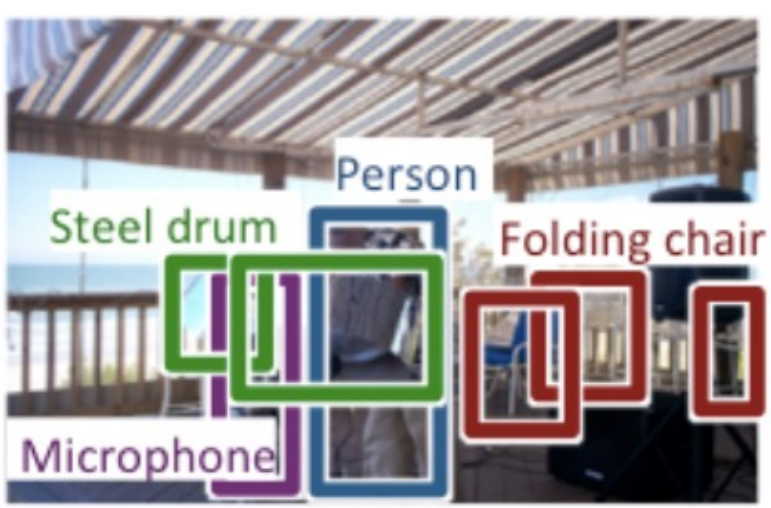
[Russakovsky et al; IJCV 2015]

https://jonathan-hui.medium.com/map-mean-average-precision-for-object-detection-45c121a31173

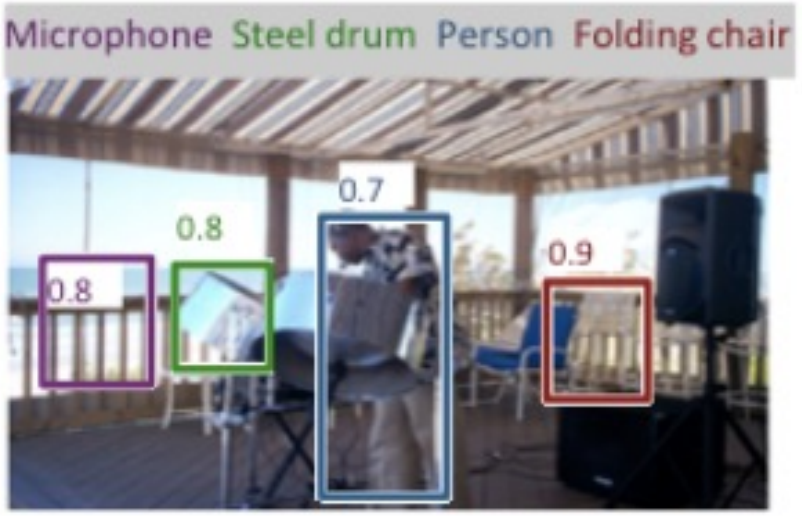
MSCOCO Challenge: Evaluation Metric (mAP)

- For each object class (e.g., cat, dog, …), compute:
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Ground truth



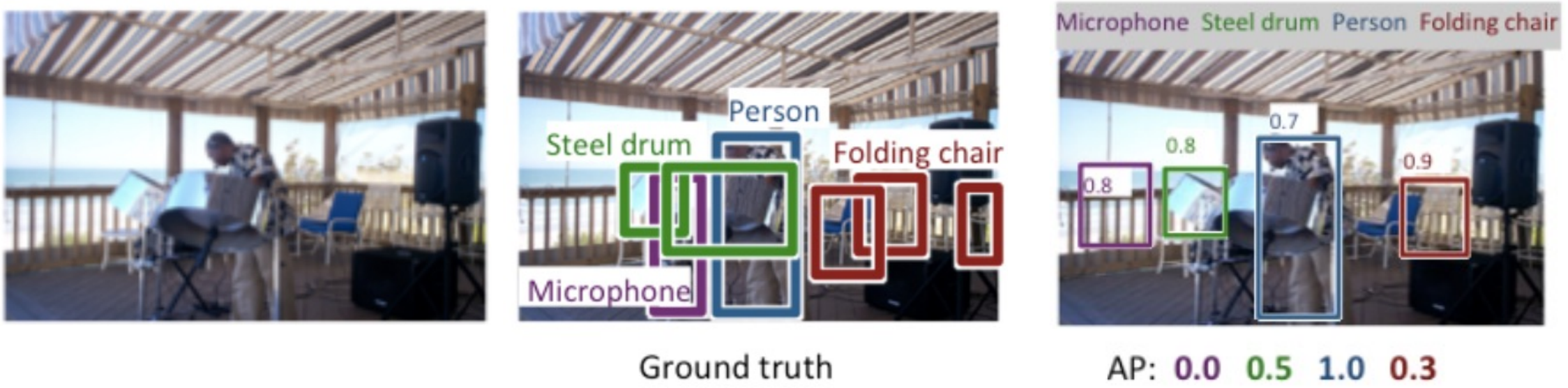
AP: ? ? ? ?

[Russakovsky et al; IJCV 2015]

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MSCOCO Challenge: Evaluation Metric (mAP)

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[Russakovsky et al; IJCV 2015]

https://jonathan-hui.medium.com/map-mean-average-precision-for-object-detection-45c121a31173

Naïve Solution: Sliding Window Approach



Image Source: https://yourboulder.com/boulder-neighborhood-downtown/

Naïve Solution: Sliding Window Approach

Car?
Car?
Car?
Car?
Car?
Car?
Car?
Car?
Car?



Image Source: https://yourboulder.com/boulder-neighborhood-downtown/

Naïve Solution: Sliding Window Approach

Would this detect the person?



Image Source: https://yourboulder.com/boulder-neighborhood-downtown/

Naïve Solution: Sliding Window Approach



Would this detect the car?

Image Source: https://yourboulder.com/boulder-neighborhood-downtown/

Naïve Solution: Sliding Window Approach

Need to test windows of different scales…

Car?
Car?
Car?
Car?
Car?
Car?
Car?
Car?
Car?
Car?
Car?



Image Source: https://yourboulder.com/boulder-neighborhood-downtown/

Naïve Solution: Sliding Window Approach

Would this scale detect the person?



Image Source: https://yourboulder.com/boulder-neighborhood-downtown/

Naïve Solution: Sliding Window Approach



Would this scale detect the car?

Image Source: https://yourboulder.com/boulder-neighborhood-downtown/

Naïve Solution: Sliding Window Approach

Need to test windows of different aspect ratios…



Image Source: https://yourboulder.com/boulder-neighborhood-downtown/

Naïve Solution: Sliding Window Approach

Would this aspect ratio detect the person?

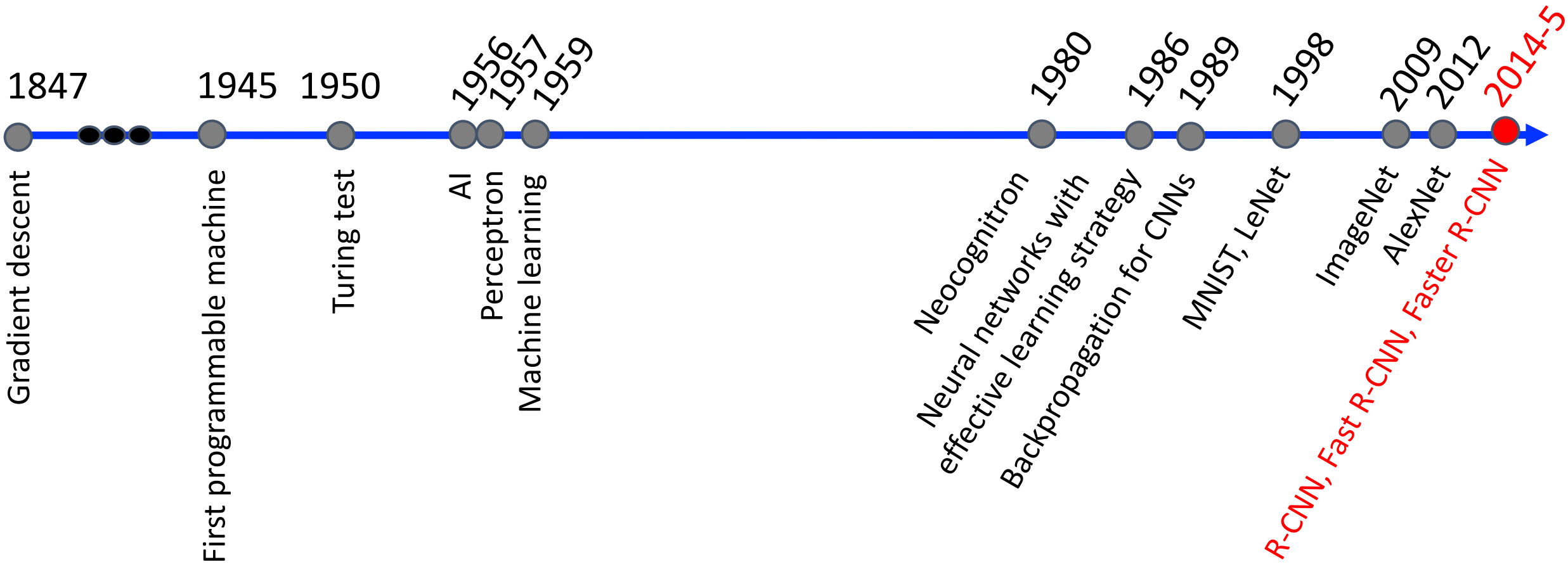


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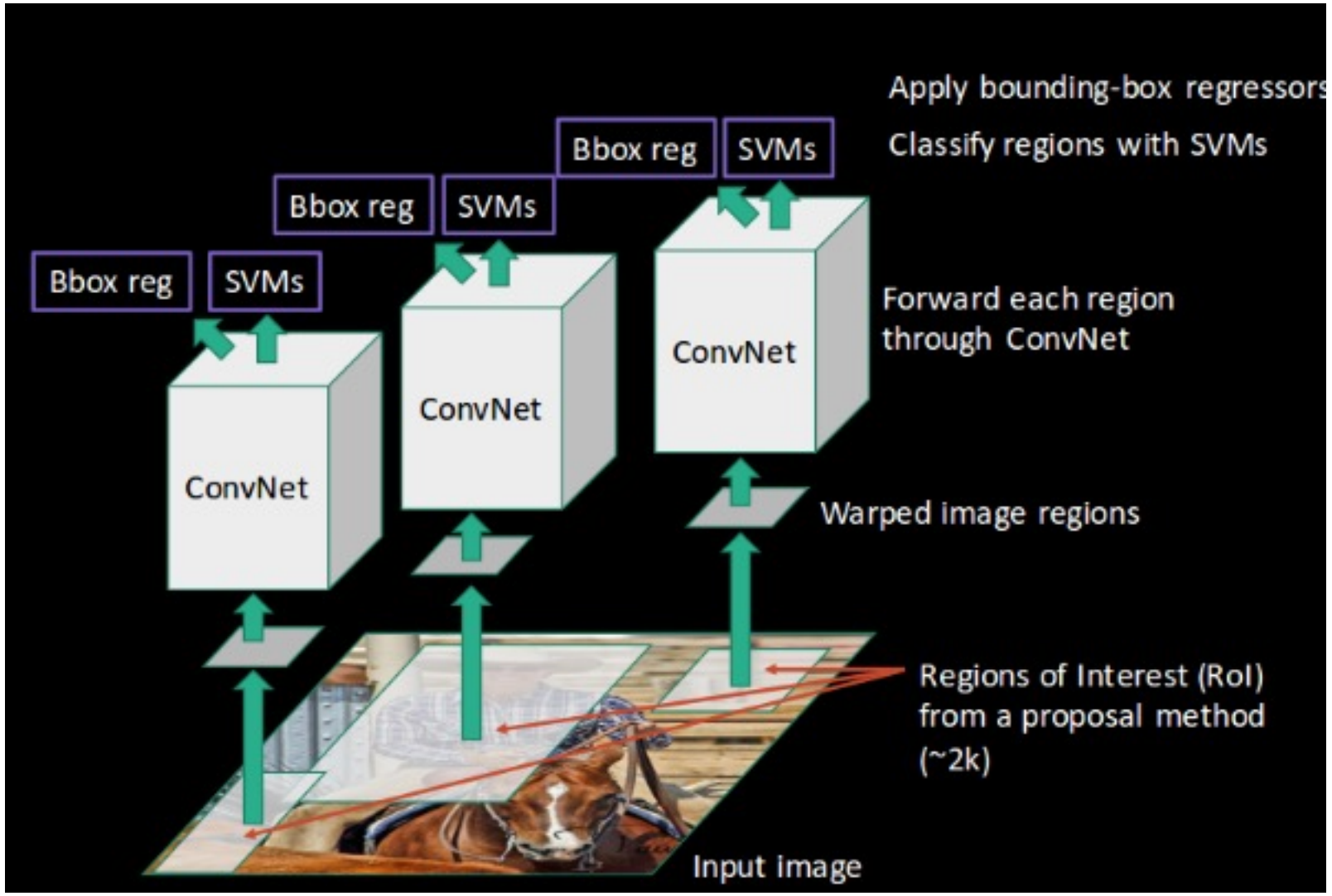
Naïve Solution: Sliding Window Approach

- Sliding window approach: must test different locations at…
  - Different scales
  - Different aspect ratios (e.g., person vs car or car taken at different angles)
- Number of regions to test? (e.g., 1920 x 1080 image)
  - Easily can explode to hundreds of thousands or millions of windows
- Key limitation
  - Very slow!

Historical Context: R-CNN Methods



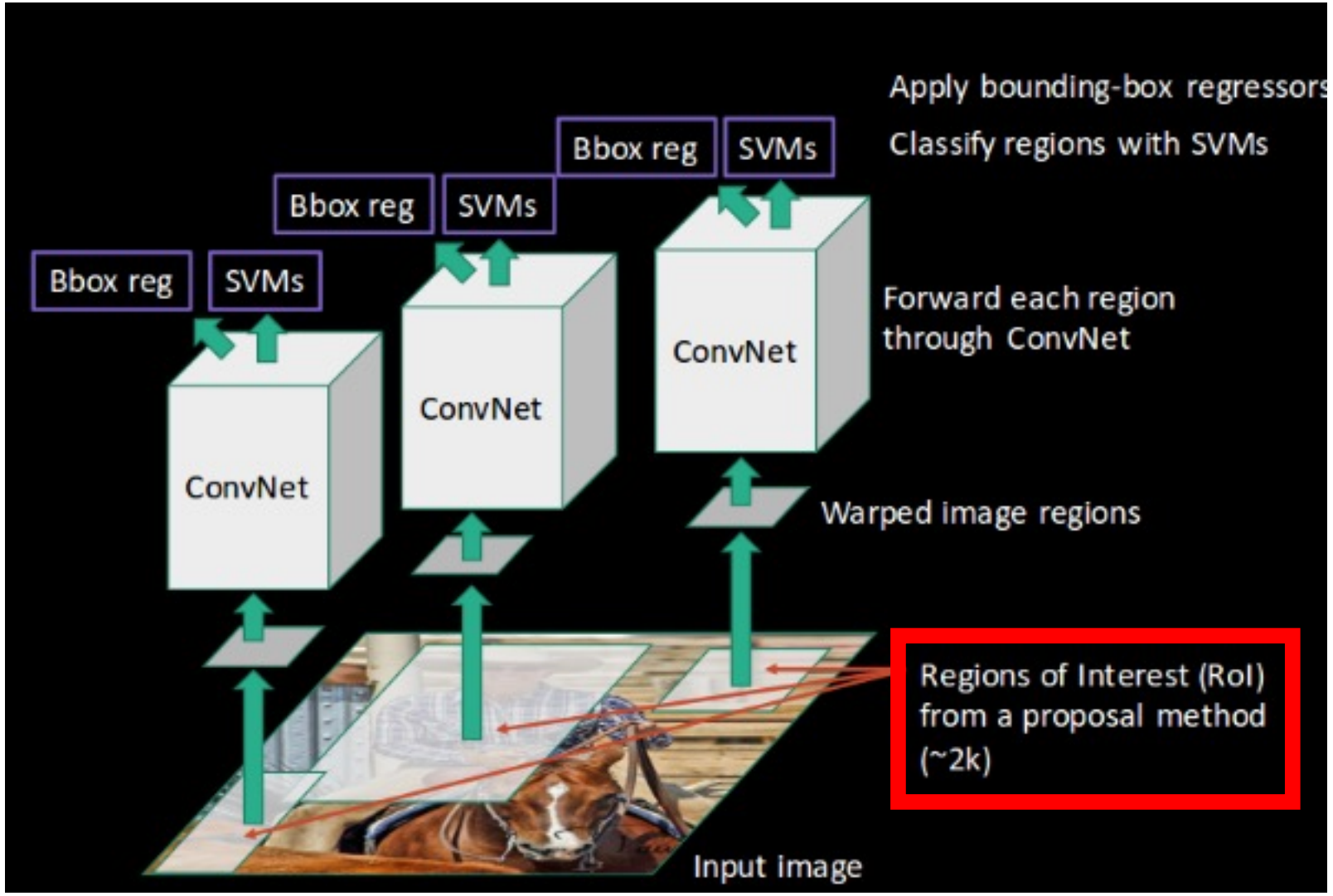
R-CNN



- First CNN to outperform hand-crafted features on detection challenges
- Named after technique: **R**egion proposals with **CNN** features

Figure Source: https://www.analyticsvidhya.com/blog/2018/10/a-step-by-step-introduction-to-the-basic-object-detection-algorithms-part-1/

R-CNN



Locate “object”-like regions using objectness methods

- Considerably fewer regions than sliding window approach
- Regions likely contain objects of interest (i.e., high recall)

Figure Source: https://www.analyticsvidhya.com/blog/2018/10/a-step-by-step-introduction-to-the-basic-object-detection-algorithms-part-1/

R-CNN

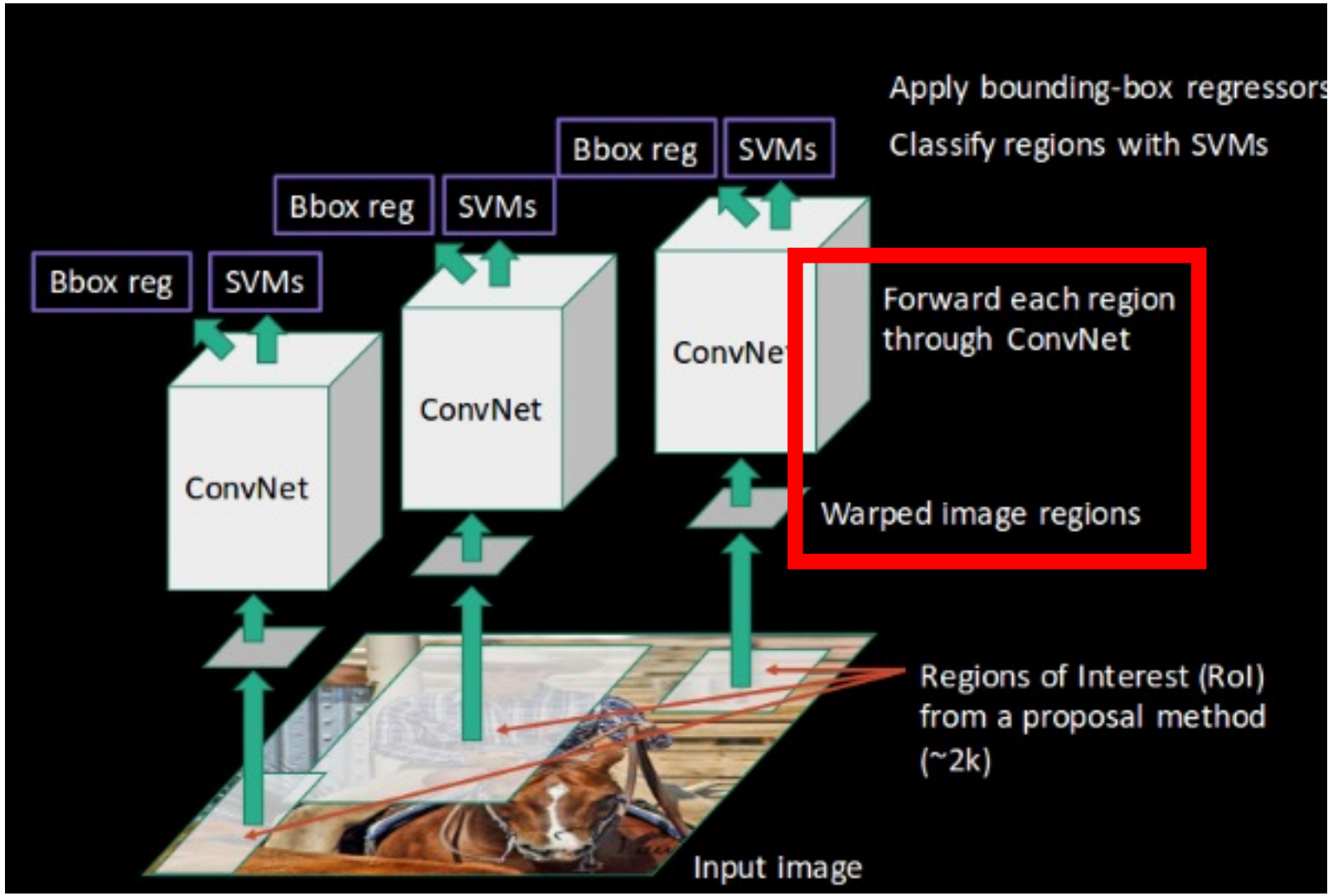


Figure Source: https://www.analyticsvidhya.com/blog/2018/10/a-step-by-step-introduction-to-the-basic-object-detection-algorithms-part-1/

Describe Each Region with Fixed-length Vector

Fine-tune: replace final layer (FC8) of pretrained AlexNet (on ImageNet) to contain the number of categories in detection dataset and train for image classification (use max IoU class, if IoU >= 0.5)

How many classes would be predicted in fine-tuning?

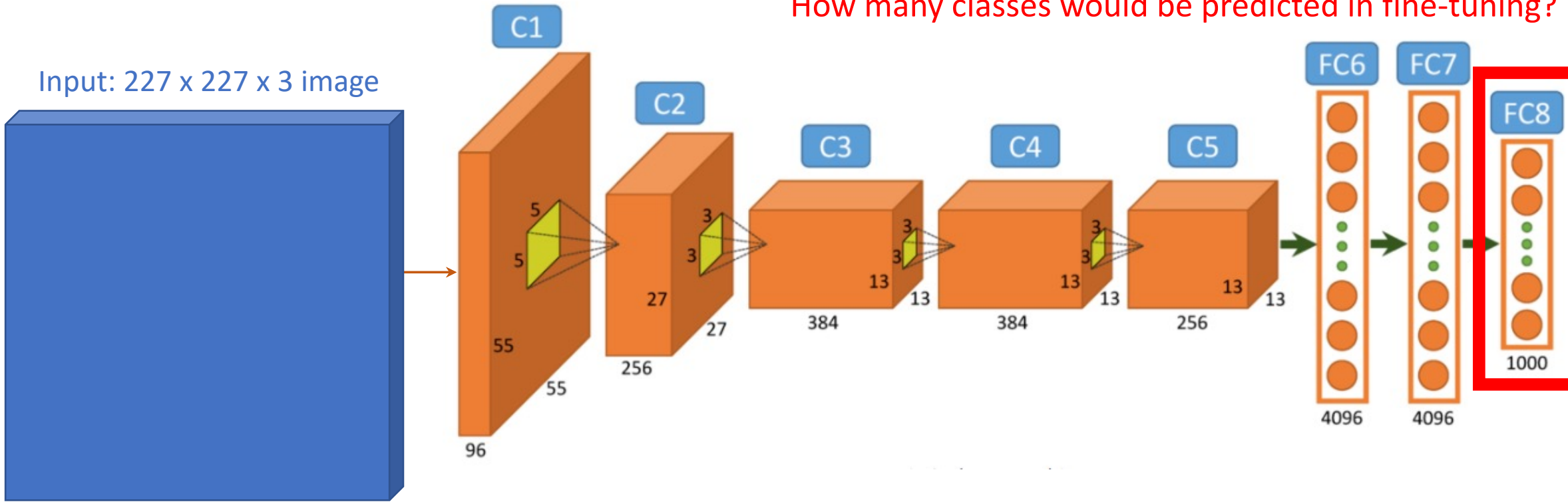


Image Source: https://www.researchgate.net/figure/Architecture-of-Alexnet-From-left-to-right-input-to-output-five-convolutional-layers_fig2_312303454

Describe Each Region with Fixed-length Vector

Use the FC7 layer from a fine-tuned AlexNet model

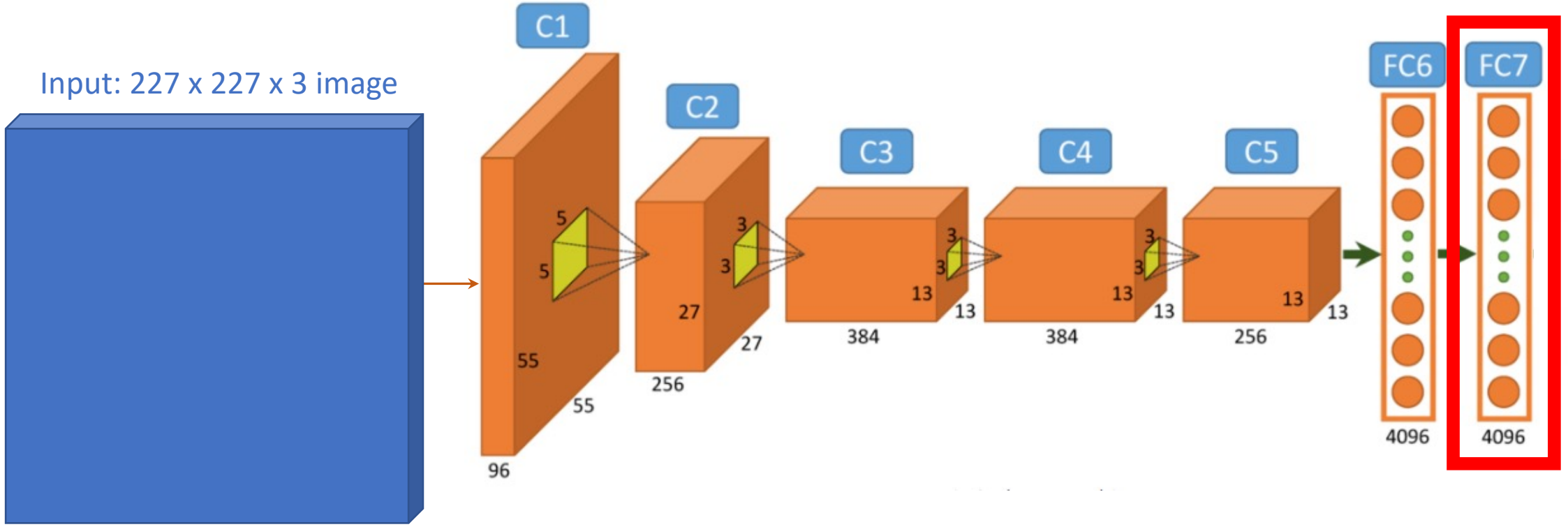


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Describe Each Region with Fixed-length Vector

Challenge: how to resize a proposed region to the required size?



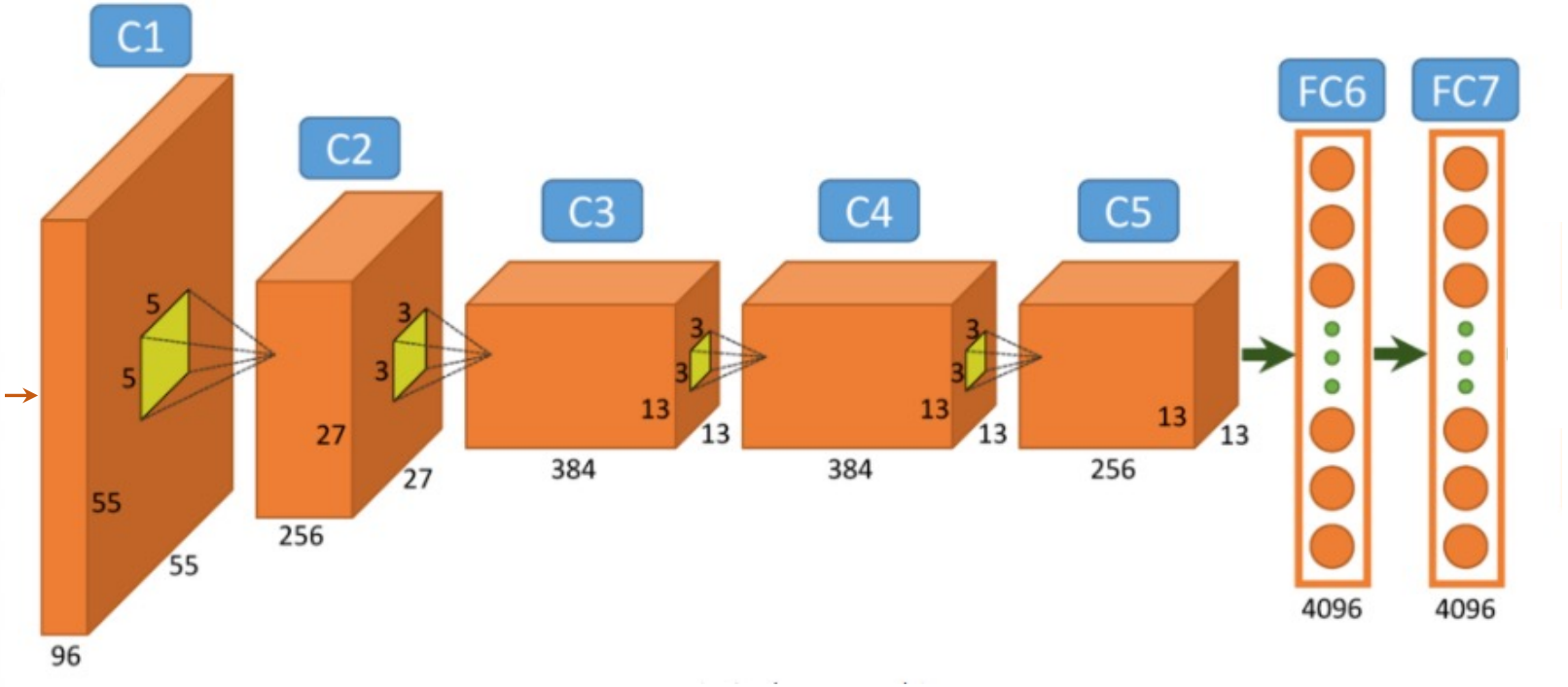


Image Source: https://www.researchgate.net/figure/Architecture-of-Alexnet-From-left-to-right-input-to-output-five-convolutional-layers_fig2_312303454

Describe Each Region with Fixed-length Vector

Region anisotropically scaled to fit the required resolution

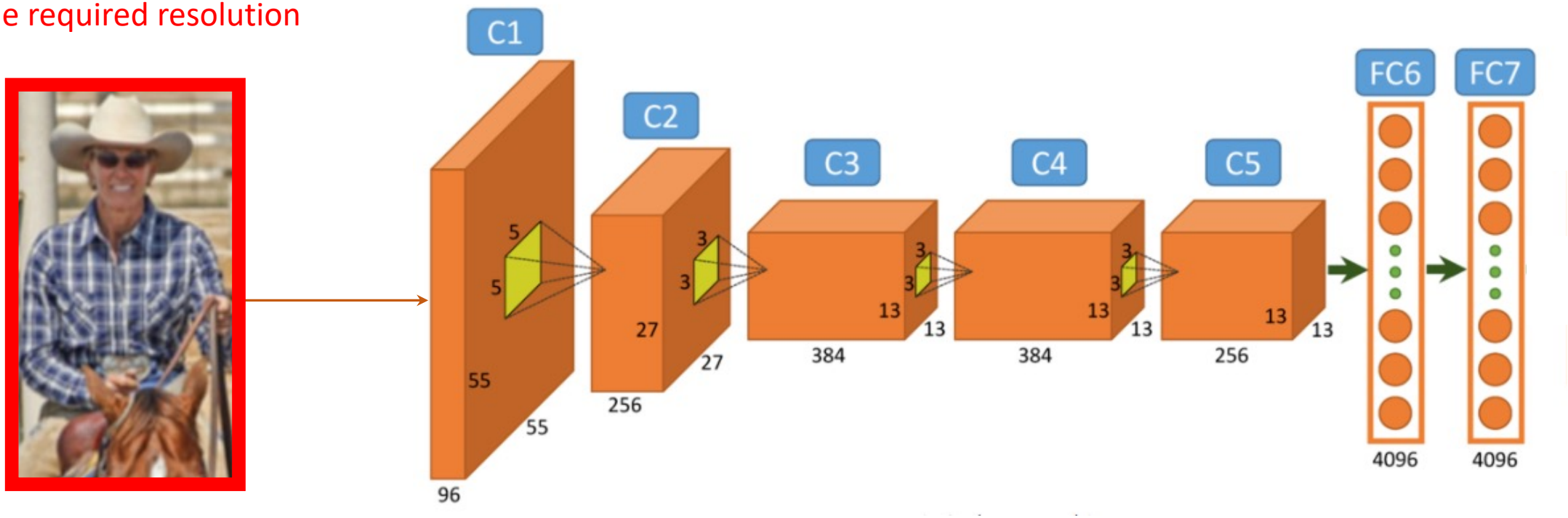


Image Source: https://www.researchgate.net/figure/Architecture-of-Alexnet-From-left-to-right-input-to-output-five-convolutional-layers_fig2_312303454

Describe Each Region with Fixed-length Vector

Input: 227 x 227 x 3 image



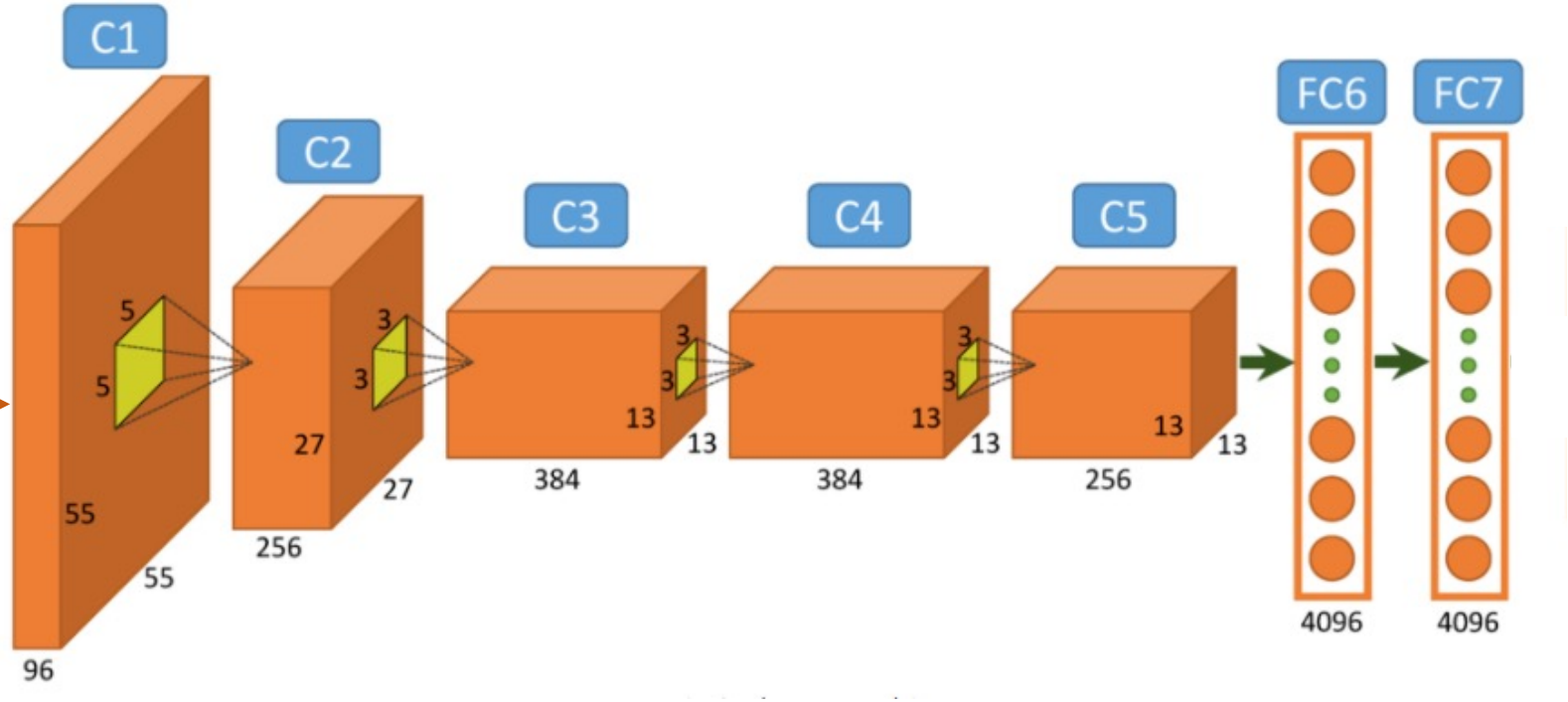
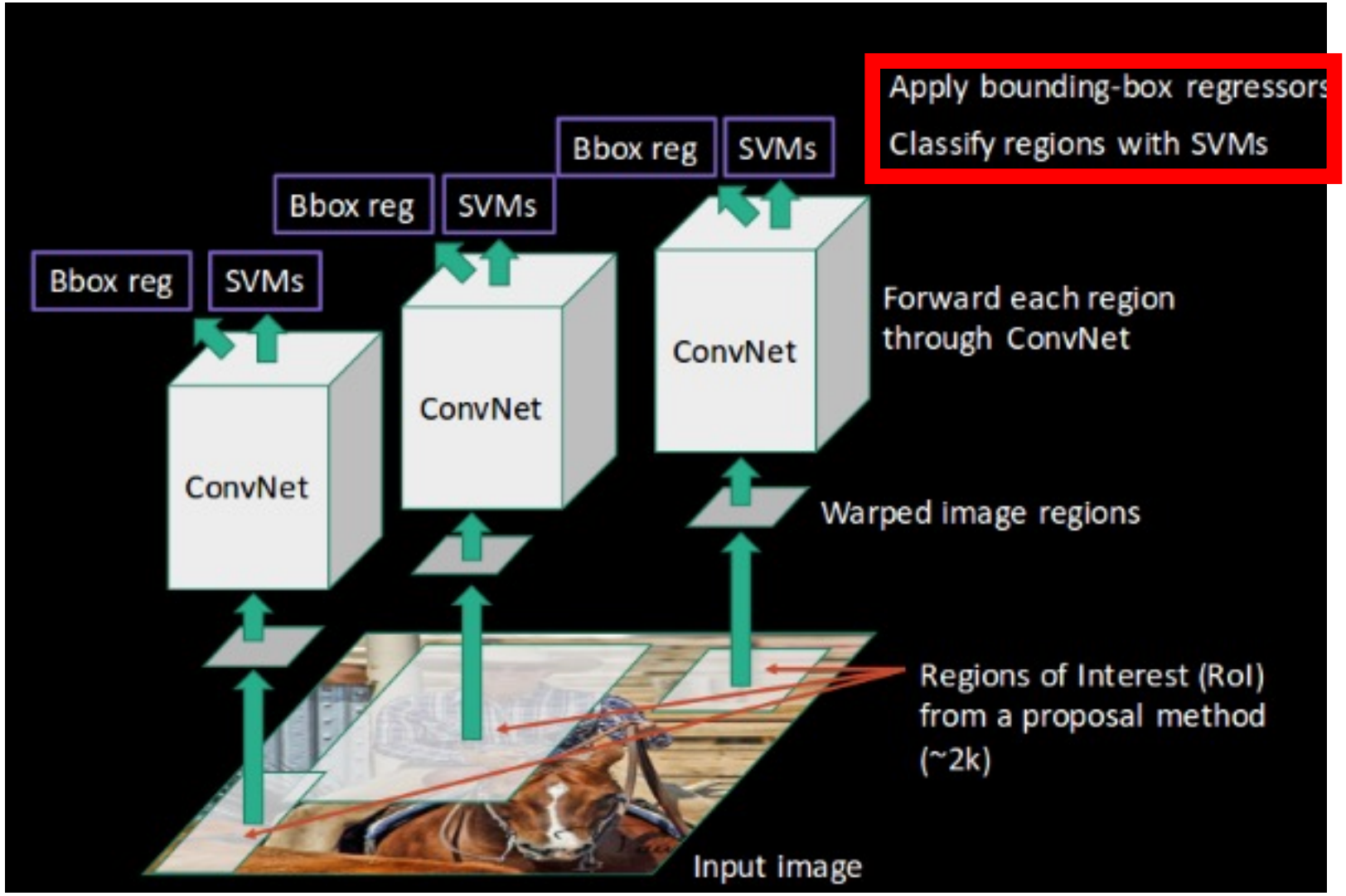


Image Source: https://www.researchgate.net/figure/Architecture-of-Alexnet-From-left-to-right-input-to-output-five-convolutional-layers_fig2_312303454

R-CNN



1. Classifier trained to use a region's CNN feature to assign a category from pre-defined set

2. Regressor trained to refine each region's position, width, and height

Figure Source: https://www.analyticsvidhya.com/blog/2018/10/a-step-by-step-introduction-to-the-basic-object-detection-algorithms-part-1/

R-CNN: Region Refinement

Original region proposal with center (p_x, p_y), width (p_w), and height (p_h) is refined using model parameters (d_x, d_y, d_w, d_y)

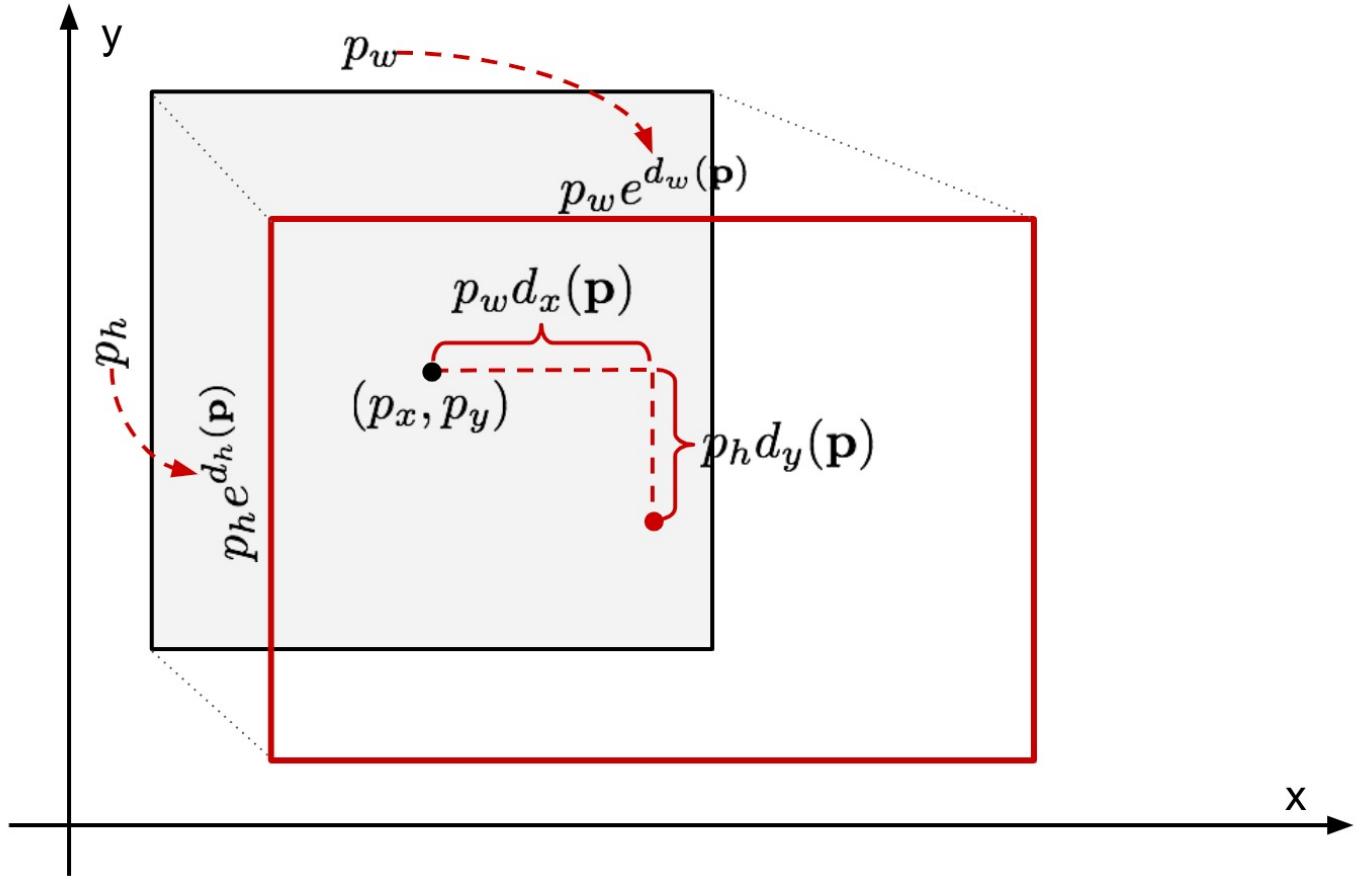
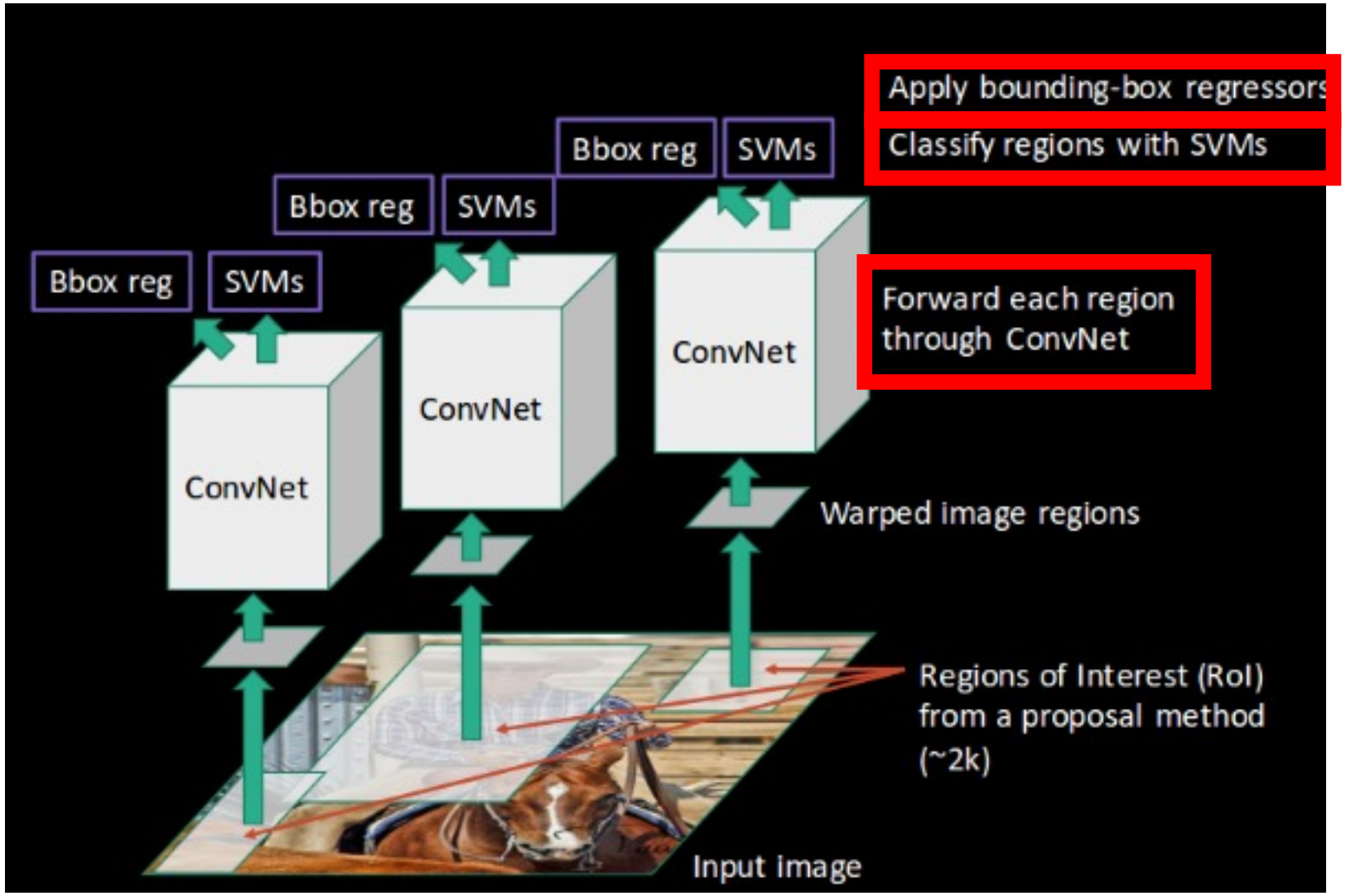


Image Source: https://lilianweng.github.io/lil-log/2017/12/31/object-recognition-for-dummies-part-3.html#bounding-box-regression

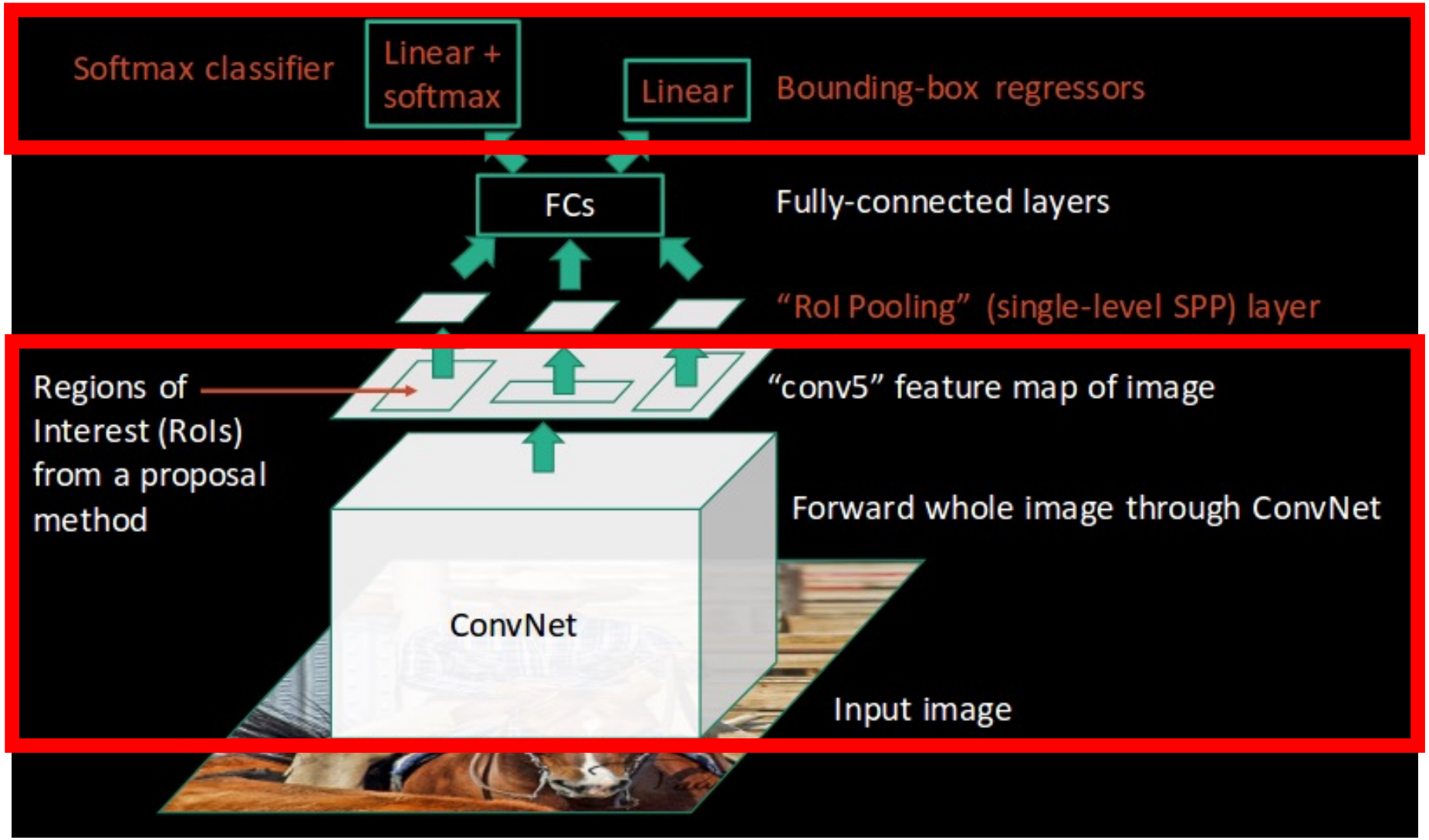
R-CNN Limitations



- Slow training procedure
  - Must train three models
- Slow at test time
  (~1 minute per image)

Figure Source: https://www.analyticsvidhya.com/blog/2018/10/a-step-by-step-introduction-to-the-basic-object-detection-algorithms-part-1/

Fast R-CNN: Single Stage Training (rather than 3)



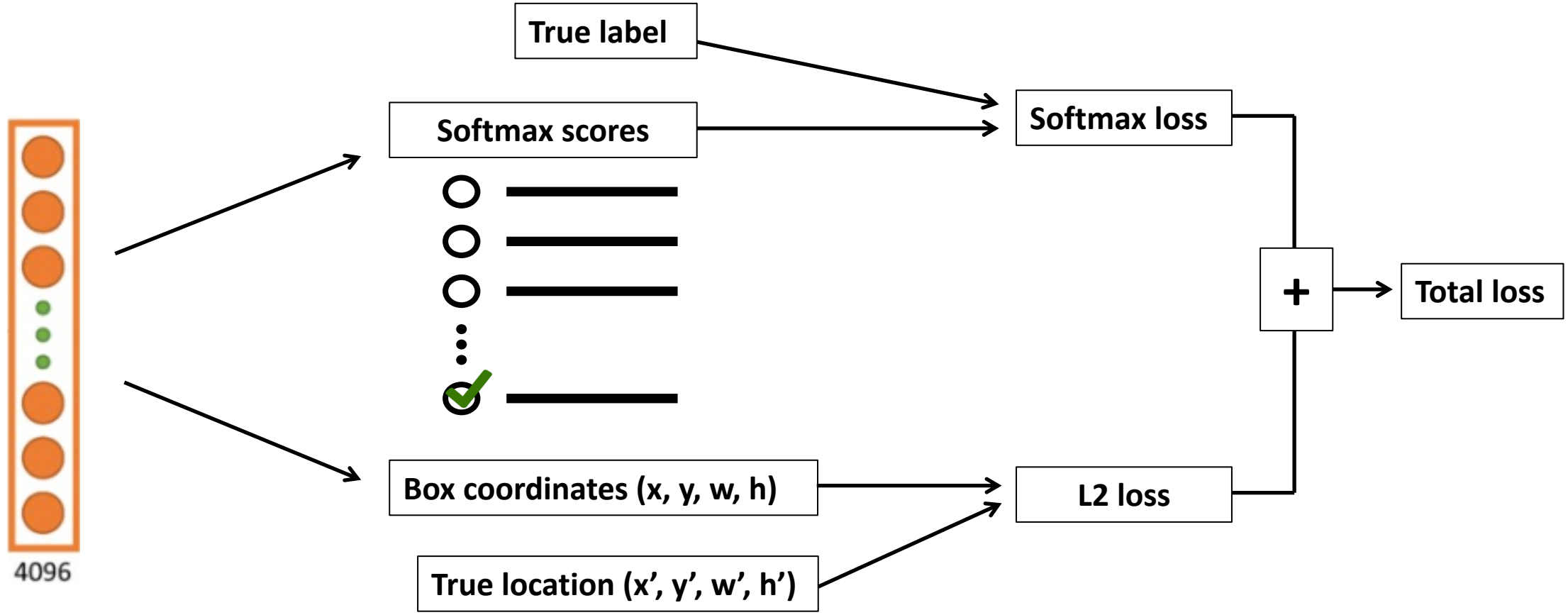
For each region, assign it to a class and refine it

Extract feature description per proposed region with section of feature map corresponding to region

Figure Source: https://www.analyticsvidhya.com/blog/2018/10/a-step-by-step-introduction-to-the-basic-object-detection-algorithms-part-1/

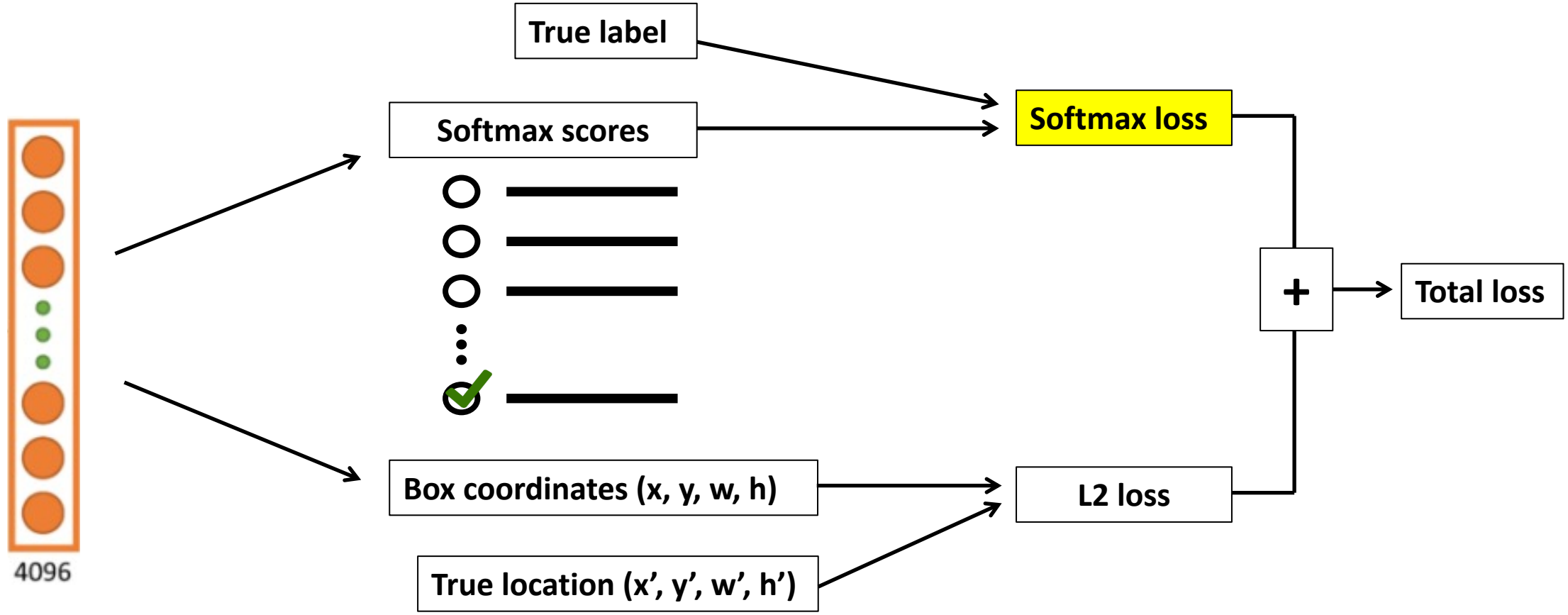
Fast R-CNN Training: Multi-task Loss

Loss for each region proposal is sum of classification and localization losses



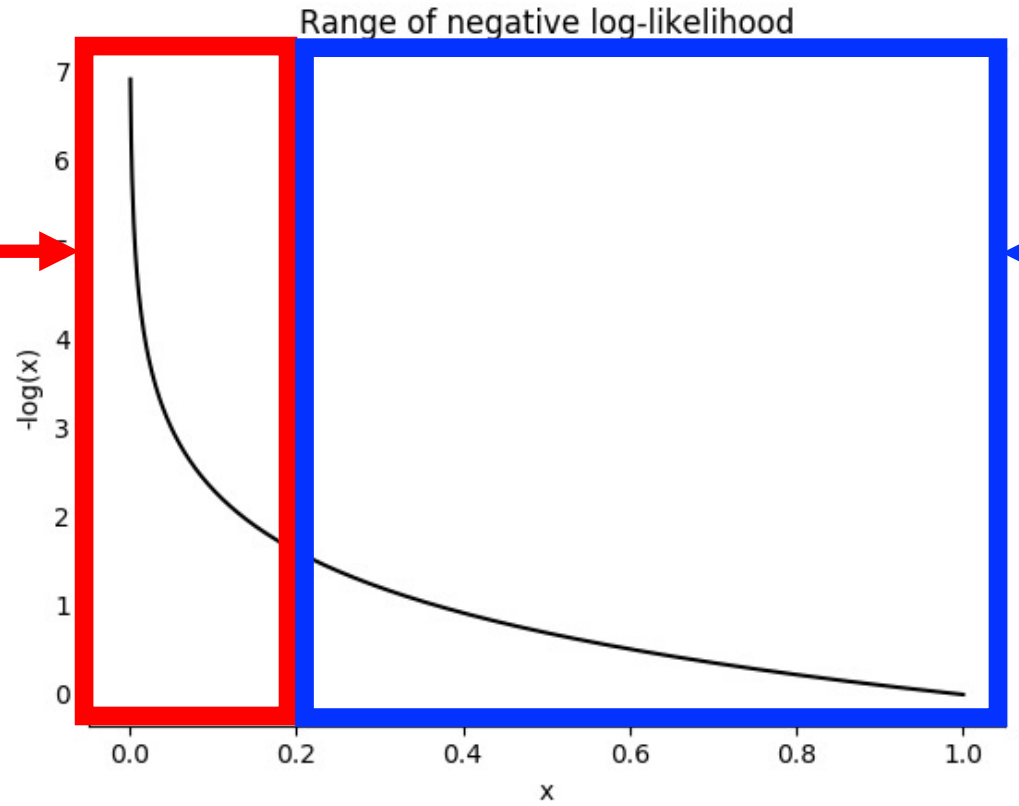
Fast R-CNN Training: Multi-task Loss

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Fast R-CNN Training: Classification Loss (Recap)

Greater penalty when predicted probability of true class is confidently wrong



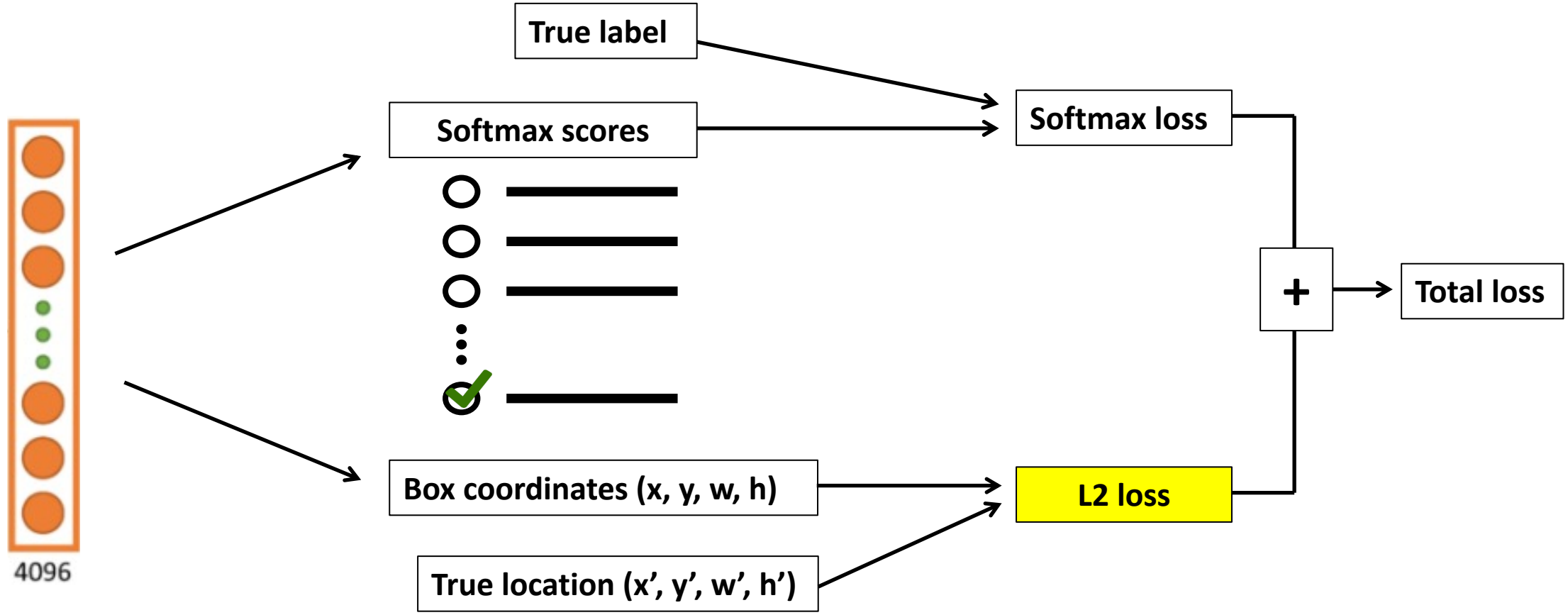
Lesser penalty otherwise

$$-\log \frac{\exp\left(w_k \cdot x + b_k\right)}{\sum_{j=1}^{K} \exp\left(w_j \cdot x + b_j\right)}$$

Figure source: https://ljvmiranda921.github.io/notebook/2017/08/13/softmax-and-the-negative-log-likelihood/

Fast R-CNN Training: Multi-task Loss

Loss for each region proposal is sum of classification and localization losses



Fast R-CNN Training: Measure Localization Loss

$$\mathcal{L}_{\text{box}}(t^u, v) = \sum_{i \in \{x,y,w,h\}} L_1^{\text{smooth}}(t_i^u - v_i)$$

$$L_1^{\text{smooth}}(x) = \begin{cases} 0.5x^2 & \text{if } |x| < 1 \\ |x| - 0.5 & \text{otherwise} \end{cases}$$

True location for true class "u"

Predicted location for class u

Less sensitive to outliers than SSE

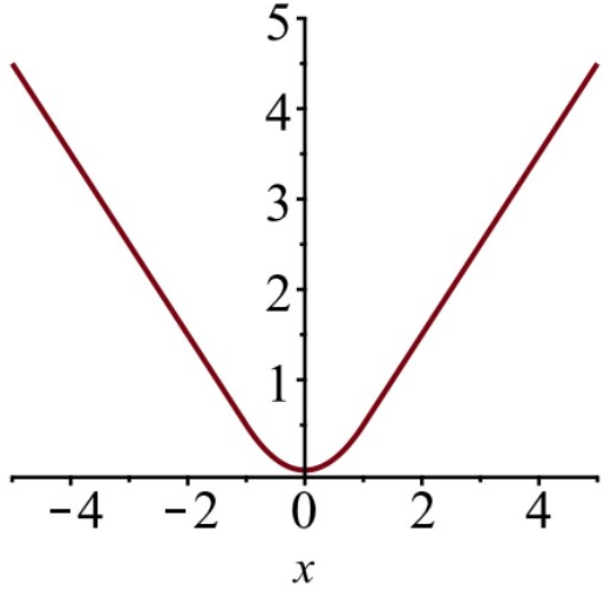
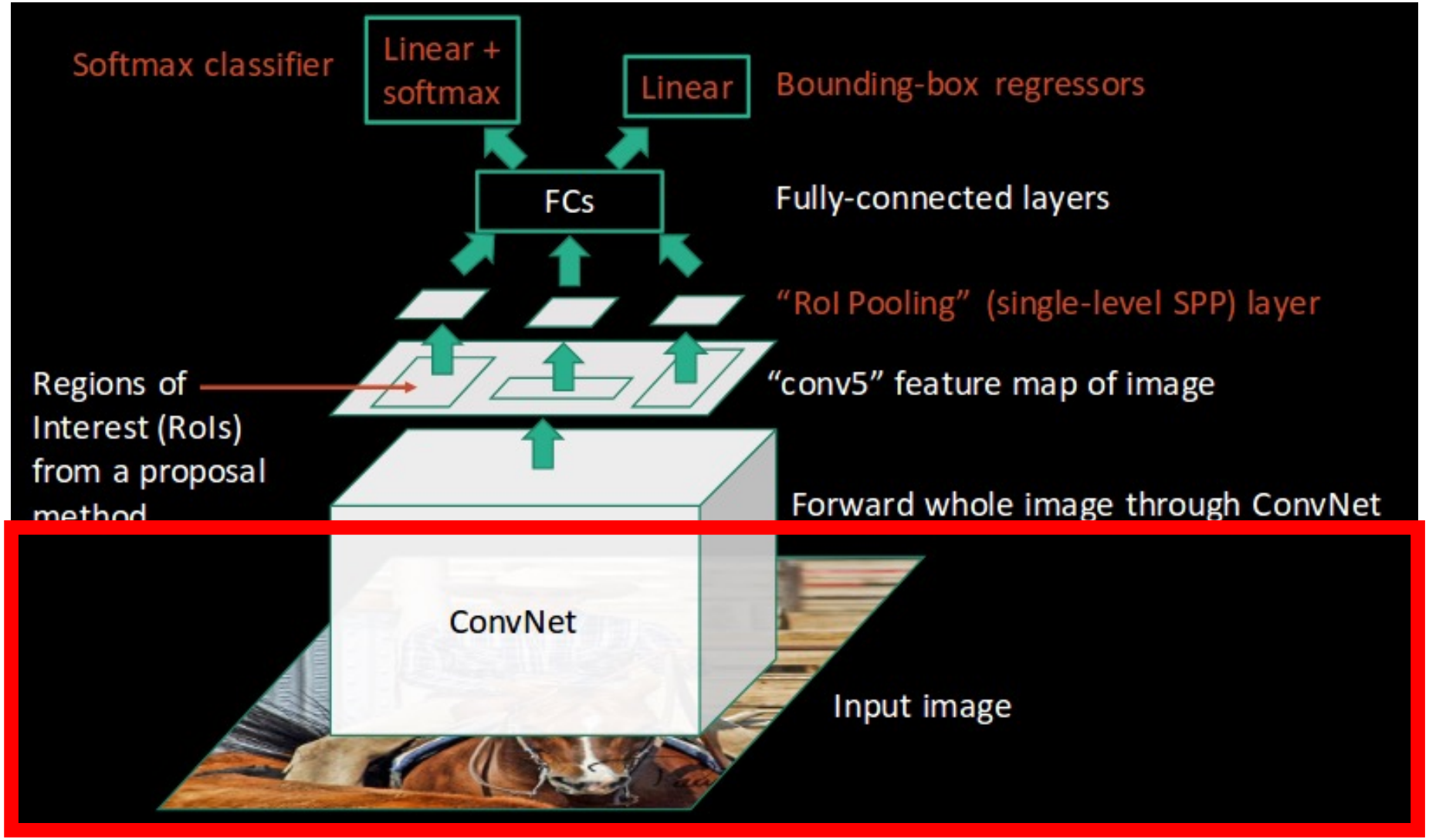


Image Source: https://lilianweng.github.io/lil-log/2017/12/31/object-recognition-for-dummies-part-3.html#bounding-box-regression

Fast R-CNN: Limitation



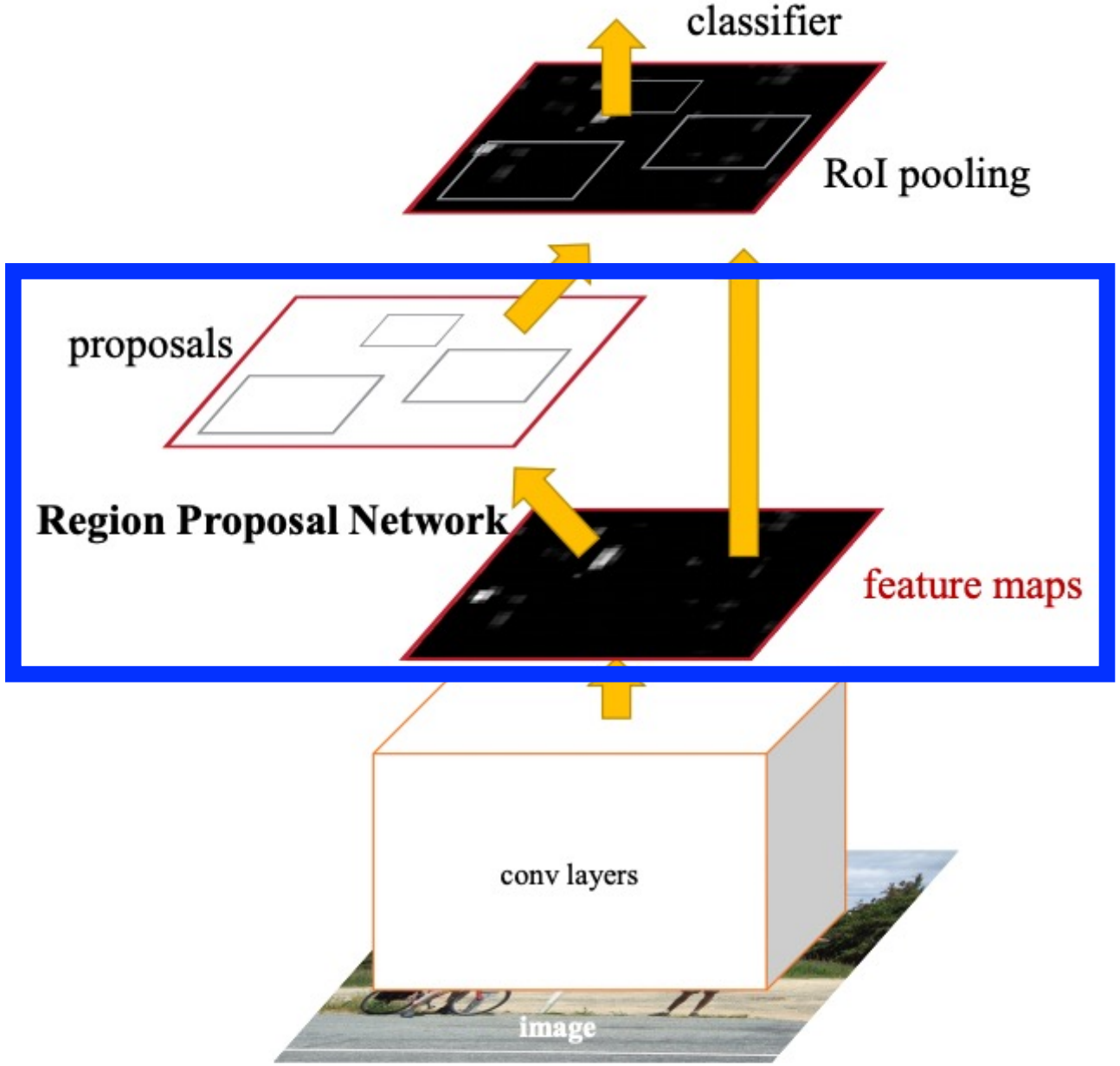
Still requires slow, initial step of generating region proposals

Figure Source: https://www.analyticsvidhya.com/blog/2018/10/a-step-by-step-introduction-to-the-basic-object-detection-algorithms-part-1/

Faster R-CNN

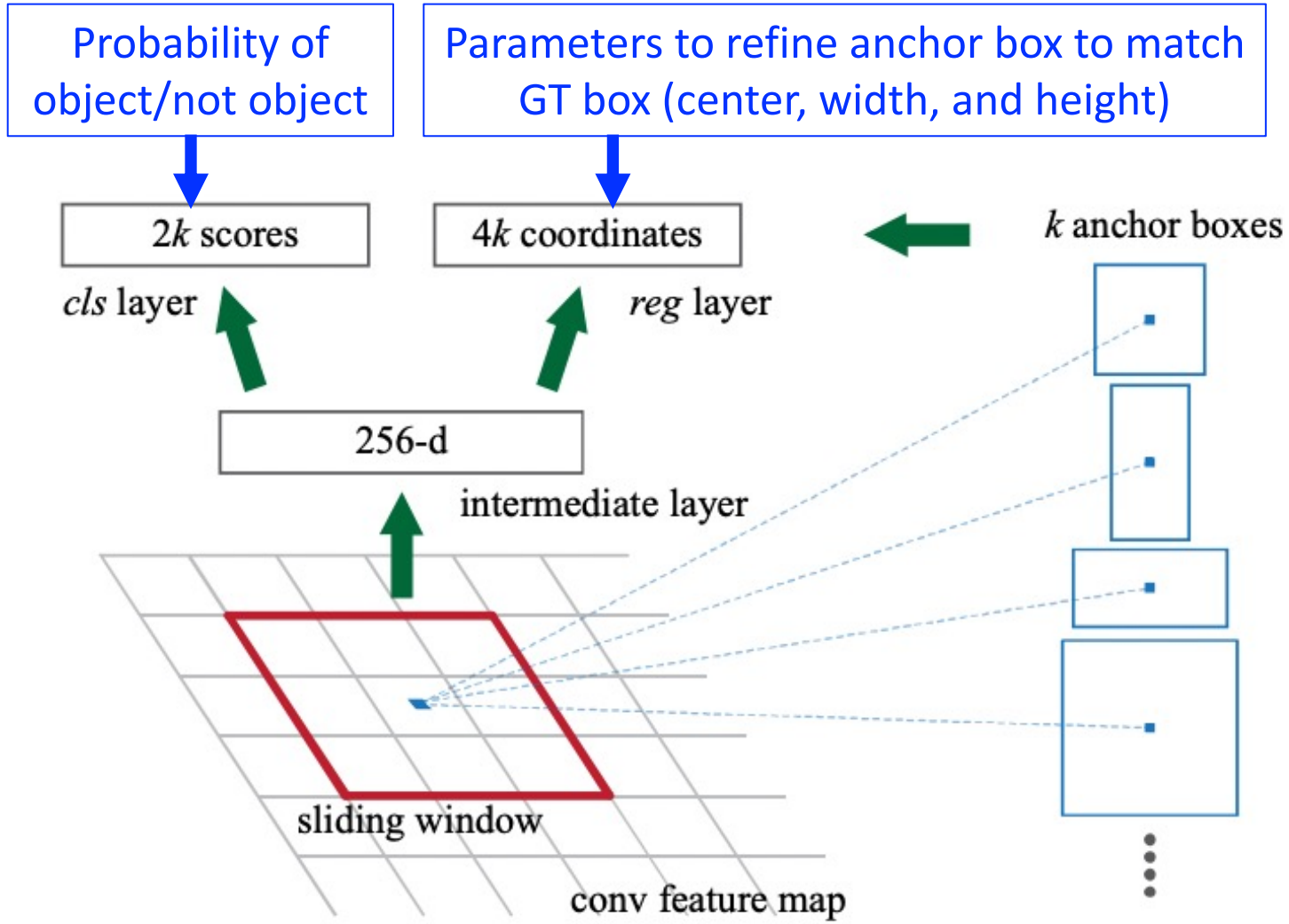
Adds finding region proposals to network so that all parts of model are learned in end-to-end fashion

Convolutional layers are shared for region proposal and detection



Ren Shaoqing Ren et al. “Faster R-CNN: Towards Real-Time Object Detection with Region Proposal Networks.” Neurips 2015.

Faster R-CNN: Region Proposal Network



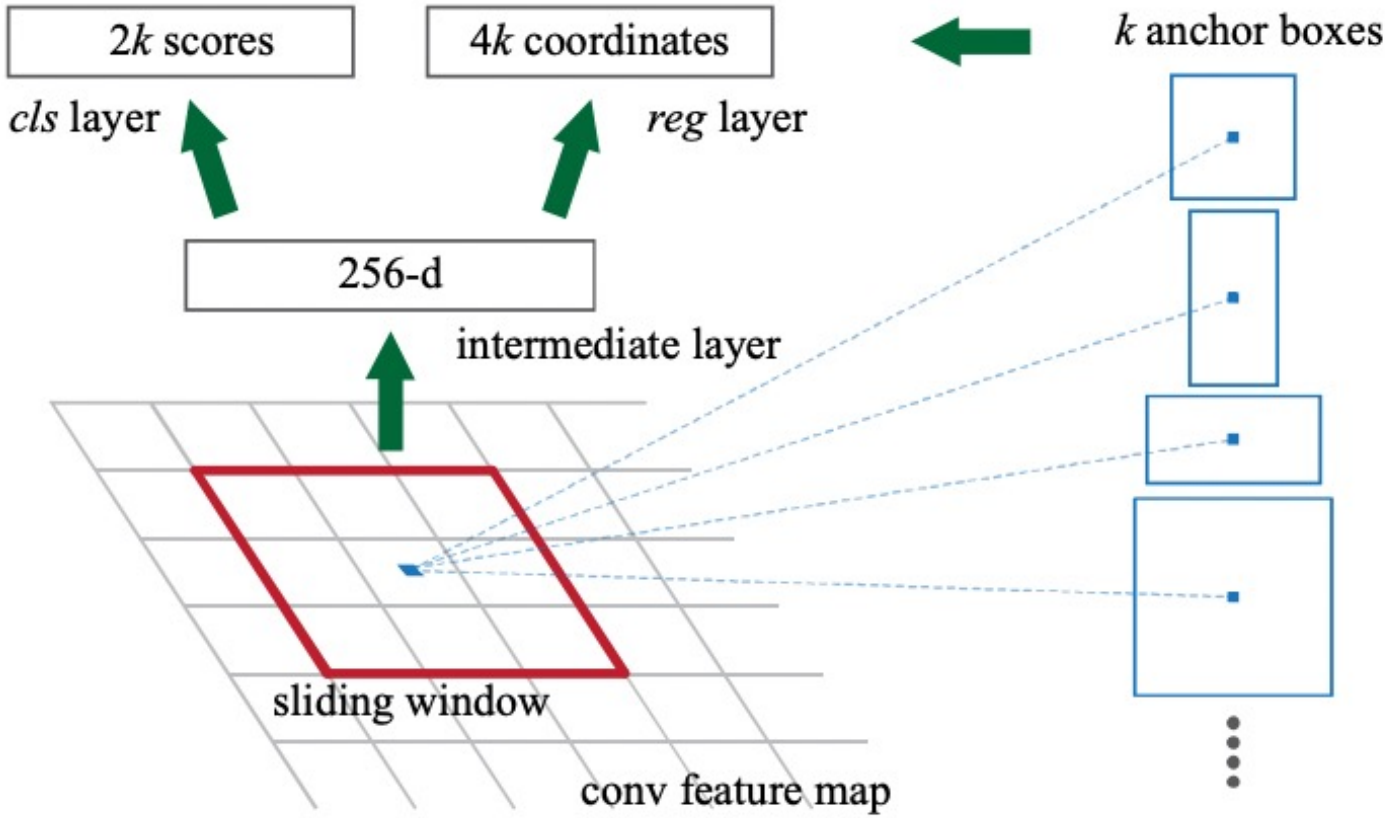
Based on convolution, so uses sliding window

- At each sliding window position, region proposals are predicted with respect to an anchor point (i.e., center of sliding window position)
- At each anchor point, k = 9 anchors are used to represent 3 scales and 3 aspect ratios

Ren Shaoqing Ren et al. “Faster R-CNN: Towards Real-Time Object Detection with Region Proposal Networks.” Neurips 2015.

Faster R-CNN: Region Proposal Network

At training, loss for each region proposal is sum of classification and localization losses



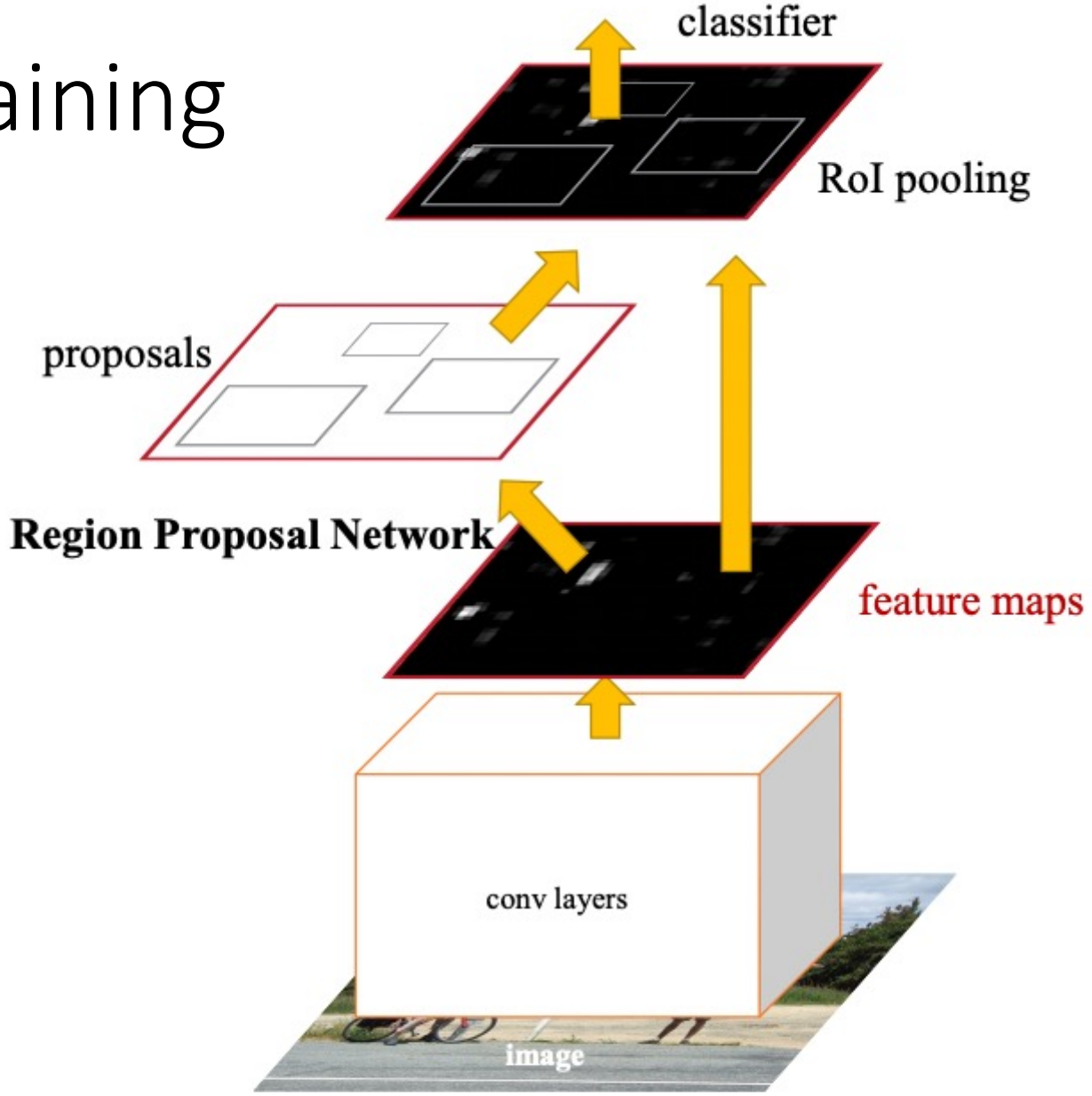
Based on convolution, so uses sliding window

- At each sliding window position, region proposals are predicted with respect to an anchor point (i.e., center of sliding window position)
- At each anchor point, k = 9 anchors are used to represent 3 scales and 3 aspect ratios

Ren Shaoqing Ren et al. “Faster R-CNN: Towards Real-Time Object Detection with Region Proposal Networks.” Neurips 2015.

Faster R-CNN Training

1. Train RPN
2. Train Fast R-CNN using proposals from pretrained RPN
3. Fine-tune layers unique to RPN
4. Fine-tune the fully connected layers of Fast R-CNN



Ren Shaoqing Ren et al. “Faster R-CNN: Towards Real-Time Object Detection with Region Proposal Networks.” Neurips 2015.

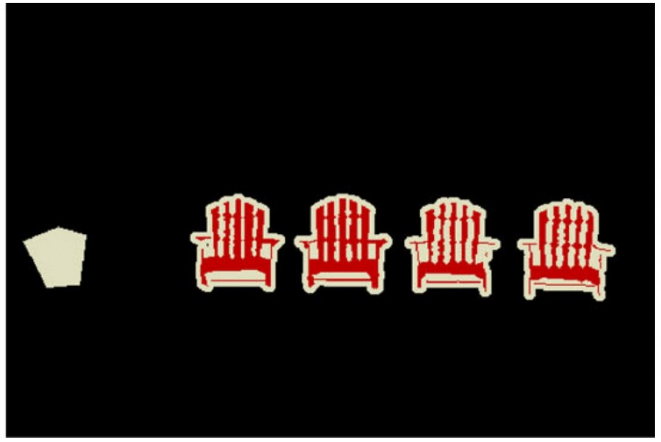
Today's Topics

- Problems
- Applications
- PASCAL VOC detection challenge: R-CNNs
- PASCAL VOC semantic segmentation challenge: fully convolutional networks

VOC Challenge

- Goal: locate all pixels belonging to 20 categories (e.g., person, cat, bus, mortorbike, potted plant, bottle) plus background
- Dataset: 11,530 images collected from Flickr and annotated by annotators at University of Leeds

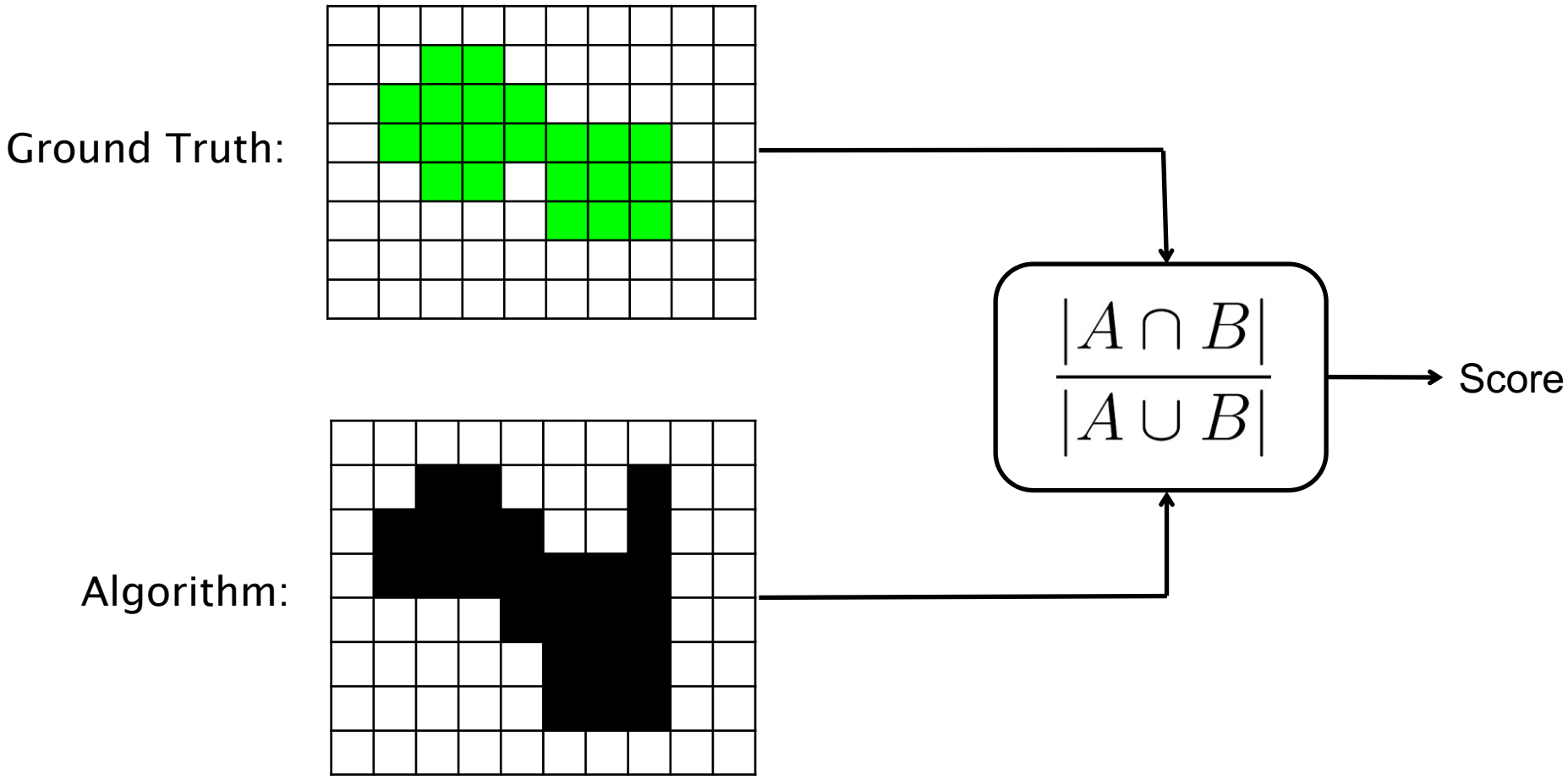


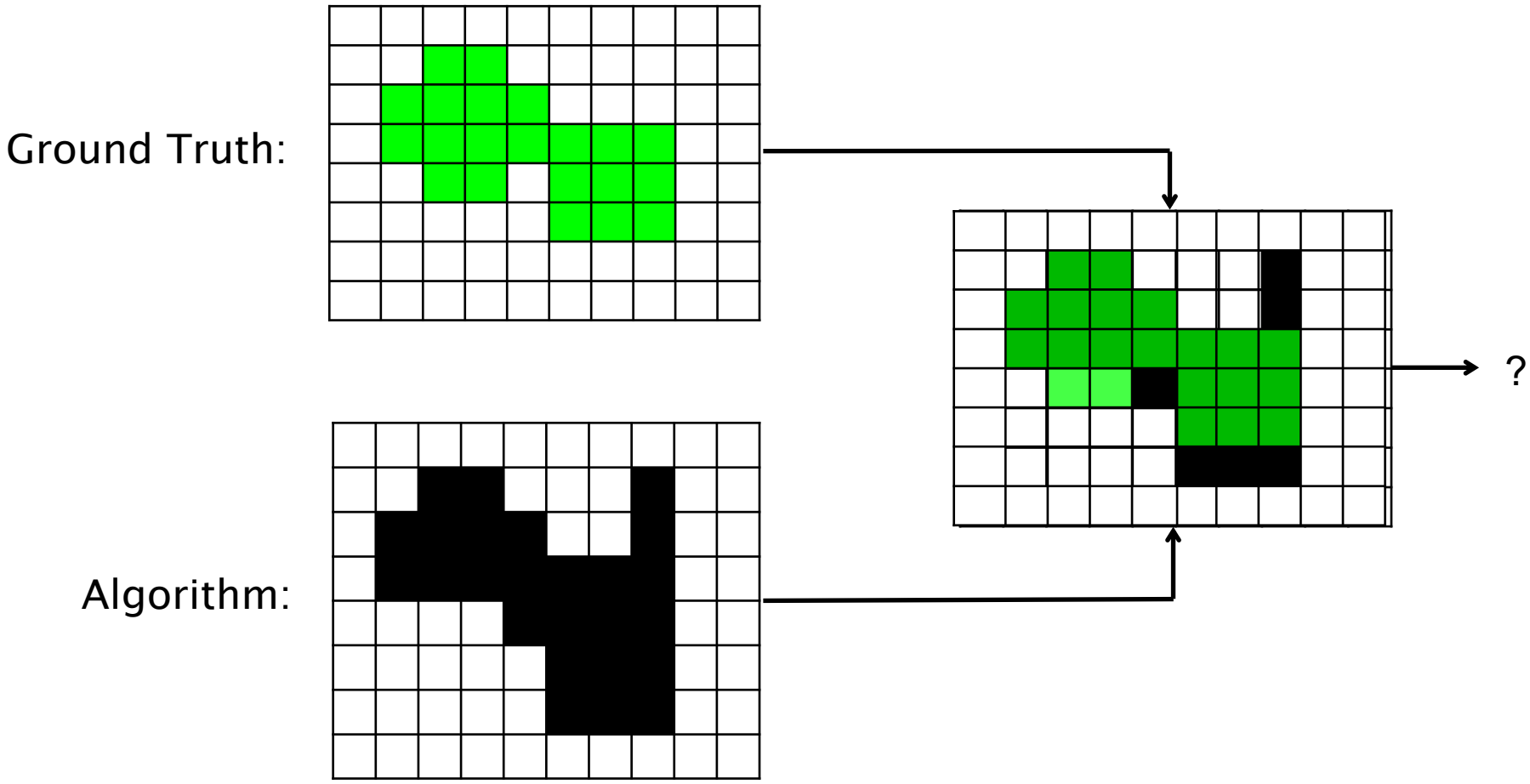
Dataset location: http://host.robots.ox.ac.uk/pascal/VOC/index.html

Mark Everingham, Luc Van Gool, Christopher K. I. Williams, John Winn, and Andrew Zisserman. The PASCAL Visual Object Classes (VOC) Challenge. IJCV 2010.

VOC Challenge: Evaluation Metric (IoU)

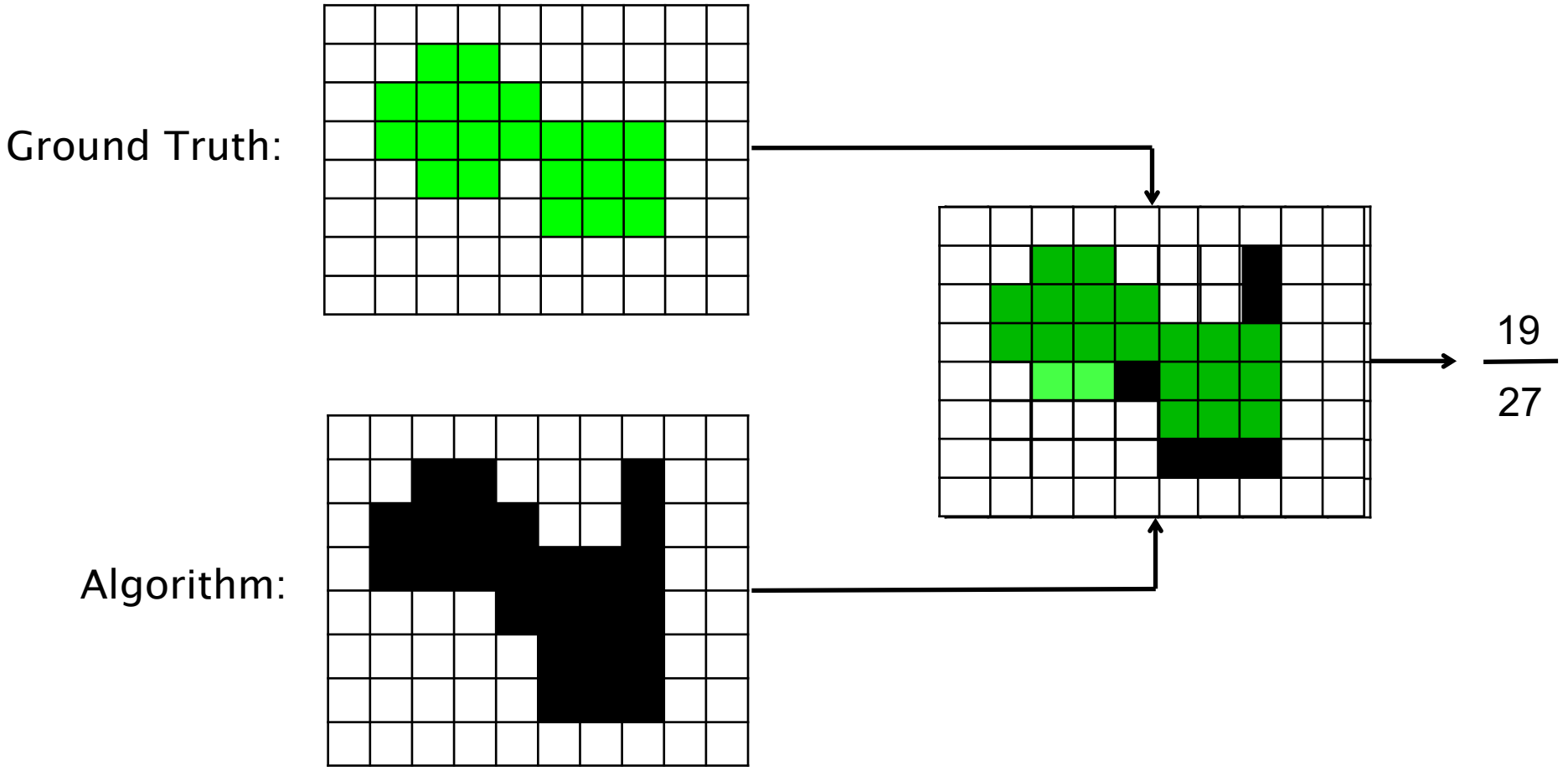


VOC Challenge: Evaluation Metric (IoU)



VOC Challenge: Evaluation Metric (IoU)

Mean IoU: IoU between predicted and ground-truth pixels, averaged over all 21 categories

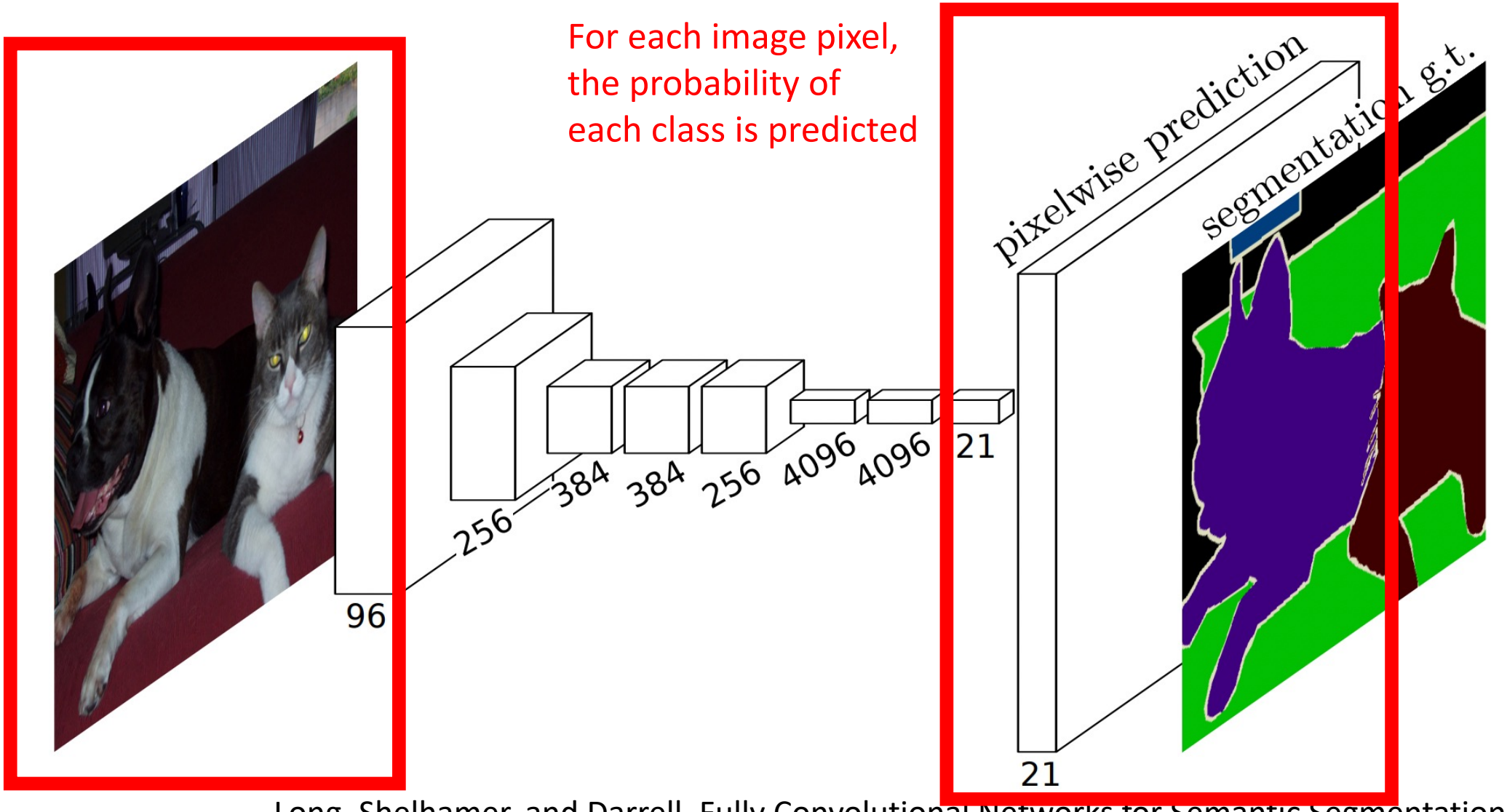


Input: RGB image of ANY size

Output: Image of same size as input

Architecture

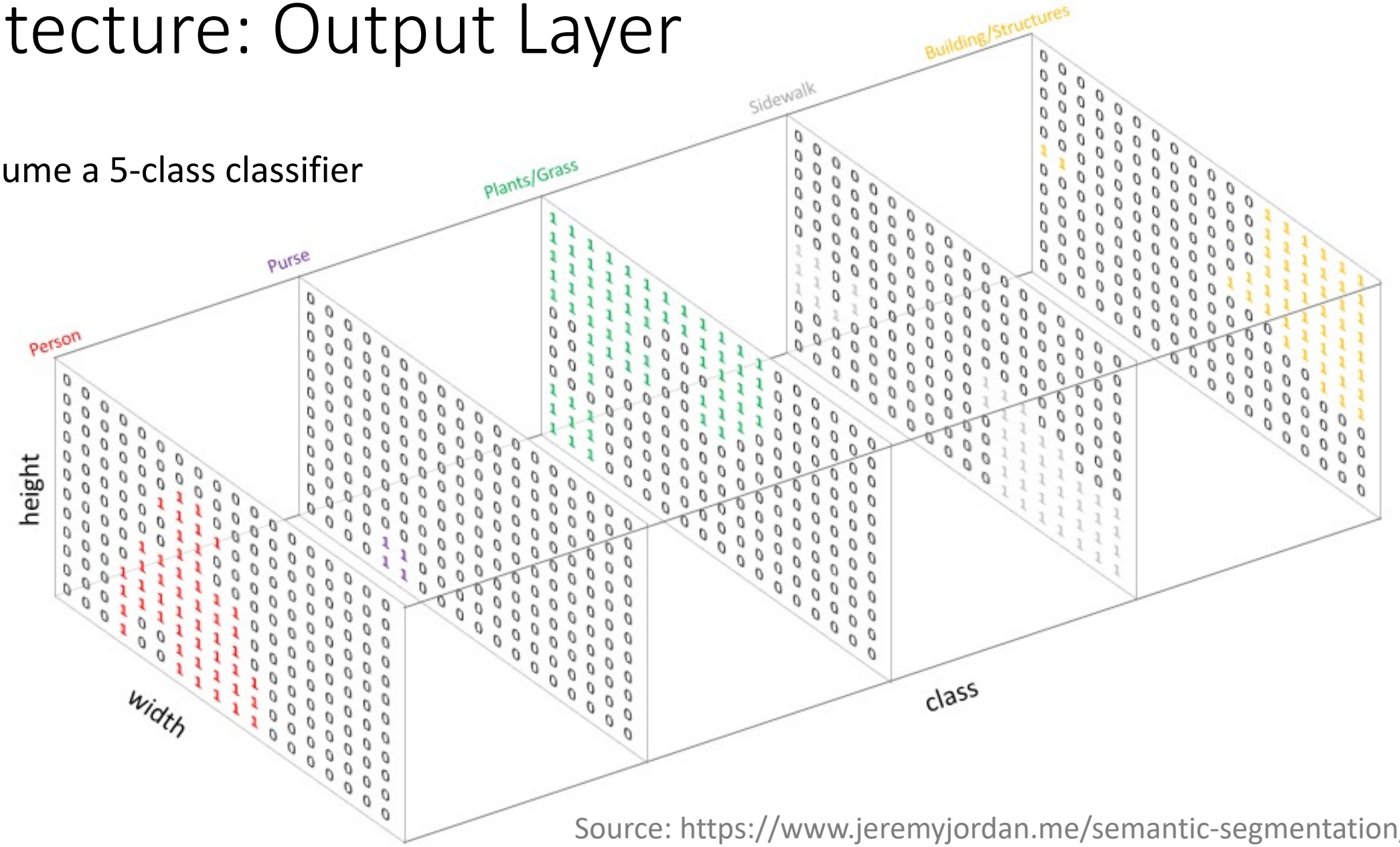
For each image pixel, the probability of each class is predicted



Long, Shelhamer, and Darrell. Fully Convolutional Networks for Semantic Segmentation. CVPR 2015.

Architecture: Output Layer

- e.g., assume a 5-class classifier



Source: https://www.jeremyjordan.me/semantic-segmentation/

Architecture: Output Layer

- e.g., assume a 5-class classifier; output 1-hot encoding collapsed into single mask image



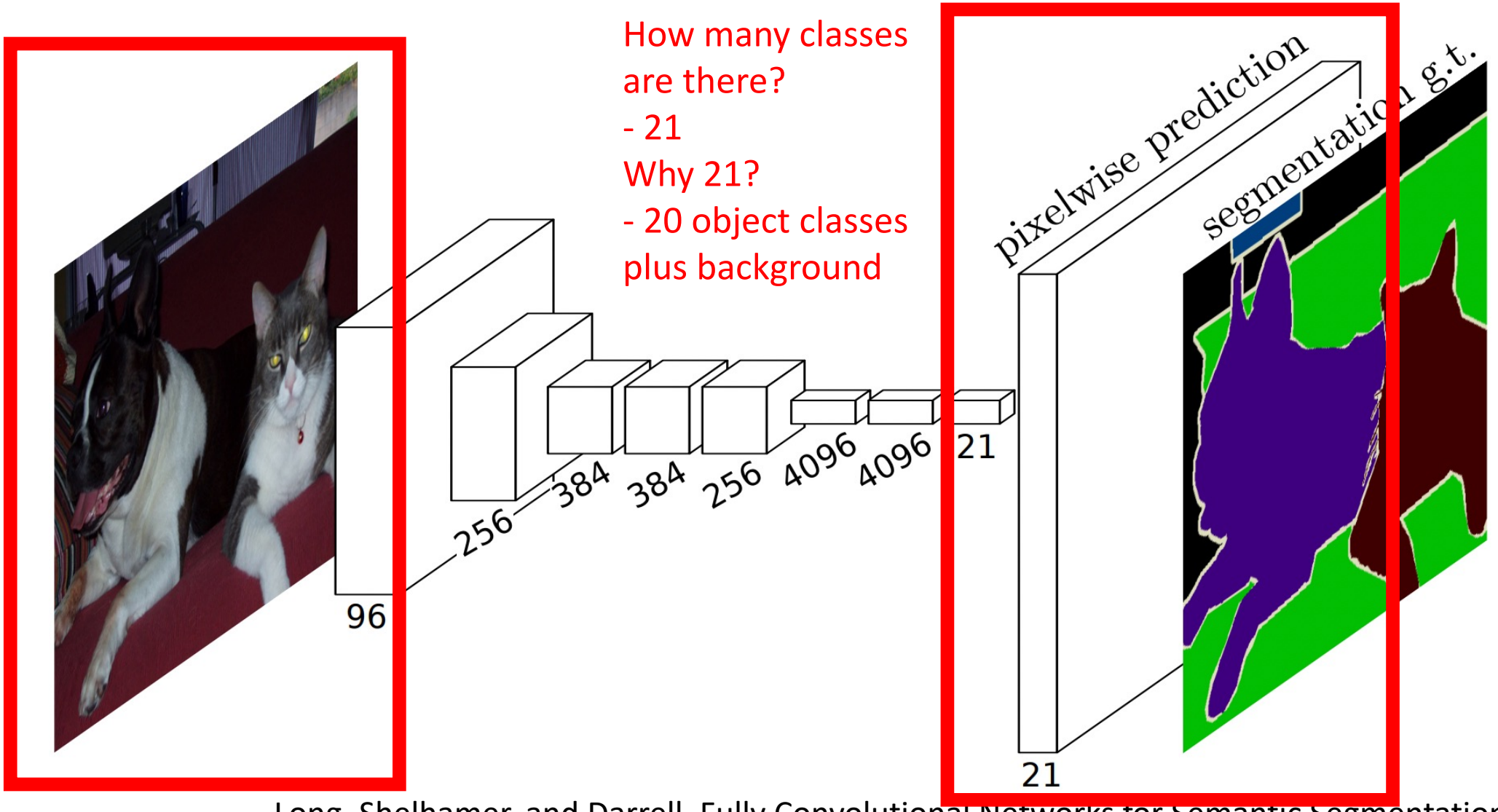
0: Background/Unknown
1: Person
2: Purse
3: Plants/Grass
4: Sidewalk
5: Building/Structures

Source: https://www.jeremyjordan.me/semantic-segmentation/

Input: RGB image of ANY size

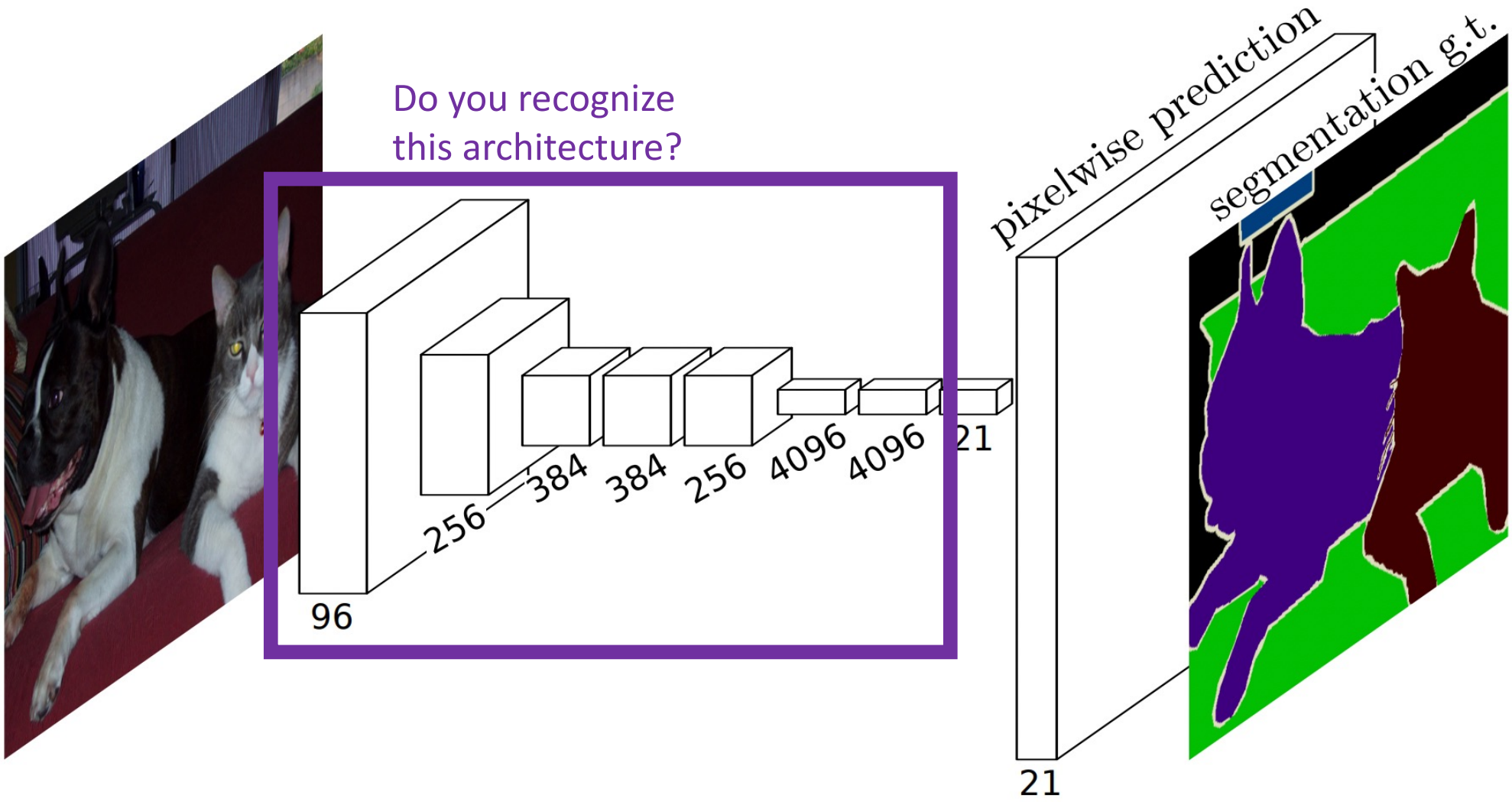
Output: Image of same size as input

Architecture



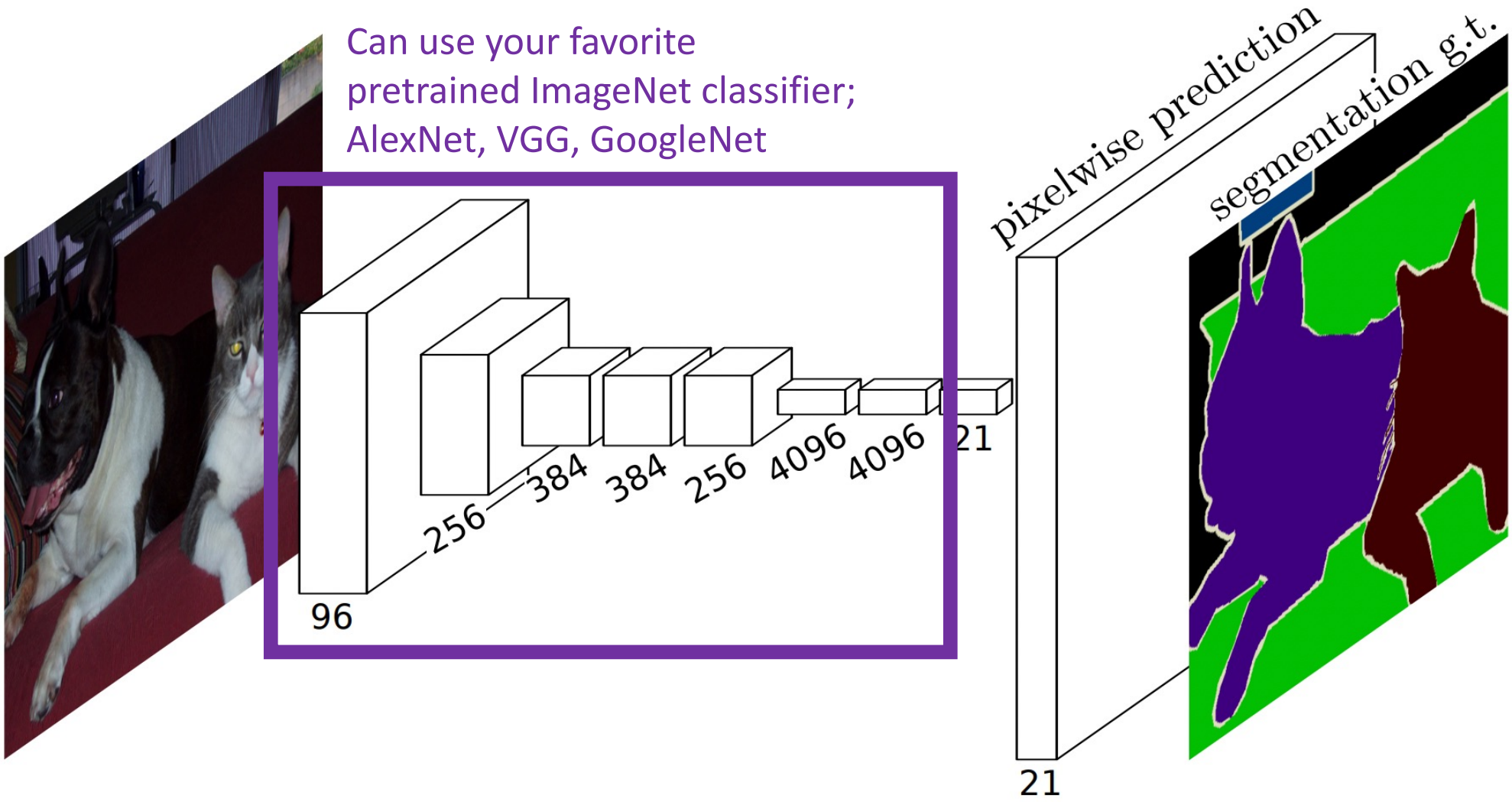
Long, Shelhamer, and Darrell. Fully Convolutional Networks for Semantic Segmentation. CVPR 2015.

Architecture



Long, Shelhamer, and Darrell. Fully Convolutional Networks for Semantic Segmentation. CVPR 2015.

Architecture

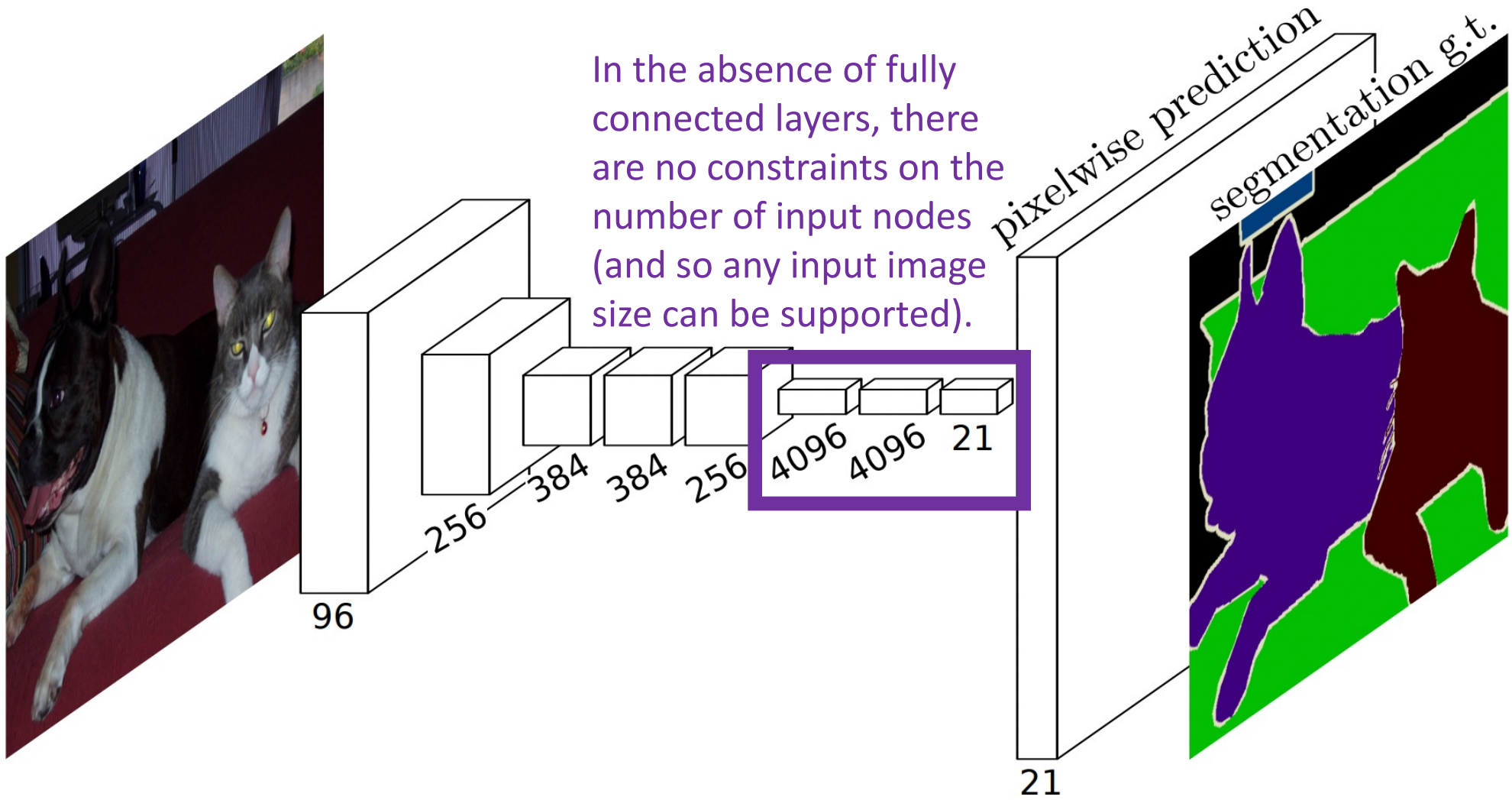


Long, Shelhamer, and Darrell. Fully Convolutional Networks for Semantic Segmentation. CVPR 2015.

Architecture

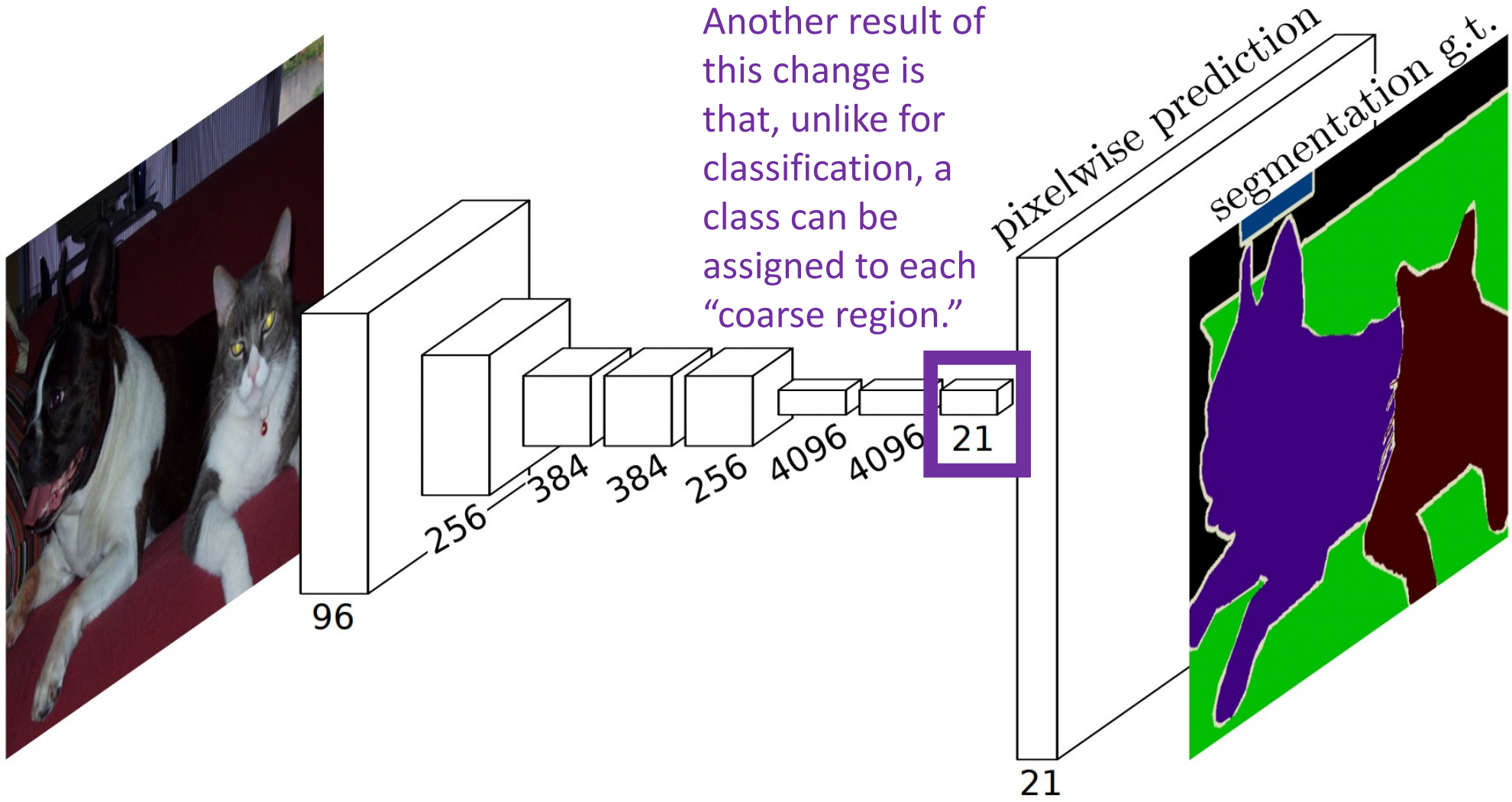
To make the architecture fully convolutional, fully connected layers are converted to convolutional layers.

In the absence of fully connected layers, there are no constraints on the number of input nodes (and so any input image size can be supported).



Long, Shelhamer, and Darrell. Fully Convolutional Networks for Semantic Segmentation. CVPR 2015.

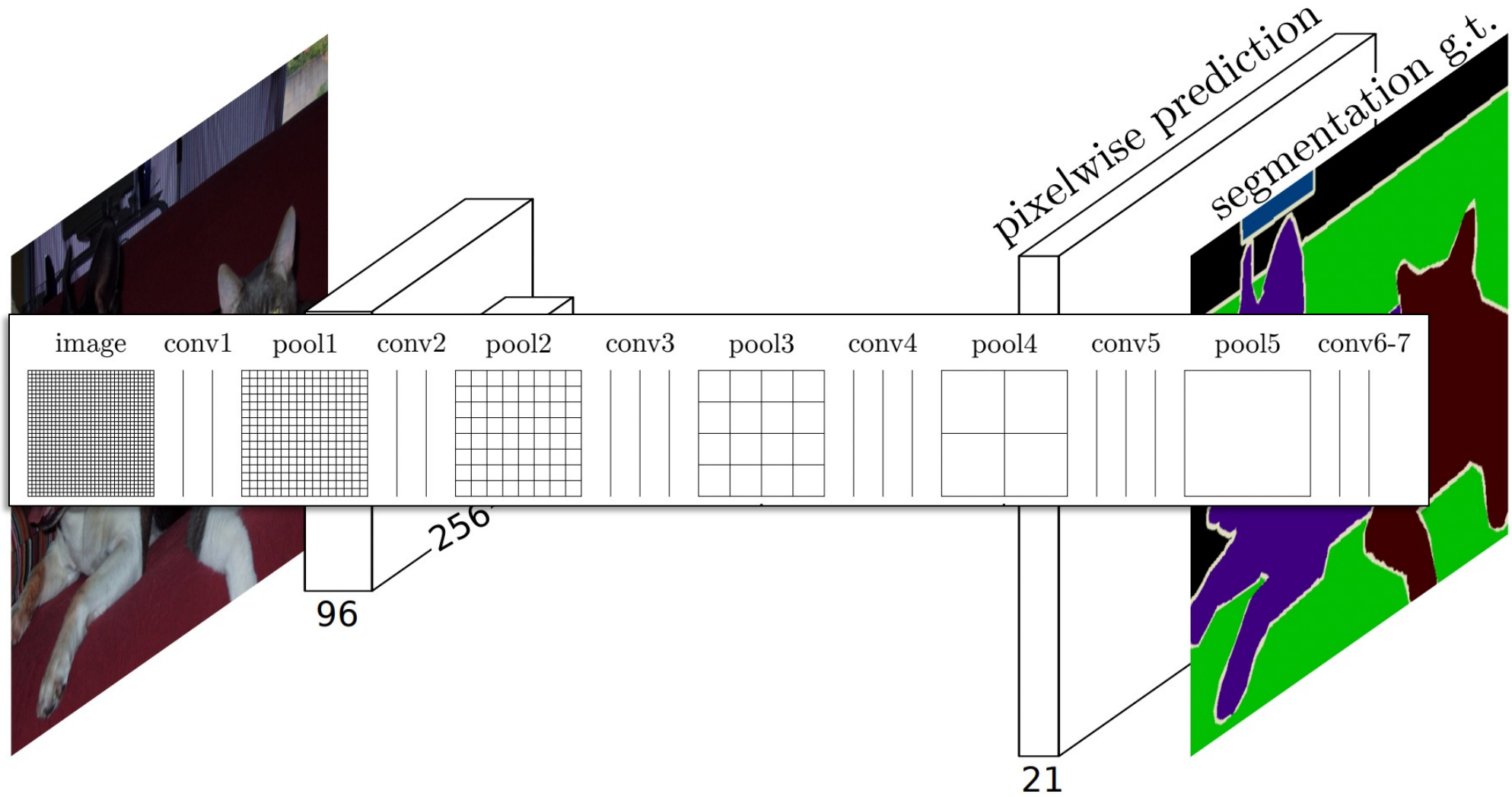
Architecture



Long, Shelhamer, and Darrell. Fully Convolutional Networks for Semantic Segmentation. CVPR 2015.

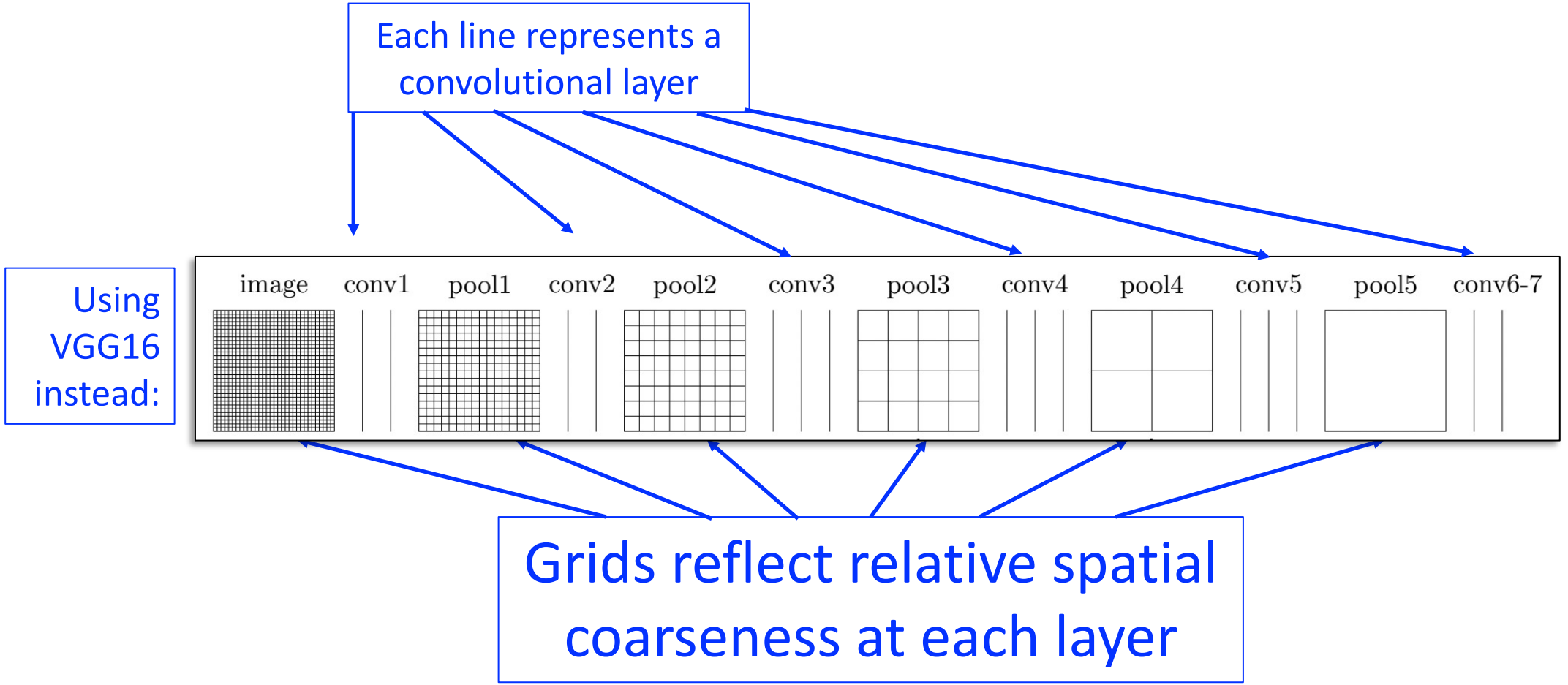
Architecture: Coarse Region Classification (Recall Intuition)

Using VGG16 instead:



Long, Shelhamer, and Darrell. Fully Convolutional Networks for Semantic Segmentation. CVPR 2015.

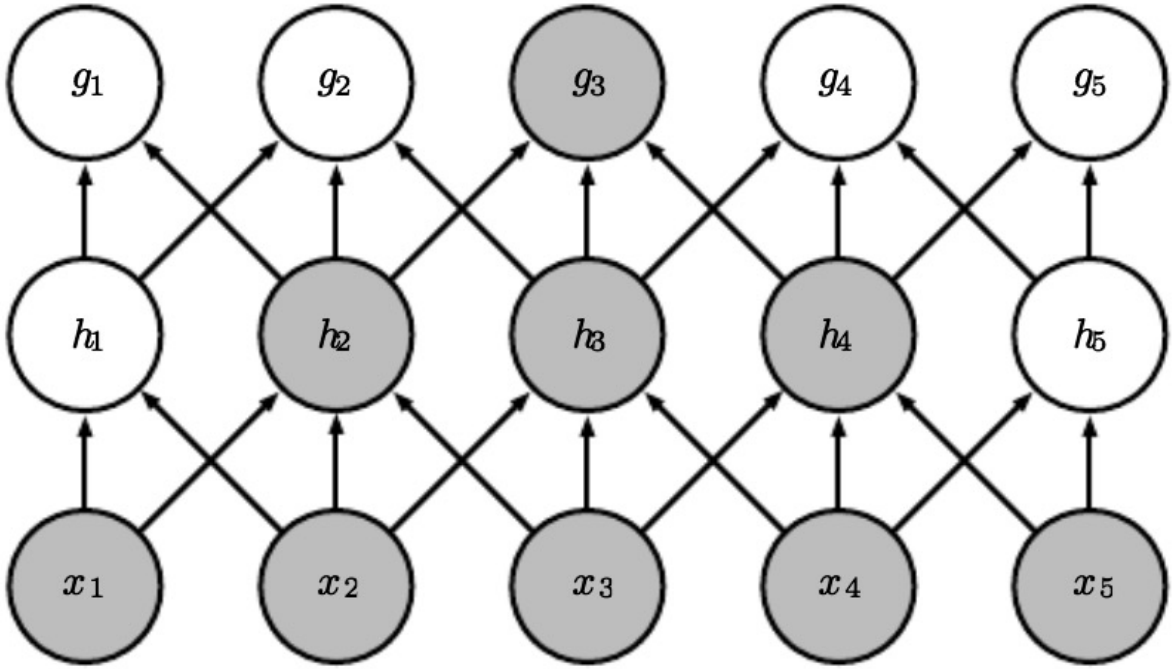
Architecture: Coarse Region Classification (Recall Intuition)



Long, Shelhamer, and Darrell. Fully Convolutional Networks for Semantic Segmentation. CVPR 2015.

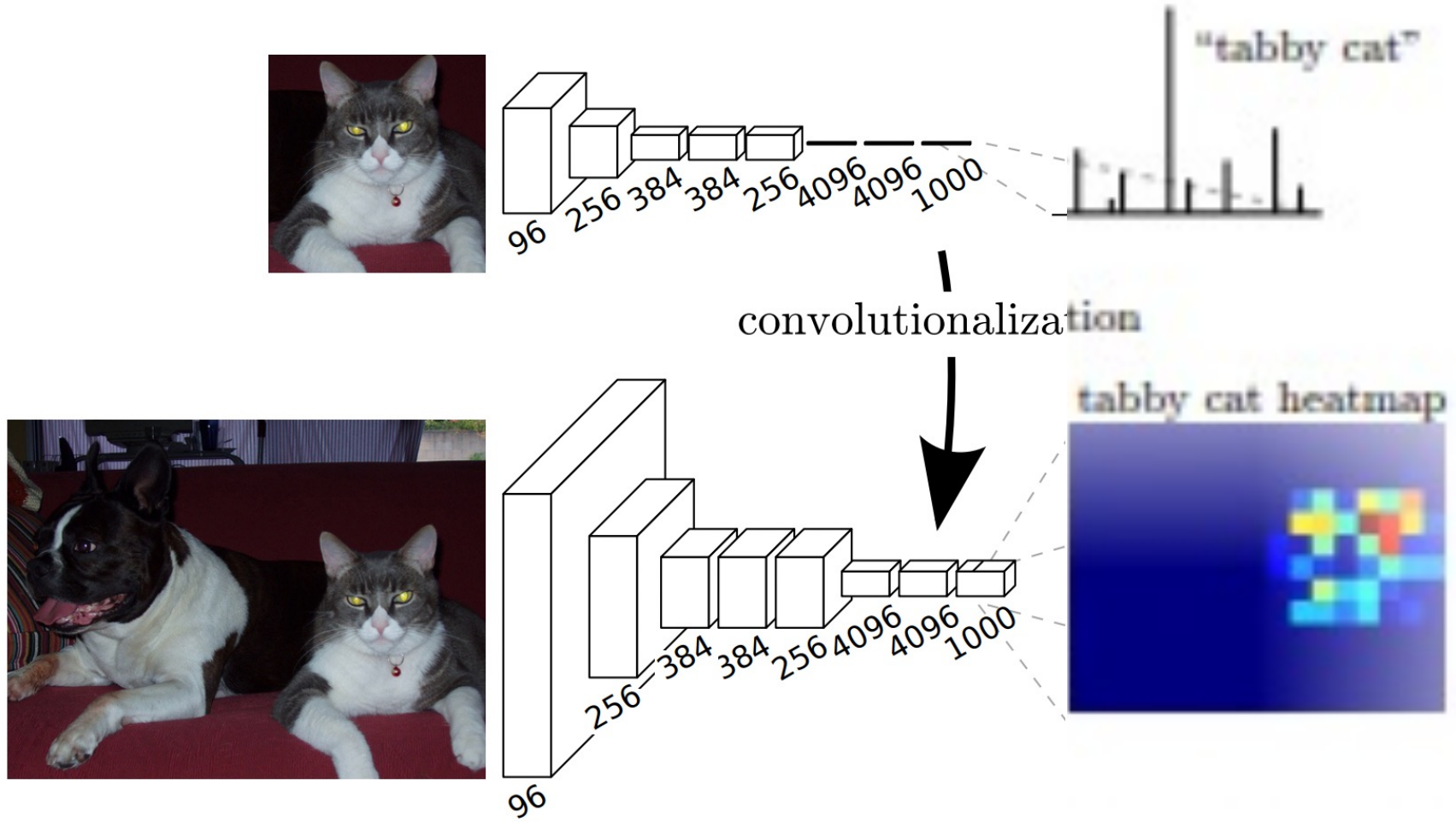
Architecture: Coarse Region Classification (Recall Intuition)

Stacking many convolutional layers leads to learning patterns in increasingly **larger regions of the input (e.g., pixel) space.**



https://www.deeplearningbook.org/contents/convnets.html

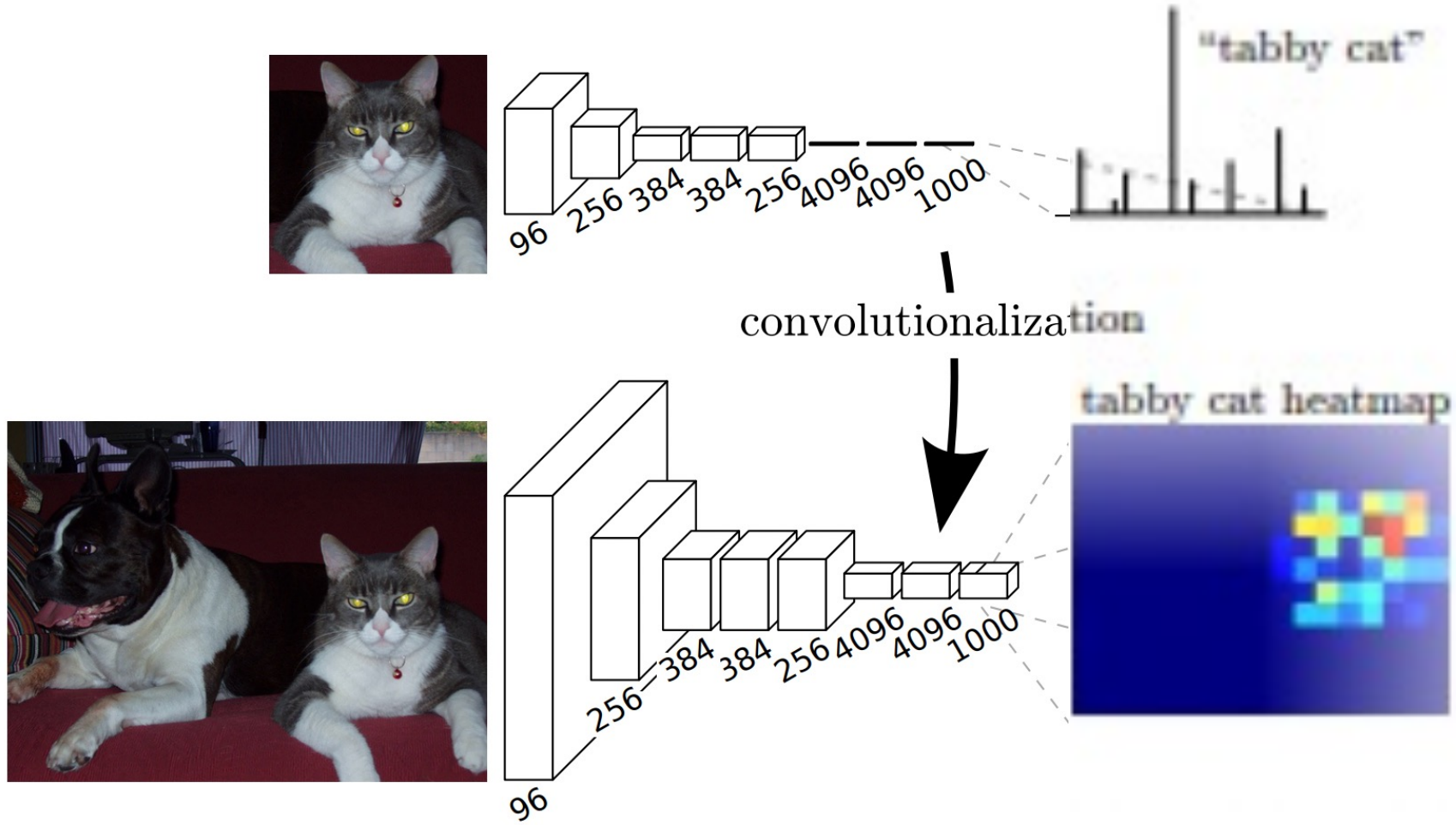
Architecture: Fully vs Convolution Layers



Each slice indicates the likelihood each pixel in the coarse region belongs to the class identified by the filter

Long, Shelhamer, and Darrell. Fully Convolutional Networks for Semantic Segmentation. CVPR 2015.

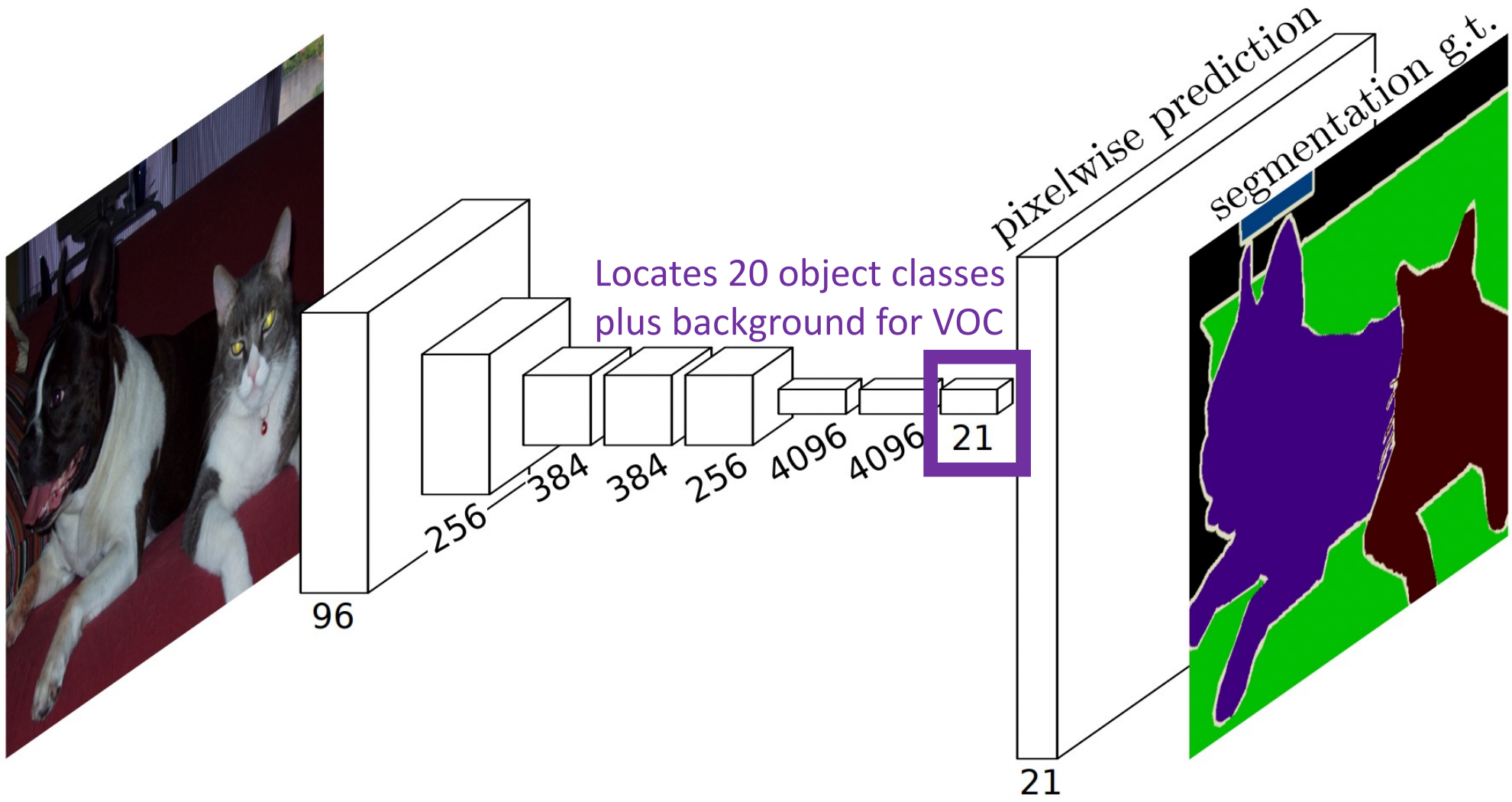
Architecture: Fully vs Convolution Layers



If convolutionizing ImageNet trained classifiers, how many classes would be predicted for each coarse region?

Long, Shelhamer, and Darrell. Fully Convolutional Networks for Semantic Segmentation. CVPR 2015.

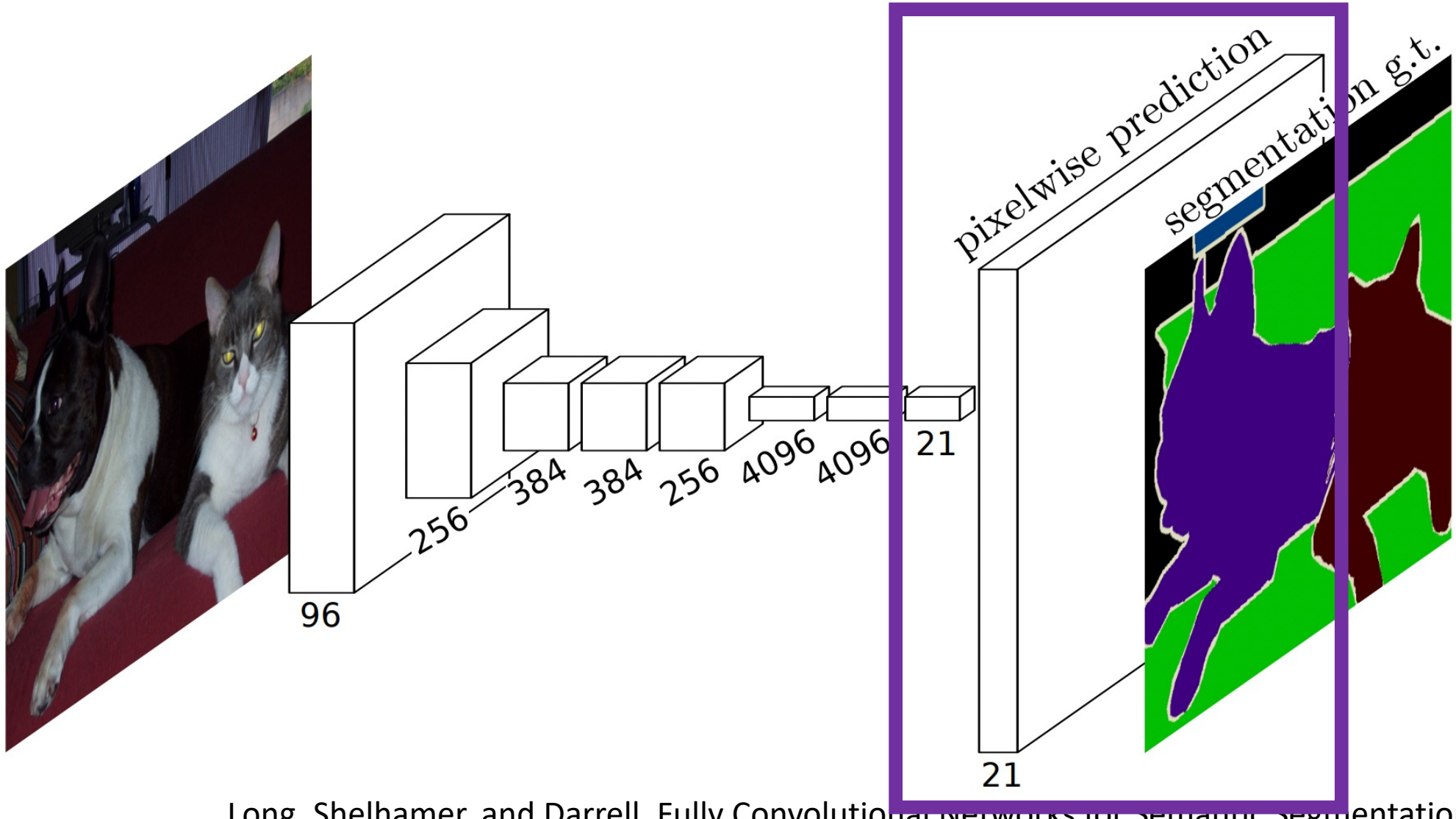
Architecture: Coarse Region Classification



Long, Shelhamer, and Darrell. Fully Convolutional Networks for Semantic Segmentation. CVPR 2015.

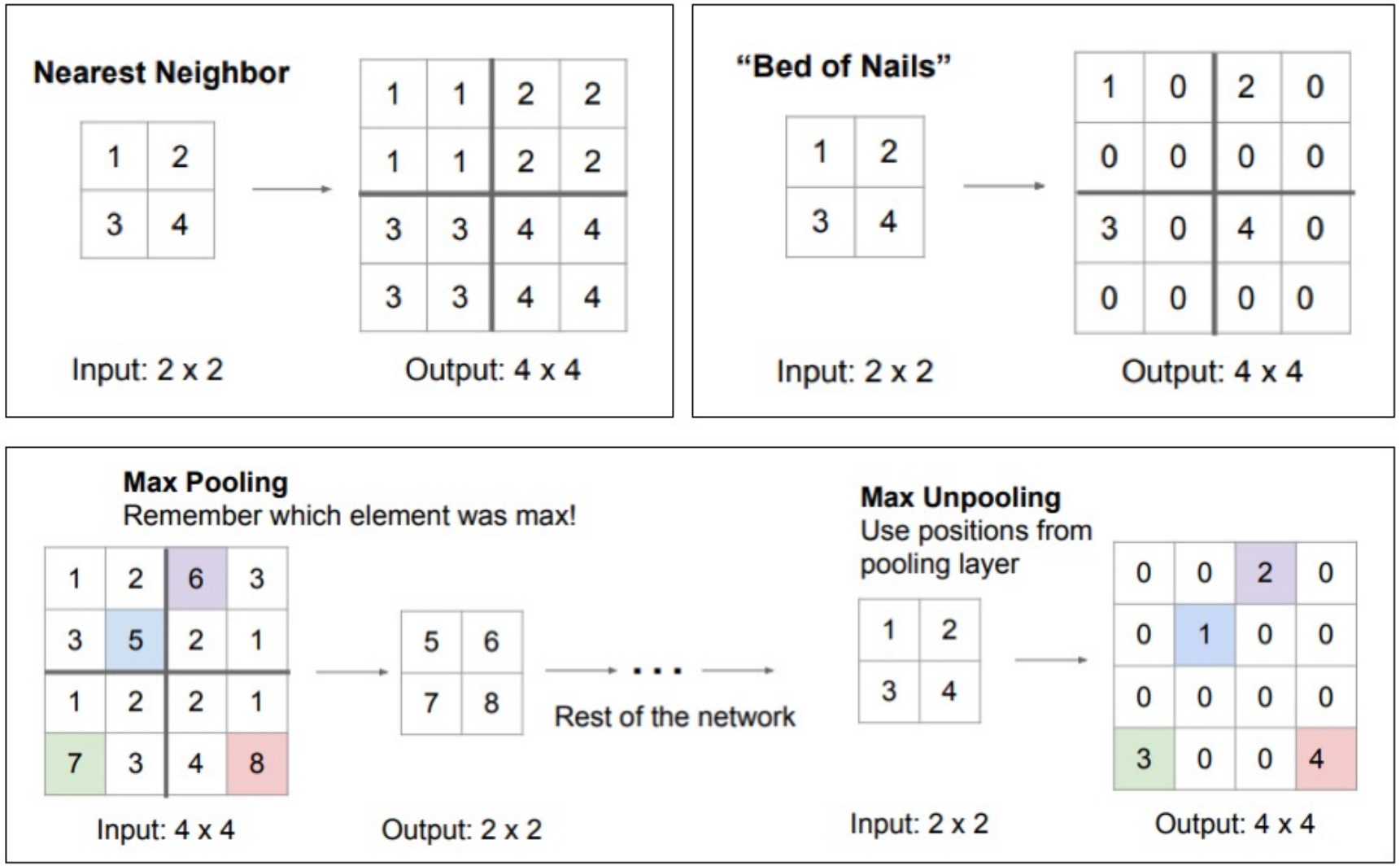
Architecture

Challenge: how to decode from coarse region classifications to per pixel classification?



Long, Shelhamer, and Darrell. Fully Convolutional Networks for Semantic Segmentation. CVPR 2015.

Architecture: Upsampling (Many Approaches)



Source: http://cs231n.stanford.edu/slides/2017/cs231n_2017_lecture11.pdf

Architecture: Upsampling (Transposed Convolutional Layer)

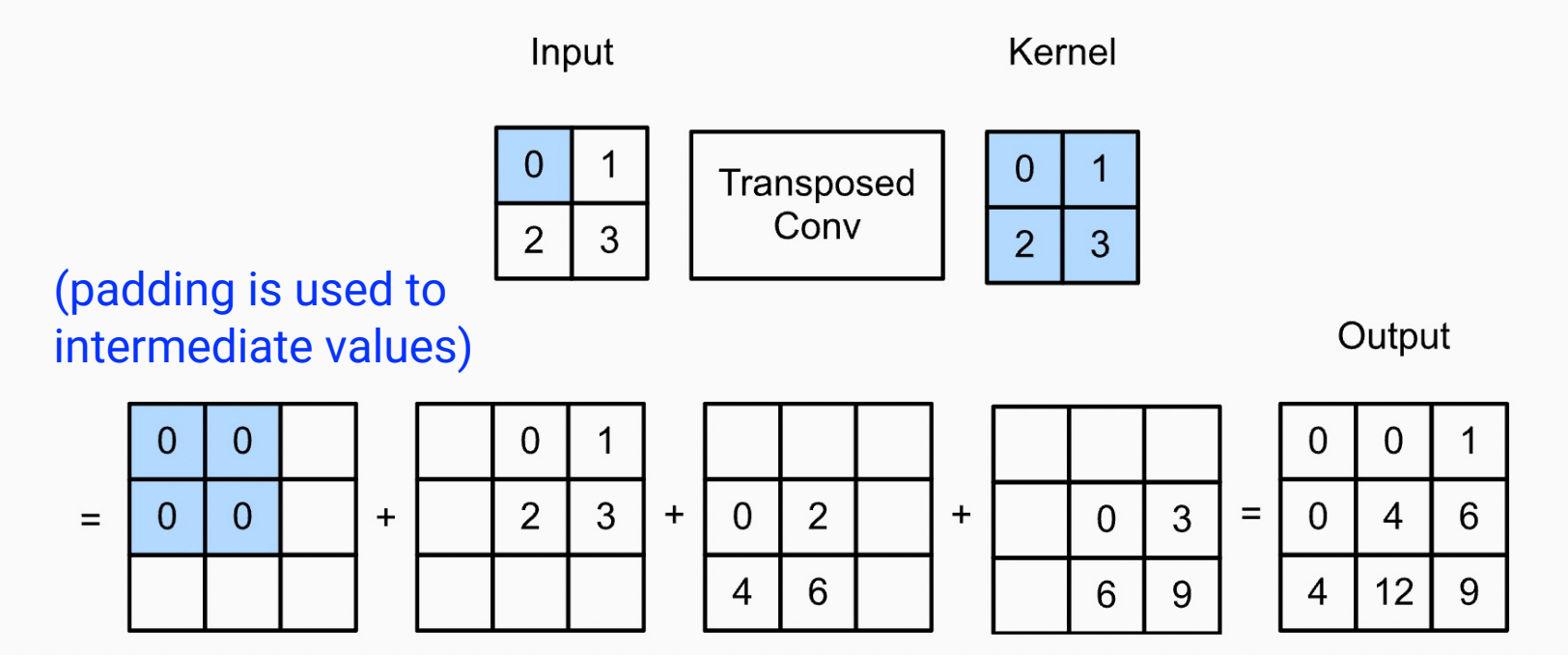
- Also called “fractional convolutional layer”, “backward convolution”, and, incorrectly, ”deconvolution layer”
- Idea: learn filters with a fractional sized stride to upsample the coarse image while refining it based on the filter values; e.g.,



https://www.machinecurve.com/index.php/2019/09/29/understanding-transposed-convolutions/#the-goal-reconstructing-the-original-input

Architecture: Upsampling (Transposed Convolutional Layer)

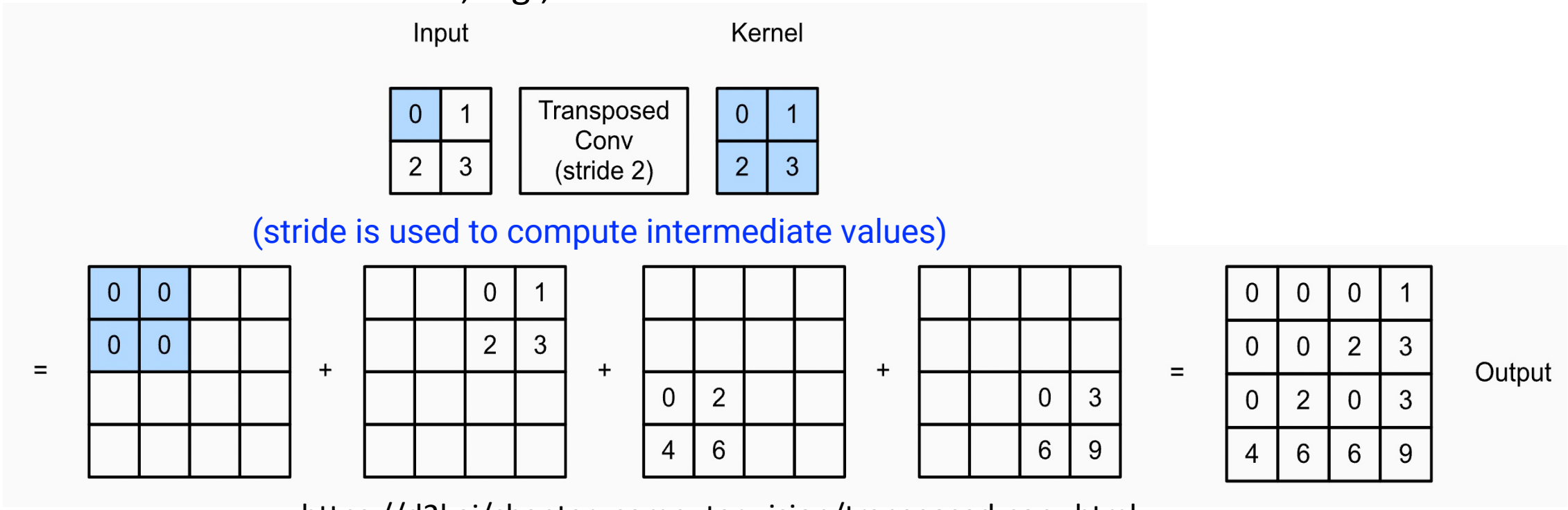
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https://d2l.ai/chapter_computer-vision/transposed-conv.html

Architecture: Upsampling (Transposed Convolutional Layer)

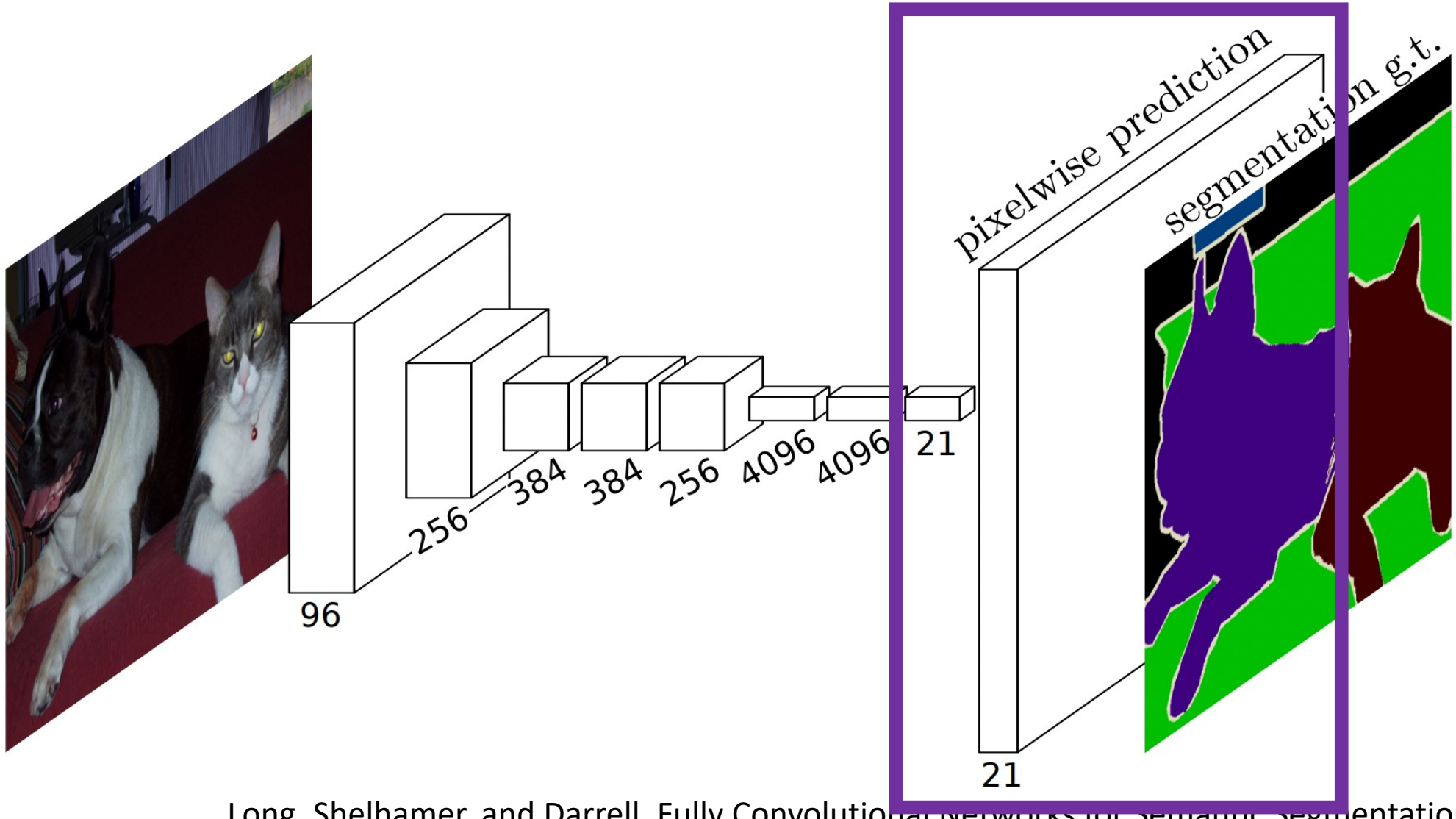
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https://d2l.ai/chapter_computer-vision/transposed-conv.html

Architecture

Next challenge: how to decode a **highly detailed** per pixel classification from the coarse region classifications?



Long, Shelhamer, and Darrell. Fully Convolutional Networks for Semantic Segmentation. CVPR 2015.

Architecture: Results

Next challenge: how to decode a **highly detailed** per pixel classification from the coarse region classifications?

Ground truth target



Predicted segmentation

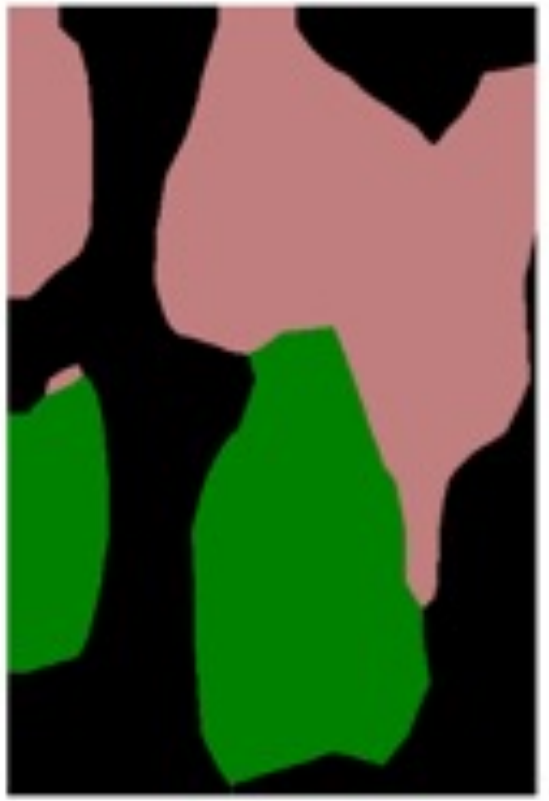
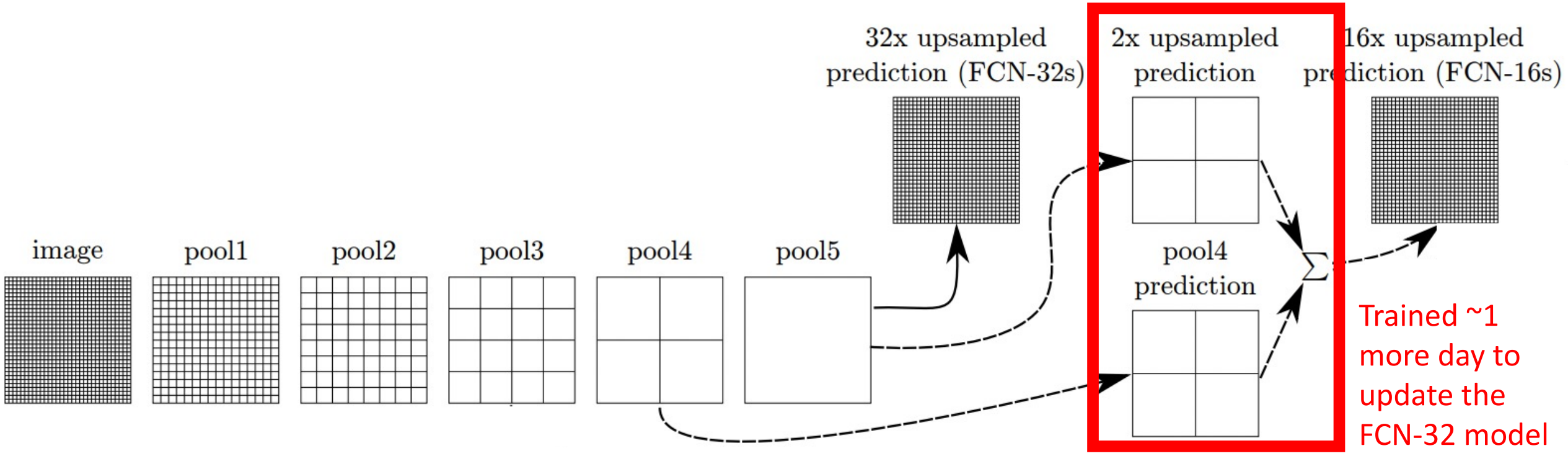


Figure source: https://www.jeremyjordan.me/semantic-segmentation/

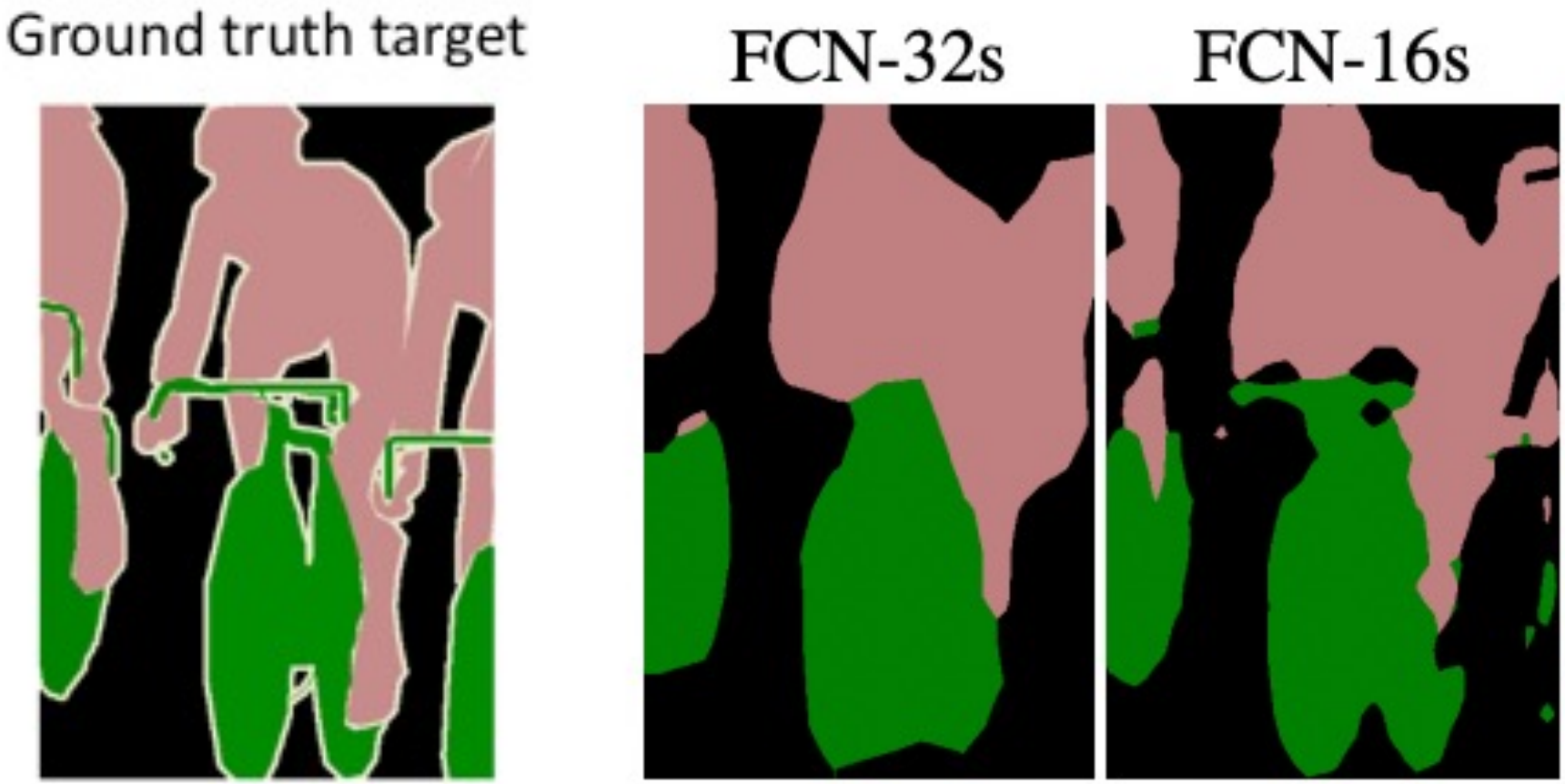
Architecture: Update to Use Skip Connections



FCN16: Sums predictions of lower-level, more fine-grained features (pool4) with the predictions at the coarser features

Long, Shelhamer, and Darrell. Fully Convolutional Networks for Semantic Segmentation. CVPR 2015.

Architecture: Results

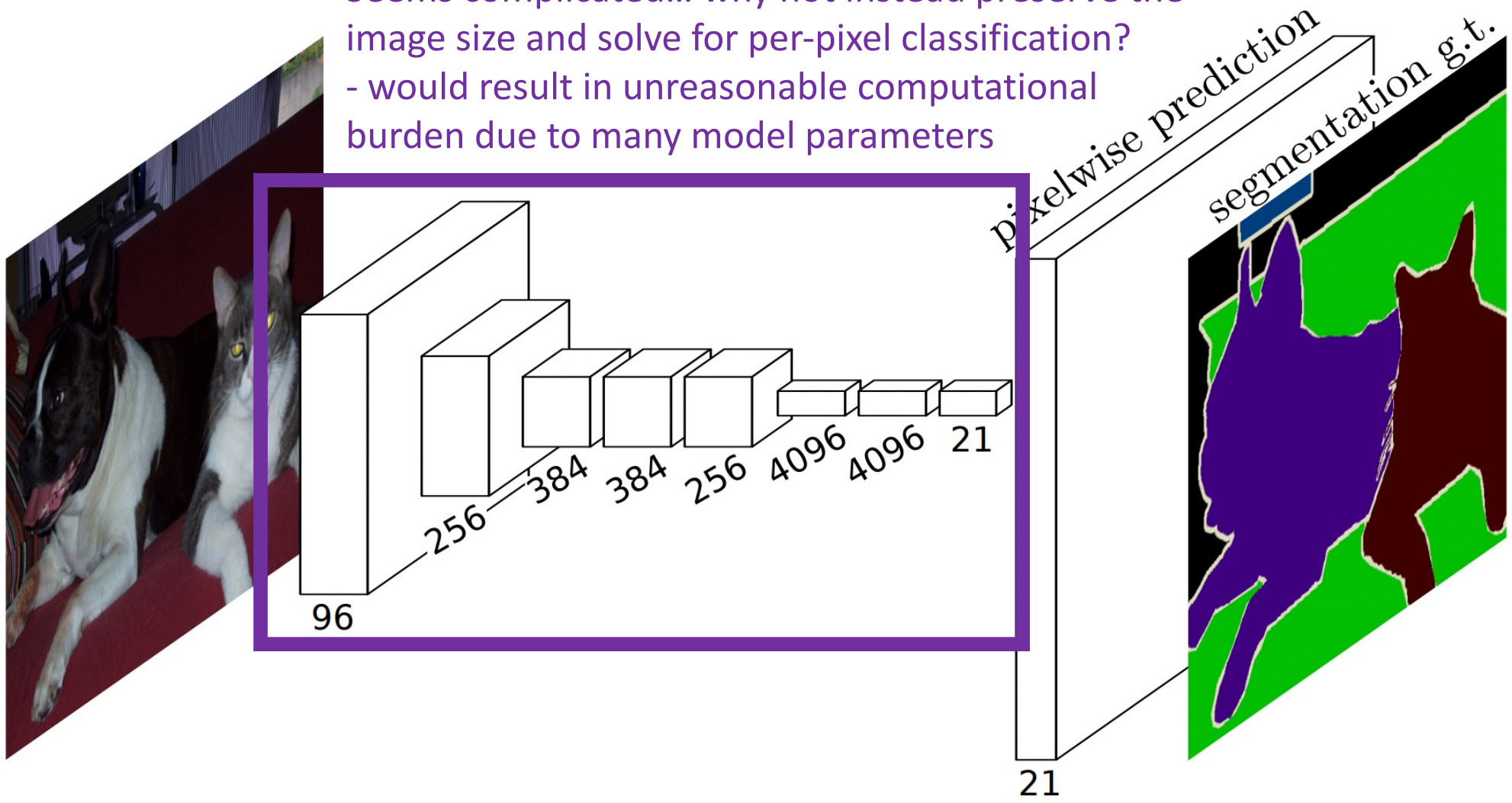


Skip connections support capturing finer-grained while retaining the correct semantic information!

Figure source: https://www.jeremyjordan.me/semantic-segmentation/

Architecture: Upsampling + Skip Connections

Seems complicated… why not instead preserve the image size and solve for per-pixel classification?
- would result in unreasonable computational burden due to many model parameters

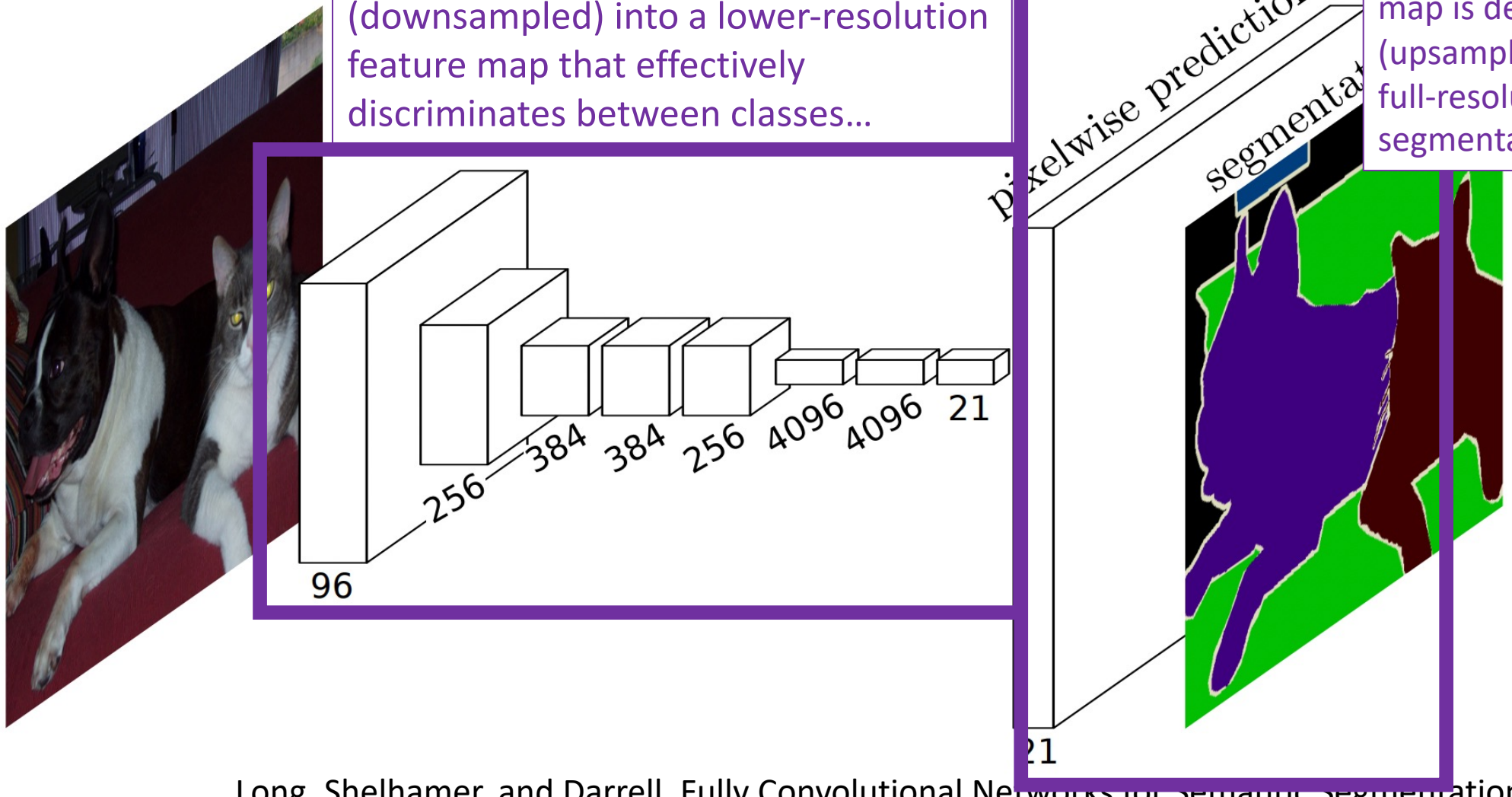


Long, Shelhamer, and Darrell. Fully Convolutional Networks for Semantic Segmentation. CVPR 2015.

Architecture: Encoder Decoder Architecture

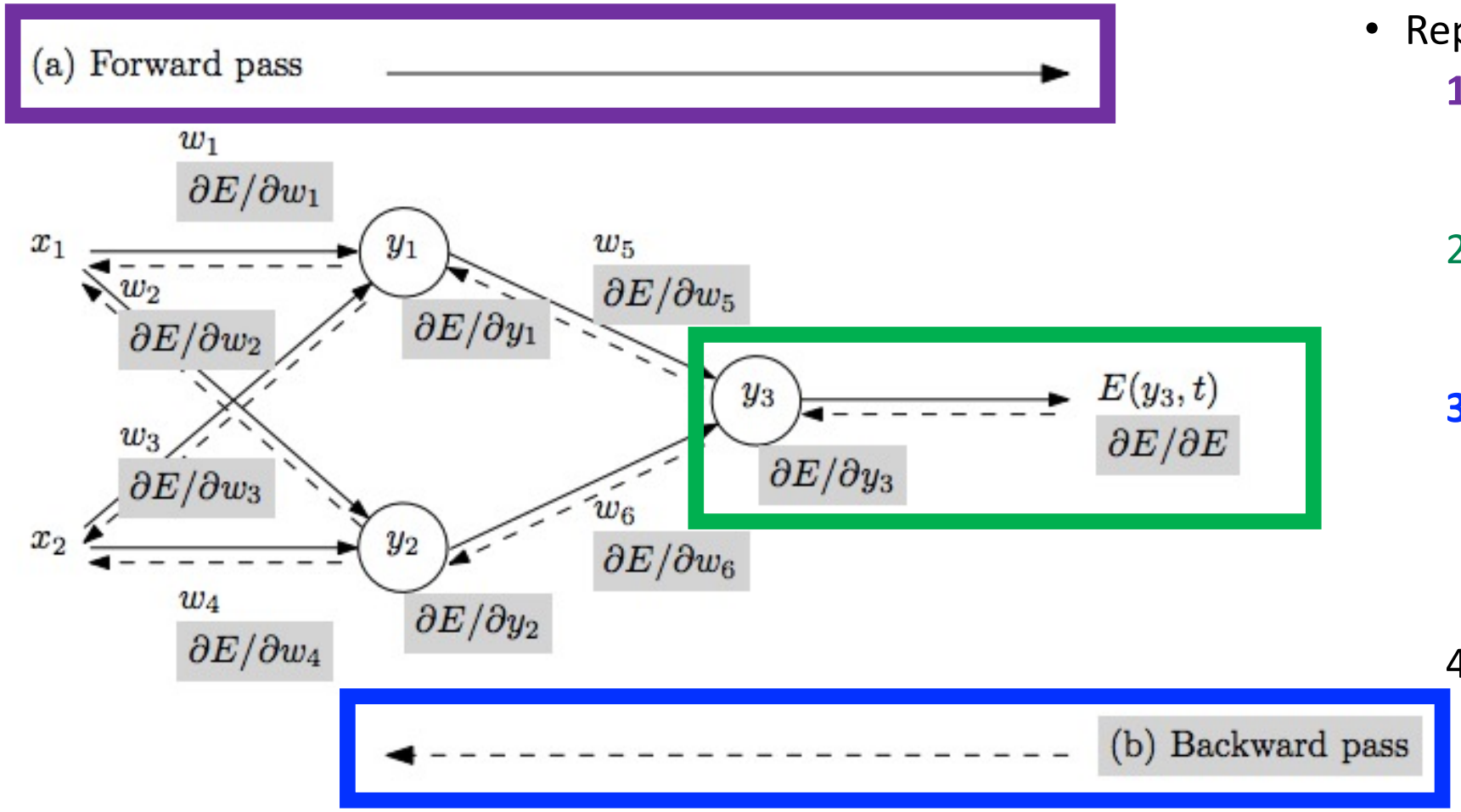
For efficiency, the image is encoded (downsampled) into a lower-resolution feature map that effectively discriminates between classes…

Then, the feature map is decoded (upsampled) into a full-resolution segmentation map.



Long, Shelhamer, and Darrell. Fully Convolutional Networks for Semantic Segmentation. CVPR 2015.

Training: Took 3 days on 1 GPU



- Repeat until stopping criterion met:
  1. **Forward pass**: propagate training data through model to make prediction
  2. Quantify the dissatisfaction with a model's results on the training data
  3. **Backward pass**: using predicted output, calculate gradients backward to assign blame to each model parameter
  4. Update each parameter using calculated gradients

Figure from: Atilim Gunes Baydin, Barak A. Pearlmutter, Alexey Andreyevich Radul, Jeffrey Mark Siskind; Automatic Differentiation in Machine Learning: a Survey; 2018

Training: How Neural Networks Learn

- Repeat until stopping criterion met:
  1. **Forward pass**: propagate training data through model to make prediction
  2. Quantify the dissatisfaction with a model's results on the training data
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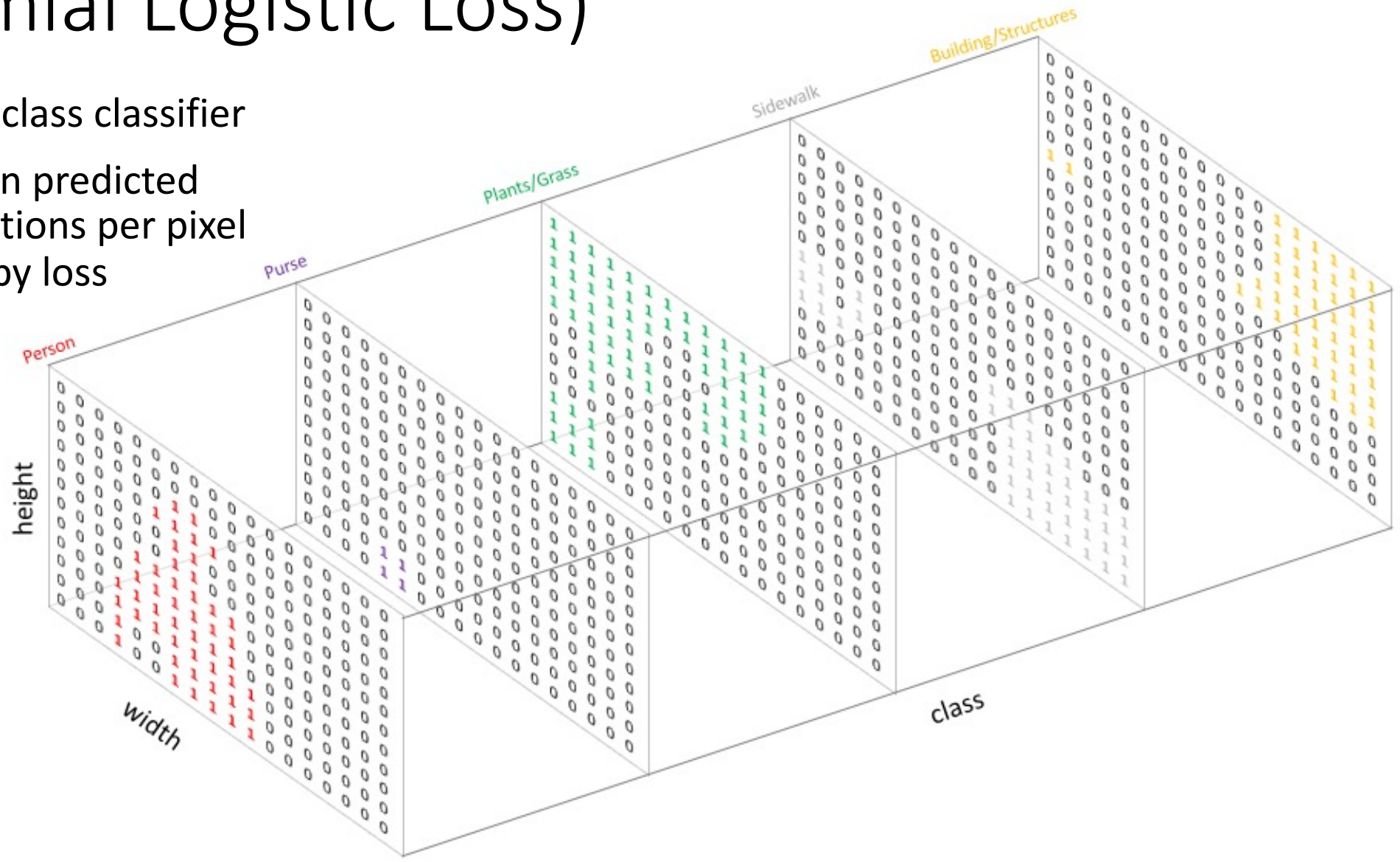
Sum across all pixels the distance between predicted and true distributions using cross entropy loss

Sum of gradients for all pixels (acts like a minibatch)

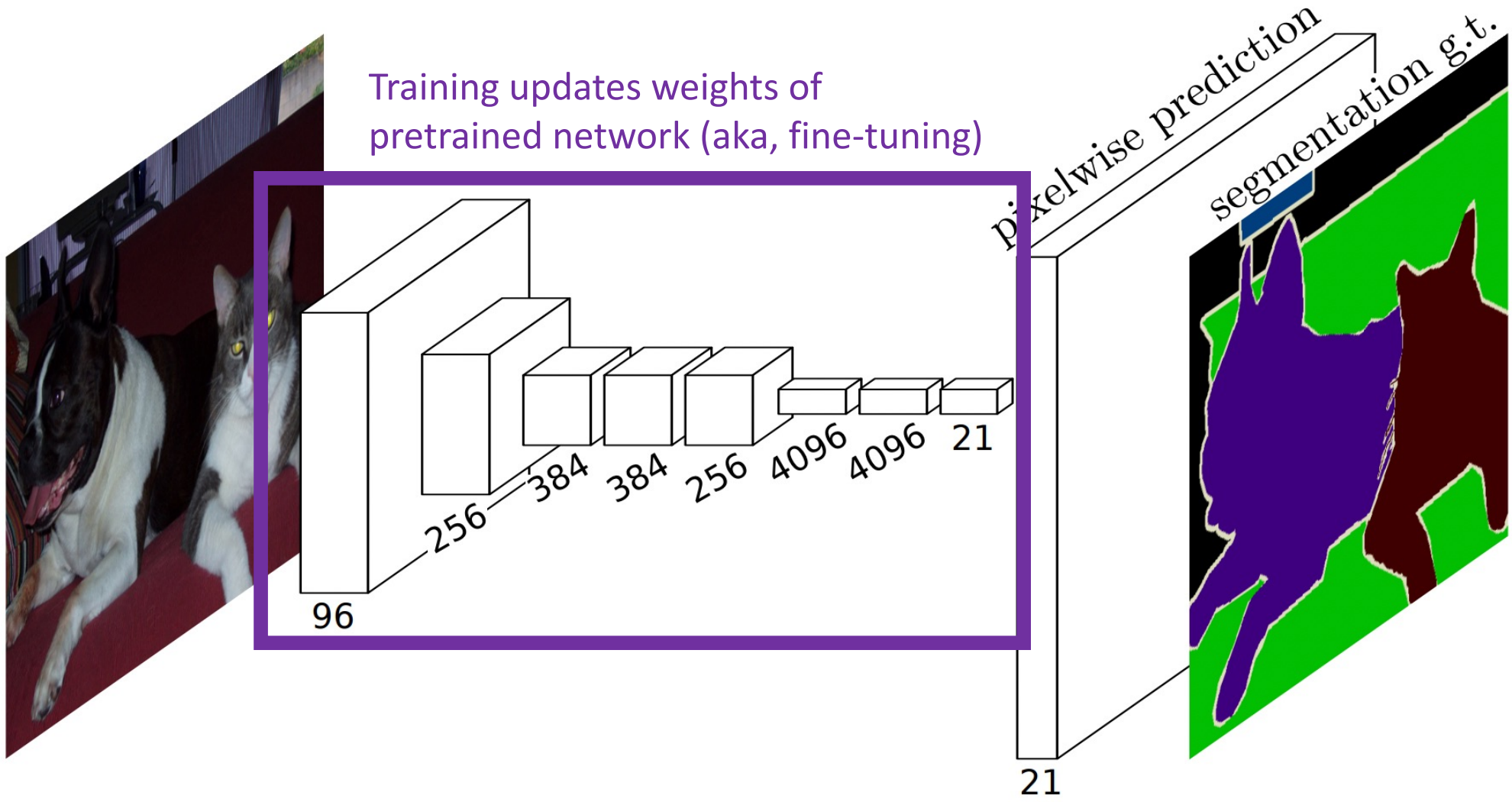
Figure from: Atilim Gunes Baydin, Barak A. Pearlmutter, Alexey Andreyevich Radul, Jeffrey Mark Siskind; Automatic Differentiation in Machine Learning: a Survey; 2018

Training: Cross Entropy Loss (Multinomial Logistic Loss)

- e.g., assume a 5-class classifier
- Distance between predicted and true distributions per pixel with cross entropy loss



Architecture: Algorithm Training



Long, Shelhamer, and Darrell. Fully Convolutional Networks for Semantic Segmentation. CVPR 2015.

Results

	mean IU VOC2011 test	mean IU VOC2012 test	inference time
R-CNN [12]	47.9	-	-
SDS [16]	52.6	51.6	$\sim$ 50 s
FCN-8s	**62.7**	**62.2**	$\sim$ **175 ms**

Compared to existing methods, produces better results at a faster speed!

Long, Shelhamer, and Darrell. Fully Convolutional Networks for Semantic Segmentation. CVPR 2015.

Today's Topics

- Problems
- Applications
- PASCAL VOC detection challenge: R-CNNs
- PASCAL VOC semantic segmentation challenge: fully convolutional networks


The End